When Near-Identity Compression Is Not Enough: An Audit of Post-Hoc Activation-Aware Uniform FFN Width Deletion with a Ridge Readout for Speculative Decoding

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Abstract

Speculative decoding rewards a draft for agreeing with the target's top-1 token. We audit a specific no-training baseline: activation-aware uniform FFN width deletion with a ridge readout, used as the target's own draft, on the evaluated raw English token population (WikiText/FineWeb/C4-style web and encyclopedic prose) under an in-sample top-2 audit at $c = 1$. On that scope the baseline is a conditional rejection no-go; the rejection layer carries the conclusion.

Retaining 99% of FFN width still changes 6.5–6.8% of top-1 decisions on WikiText/FineWeb (7.2% on C4), 27–28× the $c = 1$ FLOP break-even. The near-identity regime has a decision-boundary *resolution floor*: at 99.99% width, deleting one channel per layer still flips 1.2–1.4% of tokens, and same-corpus oracle selection reduces this floor by only 5–10%. A clean float32 rebuild removes the exact bf16 tie atom but leaves a 0.31% floor, so exact bf16 ties are not necessary for the floor, and the floor is not an artifact of the tested channel rankings.

Cost note (secondary, not a third finding). The cost layer is an implementation-conditional diagnostic. At 98–99% the measured clean kernel-time cost (a padded-kernel point estimate) is 5.004–5.28, at or above the break-even; this is an acceptance-independent diagnostic for the padded implementation, not a production-kernel claim. The rejection layer already closes these raw cells under the optimistic $c = 1$ roofline. The cost layer is load-bearing only for a few in-domain MATH cells (9B 50%, 27B 50–90%, Llama 50–70%).

The margin-mass rule is an exploratory triage diagnostic, not a certificate. In an exploratory stress test, a best-of-budget global search (under one A100 GPU-hour) on the tested Qwen3.5-9B WikiText/FineWeb cells over channel subsets, per-layer widths, and an OBS-style score does not reopen a raw cell. OBS and random lower point q but do not improve the certified lower bound; the default activation-aware uniform selector retains a larger certified margin than both. For the default, OBS, and random selectors, the single-cell lower endpoints are 1.4–18× $q^\star$, not a family-wise bound. Confirmatory evidence is 36 Qwen3.5-9B raw cells plus 9 independent Llama/K2 cells. The claim is limited to the evaluated token population, the measured operator, $c = 1$, and the measured batch-1 kernel; trained/adapted drafts, weight-only quantization, tree verification, arbitrary raw distributions, and compute-bound serving are outside it.

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Table 1: Key symbols. The full table, including batch-dependent quantities, is Table A.2.

Symbol	Meaning
q_g	greedy top-1 rejection $\Pr(\arg\max Q_x \neq \arg\max P_x)$
q_s	standard-sampling rejection $\mathbb{E}[\mathrm{TV}(P_x, Q_x)]$
q_{lb}	post-hoc top-2 lower bound $\Pr(m + \Delta_m < 0)$
$q^\star$	break-even rejection from the cost model
$q^\star_{\mathrm{FLOP}}, q^\star_{\mathrm{mem}}$	roofline break-even from nominal FLOP and byte ratios
$q^\star_{\mathrm{meas}}, q^\star_{\mathrm{free}}$	measured-kernel and free-FFN break-even diagnostics
c_{shape}	width-only shape ceiling $1/\rho_{\mathrm{FLOP}}$
ρ_1, τ_1	batch-1 draft pass ratio and target verification ratio
$\rho_{\mathrm{FLOP}}, \rho_{\mathrm{mem}}$	nominal FLOP and byte ratios of draft to target
γ	number of draft tokens per speculative step
$c, c^\star$	draft-kernel throughput advantage; minimum advantage that reopens a cell
m, Δ_m	target top-1/top-2 log-margin; draft perturbation of that margin
L	low-margin mass $\Pr(m < 0.5)$
π_{tie}	fraction of exact target top-1/top-2 ties
$\alpha, \bar{\ell}$	per-token acceptance $1 - q$; mean accepted length

Table 2: Abbreviations.

Abbreviation	Meaning
SD	speculative decoding
FFN	feed-forward network
MoE	mixture of experts
MLP	multilayer perceptron (the FFN block)
FLOP	floating-point operation
KV	key/value cache
LoCO	leave-one-corpus-out
TOST	two one-sided tests (equivalence)
OOD	out of distribution
bf16	bfloat16
K2-3.7B	K2-Horizon-3.7B (table abbreviation)
Qwen3.5-4B	hybrid 4B target; distinct from Qwen3-4B-Instruct-2507
GEMM	general matrix multiply

1 Introduction

The baseline. Speculative decoding accelerates a target model by proposing tokens with a cheaper draft and verifying them with the target [21, 5]. The draft is useful only if it is cheap *and* agrees with the target often enough. The natural no-training baseline is to compress the target itself: delete FFN channels post hoc, keep a single weight family, and use the compressed model as its own draft. This is the first step of trained self-draft pipelines—SD2 [20], T2D [32], and DistillSpec [44] prune or distill the target and then adapt on target responses—and it is the only option when target responses are unavailable. We audit it on measured raw English prose under an in-sample top-2 audit at $c = 1$ and a clean kernel-time batch-1 cost measurement.

The baseline is usually discussed as a preliminary step and rarely audited on its own terms. That is the gap this paper fills. We do not propose a new compressor or a new speculative-decoding theorem. We ask a narrower and more useful question: *if a practitioner skips adaptation and deploys the compressed target as a draft, what fails first, and under which conditions?*

Two layers, not one. A draft can fail for two logically independent reasons, and conflating them produces both overclaims and underclaims. The *rejection layer* fixes same-kernel efficiency ($c = 1$) and asks whether top-1 agreement is good enough to pay for the draft pass at all. This layer is a property of the operator: it is measured on tokens and does not depend on batch size or on which kernel implements the matmul. The *cost layer* asks whether the measured draft pass is cheaper than the target pass in a stated serving regime. This layer is a property of an implementation. We audit both, report them separately, and state what would change each one (Table 3).

Table 3: Claim–evidence–condition and protocol matrix. The top two rows separate the greedy top-1 audit from the exact-sampling audit: q_{lb} is an in-sample lower bound on greedy rejection, while $q_s = \mathbb{E}[\mathrm{TV}]$ is the evidence for standard speculative sampling. Both are measured on the evaluation tokens and both exceed $q^\star$ on measured raw English prose; tree verification is outside the claim. The full map is Table A.3 in Appendix A.2.

Claim	Evidence	Protocol	Condition / scope
Greedy rejection floor. *thm+meas*	Measured $q_g \geq q_{\mathrm{lb}} > q^\star_{\mathrm{FLOP}}$ on every raw cell. At 99% width, 16–19× the point threshold (14× at the family-adjusted lower endpoint).	Greedy top-1; in-sample.	Same-kernel $c = 1$; measured tokens; measured raw English prose; nominal $\phi_{\mathrm{MLP}} = 0.609$.
Exact-sampling rejection floor. *thm+meas*	Measured $q_s = \mathbb{E}[\mathrm{TV}] > q^\star_{\mathrm{FLOP}}$ on every raw cell. At 99% width, 21–24× the threshold.	Standard SD; in-sample.	Exact token-level rule; tree verification outside.
Batch-1 cost audit failure. *meas*	$4\rho_1 + \tau_1 = 5.004$–$5.28 \geq 5$ at 98% on every target (99% on 9B; one cell at 5.004). Zero-FFN 3.1–4.1 < 5, not a ceiling.	Measured clean kernel-time point estimate (padded narrow-MLP implementation).	Measured regime, $B = 1$; not a wall-clock theorem.
High-width reopening. *thm*	$c^\star \approx 1.2$–1.4 at 98–99%. Mid width 3–6× at $\tau = 1$.	Draft-kernel advantage c; bf16 tie floor.	Arithmetic conditional on a whole-pass draft-specific kernel; not realizable when the pass is shared. A free-FFN counterfactual would reopen all six targets.
Mechanism and screen. *explor.*	Low-margin entropy mass; LOCO MAE 0.0138 at 98%. No observed raw false negatives.	One target forward pass.	Triage, not a guarantee; below-threshold ⇒ direct audit.
Global search stress test.	Global selectors lower point q but do not increase the certified margin; OBS and random lower point q but not q_{lb}, and the default retains a larger margin than both. For the default, OBS, and random selectors, $q^{\mathrm{lo}}_{\mathrm{lb}}$ remains 1.4–18 × $q^\star$ at 30–99% width (single-cell intervals).	Global subset, per-layer DP, sensitivity DP, random allocation check; same caches and audit.	Best-of-budget heuristic on Qwen3.5-9B WikiText/FineWeb; not a global-optimum proof.
Out of scope. *lit+meas*	Trained drafts (MTP control), weight-only quantization, retrieval, compute-bound serving.	—	Different objects; the audit failure does not cover them.

What we find. Two rejection results organize the paper, followed by a nested cost note.

(1) Near-identity compression already fails the rejection test. On the evaluated raw corpora, retaining 99% of FFN width—deleting 1% of the intermediate channels—still changes 6.5–6.8% of greedy top-1 decisions, 27–28× the $c = 1$ FLOP break-even. The gap is not a bad channel ranking: same-corpus held-out selection reaches 4.9–5.6%, and an evaluation-corpus oracle upper bound reaches 4.3–5.3%, both 9–11× the threshold.

(2) The high-width residual is a resolution floor, not bulk information loss. At 99.99% width—one deleted channel per layer—rejection is still 1.2–1.4%, 500–550× the break-even, and oracle selection reduces it by only 5–10%. This is a mechanism diagnostic, not a deployable configuration: the FLOP saving at one channel per layer is negligible; the point is that near-identity perturbation still flips low-margin tokens. The target's top-1/top-2 margin has a 3.1% atom of exact bf16 ties and a 3.7% near-boundary mass below 0.1; any nonzero perturbation flips part of this mass, and the flip is concentrated in the tie and small-margin bins. A clean float32 rebuild removes the exact tie atom but leaves a 0.31% floor (129× the float32 break-even), so the floor is a near-boundary mass phenomenon, not a bf16 tie-order artifact. Channel ranking matters at 99% (oracle 3.7% versus default 6.8% on the main target), but it cannot remove the floor. This is the general lesson of the paper: *top-1 agreement under a compression operator does not degrade continuously with reconstruction error; it has a floor set by the target's decision-boundary mass.*

Cost note. The cost layer is nested inside the rejection layer. Under the optimistic $c = 1$ roofline, width deletion always leaves a positive break-even threshold $q^\star_{\mathrm{roof}} > 0$, so a frontier no-go is a rejection statement. At 98–99% width the measured raw rejection is 13–20× the optimistic FLOP threshold, so every target is roofline-bound at 98–99% width; the measured clean kernel-time cost is also 5.004–$5.28 \geq 5$, an additional acceptance-independent implementation diagnostic (point estimate), while a zero-FFN counterfactual is 3.1–4.1. At 50% the cells are R-free: even a free FFN fails. The cost-load-bearing cells are the 9B MATH 50% point (G), the 27B MATH 50–90% points (K), and the Llama MATH 50–70% points (K); the Llama margin is the tightest, with a 0.4–2.3% shape margin inside the timing spread. A portable fused-MLP kernel (1.2–2.7×) moves the relative ratio toward one and does not reopen an R cell; only a draft-exclusive kernel can move a K cell. A best-of-budget global stress test on the tested Qwen3.5-9B WikiText/FineWeb cells over channel subsets, per-layer widths, and an OBS-style score does not reopen a raw cell (Section 5.5); it lowers

point q but does not increase the certified lower bound. OBS and random lower point q without improving the certified lower bound; the default retains a larger certified margin than both. For the default, OBS, and random selectors the single-cell lower endpoints are 1.4–18× $q^\star$.

From mechanism to a deployment screen. The binding variable is the target's *low-margin entropy mass*. Measured raw English prose has 22–28% of tokens with top-1/top-2 log-margin below 0.5, versus 3.3% on MATH; reweighting MATH to the WikiText entropy histogram raises its low-margin mass from 0.033 to 0.343, and the reverse reweighting lowers WikiText from 0.219 to 0.008. The same quantity yields a cheap triage rule: estimate $L = \Pr(m < 0.5)$ on a deployment prefix, predict $q = a_w + b_w L$, and measure the certificate directly when the prediction is below threshold. The screen has leave-one-corpus-out MAE 0.0138 at 98% width and no observed false negatives on eleven raw corpora, but the coefficients are per-target and the rule is deliberately asymmetric: a prediction above threshold is evidence of a audit failure, while a prediction below threshold only triggers a direct audit.

Contributions. In one sentence: the cheapest post-hoc self-draft is rejected by the target's own decision boundary on measured raw English prose, and that operator failure is logically separate from implementation cost. Concretely: (1) A rejection-layer audit of the measured operator: on measured raw English prose (WikiText/FineWeb/C4-style web and encyclopedic prose) its top-1 rejection is above the $c = 1$ FLOP break-even at every compressed width, and the near-identity floor is a decision-boundary resolution effect (bf16 ties plus near-boundary mass), not a channel-ranking artifact. This is the confirmatory claim. (2) A cost-layer taxonomy: a batch- and hardware-invariant roofline bound, a portability/Amdahl bound, and ordered W/K/G/R thresholds. The measured cost is an implementation-conditional diagnostic; at 98–99% the cells are already R by rejection, and the K cells we identify are Llama MATH 50–70% and 27B MATH 50–90%; the 27B 90% point has the larger margin. (3) An exploratory margin-mass triage diagnostic with a mandatory direct audit; it is not a deployment certificate. (4) A descriptive breadth study across eleven raw corpora, SWE-bench and MT-Bench, six model families, and FFN/depth/head/MoE operators, together with the negative results that rule out channel ranking, readout refitting, non-uniform allocation, and several adaptation controls as repairs.

Scope and what we do not claim. The confirmatory rejection no-go is for the measured activation-aware uniform FFN width-deletion operator with a ridge readout on the measured raw English prose suite, under $c = 1$ and the measured batch-1 kernel. Depth, head, and MoE operators are descriptive operator-zoo checks, not members of the confirmatory family. Confirmatory evidence is the $36 + 9$ family; the 200-cell raw count, 218-cell audit, 55-cell breadth suite, and 99/135 cross-family counts are descriptive or exploratory. The claim is not that compression is useless, that structured pruning is bad for perplexity, that trained drafts fail, or that every engine sees the same wall clock. Trained or pretrained drafts, weight-only quantization, tree/top-k drafting, retrieval, arbitrary raw distributions, and compute-bound serving are different objects; Section 2 positions them and Section 10.4 lists the conditions that would move each layer. "Raw" is operational: pretraining data are not public, so we report target NLL and calibration n-gram overlap rather than a contamination certificate (Appendix A.30). The lower bound is in-sample; its formal transfer statement is the TV-Lipschitz bound of Proposition 3. The cost numbers are measured on one A100 with a padded narrow-MLP implementation and are reported as an upper bound on relative cost, not as a wall-clock theorem.

Reader's map. Table 3 is the claim map. The main width sweep is Table 7; the terminology is Table 5; the floor mechanism is Tables 9–8; the global-search stress test is Table 10; the cost layer is Tables A.11–A.13; the fitted screen coefficients are in Table A.82, the leave-one-corpus-out validation in Table A.79, and the cross-model transfer in Table A.8. All remaining result tables are supporting or appendix evidence, and only the $36 + 9$ cells are confirmatory.

Roadmap. Section 2 positions the audit against trained drafts, structured pruning, and quantization. Section 3 defines the two layers and the resolution model. Section 4 gives the protocol and the statistical plan. Section 5 presents the rejection layer; Section 6 the cost layer; Section 7 the deployment screen. Section 8 collects the breadth and negative results, Section 10 the implications and the conditions that would change

the conclusion, and Section 11 the limitations. The appendix contains proofs, full width sweeps, extended cost protocols, and reproduction notes.

2 Related Work

Speculative sampling preserves the target distribution while accelerating decoding [21, 5]; variants include Medusa [4], EAGLE [22, 23, 24], multi-token prediction [8], tree verification [31, 7], and adaptive/online draft selection [15, 26]. Our γ scan covers the draft-length axis; online adaptation is a trained-draft method outside the post-hoc operator. The cost of SD is a draft/verification trade-off [21, 36]; Su et al. [36] make the batch dependence explicit, and Liu et al. [27] measure it in a production engine.

Self-speculative decoding reuses the target: Draft & Verify, LayerSkip, Kangaroo, SWIFT, and BiLD [41, 10, 25, 39, 19] are trained or layer-selection methods, outside our post-hoc width operator; our depth operator drops trailing blocks and is a compression baseline, not an early-exit schedule. Retrieval and prompt-lookup drafts [13, 35] are the $\rho \to 0$ limit and have no compressed target weights to certify. Structured pruning and low-rank factorization [11, 28, 37, 2, 29, 40] are usually evaluated by perplexity; we evaluate top-1 agreement and its cost. Weight-only quantization is a memory-only axis: QSpec [43], QuantSpec [38], ML-SpecQD [12], Quasar [14], and W4A16 [42] report 1.2–2.7× speedups at 80–98% acceptance; they change precision, not the FLOP ratio of an FFN/depth/head/MoE operator, so our audit failure does not cover them. The closest trained alternatives are SD2 [20], DistillSpec [44], and T2D [32]: they prune or distill the target and then adapt on target responses, reporting strong in-domain acceptance. Their objective is empirical risk on a target-response dataset, which controls the target only on that dataset's prefix distribution; Section 3.4 makes the deployment-distribution gap explicit.

Positioning. Liu et al. [27] measure batch dependence in a production engine; we measure the distribution robustness of a post-hoc operator. The two are complementary. This is not a new compression algorithm and not a new speculative-decoding theorem: it is a controlled audit of a natural baseline that the trained-draft literature starts from. The ingredients are standard; the deliverable is the composition (a lower bound that survives readout refitting, a batch-independent roofline test, and a deployment screen), the negative result on measured raw English prose, and the boundary between the operator-specific rejection floor and the measured-regime cost. We report the audit and its conditions rather than a single wall-clock number.

2.1 Extended positioning and measurement methodology

Self-speculative decoding. The closest structural relatives reuse the target instead of training an independent draft. Draft & Verify [41] trains a smaller model on target outputs; LayerSkip [10] exits early and verifies with the full model; Kangaroo [25] uses a shallow subnetwork and a trained adapter; SWIFT [39] combines a small draft with a trained head; BiLD [19] uses a small model and a fallback rule. These methods either train a component or change the inference schedule. Our operator changes weights only, deletes FFN channels, and does not add a head. The depth operator in our zoo is a compression baseline (drop trailing blocks), not an early-exit schedule, because an early exit needs a confidence rule and a rollback policy that our post-hoc operator does not have.

Structured pruning and low-rank factorization. SparseGPT [11], LLM-Pruner [28], SliceGPT [2], ShortGPT [29], Sheared LLaMA [40], and simple low-rank baselines [37] are usually evaluated by perplexity or downstream accuracy. Those metrics reward preserving the language model; speculative decoding rewards top-1 agreement with the target and pays for a draft pass. A compression that is excellent by perplexity can still be a poor draft if it moves the top-1 decision on low-margin tokens. We therefore evaluate the operator on the metric it would be deployed under, and we report the reconstruction and oracle controls that separate "the compressor is bad" from "the metric is different". The shared-rotation baseline in our zoo is not a fully optimized SliceGPT pipeline; it is an unfused rotation that gives $q = 1$ at every rank, and we say so rather than treating it as a strong pruning baseline.

Quantization and speculative decoding. Weight-only quantization is a different axis. QSpec [43], QuantSpec [38], ML-SpecQD [12], Quasar [14], and W4A16 [42] keep the architecture and change numerical precision, often with a quantized draft or a quantized KV cache. They report 1.2–2.7× speedups at 80–98% acceptance. Our FLOP-ratio audit failure does not cover them, and our own packed INT4/INT8 kernels are not faster than bf16 at batch 1. We treat quantization as a different, memory-only axis, not as a failed instance of our operator, and we do not claim a measured end-to-end win here.

Draft scaling and negative results. The earlier off-the-shelf capacity scan (Appendix A.18) found shallow out-of-domain returns when scaling pretrained drafts from 0.8B to 27B: in-domain acceptance improves, but the measured raw-text rejection floor falls slowly and the cost ratio grows with draft size. That scan is the complementary direction to this paper. It also motivates the negative result: if scaling a pretrained draft has shallow OOD returns, compressing a target into a draft should not be assumed to work just because the draft shares the target's weights. Negative results are useful when they identify the binding variable; here the binding variable is the target's low-margin token mass, not the reconstruction error of the compressor.

Measurement methodology. We separate a certificate layer (same-kernel, $c = 1$) from a cost-only layer (realized kernel), report confidence intervals and family-wise corrections, and state what would change each layer. This is a response to a common failure mode in efficiency papers: a wall-clock number measured on one engine is presented as a property of an operator. The two-layer structure is our attempt to make the claim falsifiable at both levels. The claim–evidence–condition map in Appendix Table A.3 is the compact form of the same idea.

3 The audit

We separate two statements that are often conflated. The *rejection layer* fixes same-kernel efficiency $c = 1$ and asks whether top-1 agreement is good enough. The *cost layer* asks whether the measured draft pass is cheap enough at batch 1. The rejection floor is operator-specific: q is a top-1 agreement property and does not depend on batch size or kernel. The cost failure is regime-specific: it is a statement about a measured implementation, not a wall-clock theorem for every engine. This section defines both layers and a resolution model for the near-identity regime; proofs and the full assumptions are in Appendix A.1.

3.1 Operator and perturbation

A post-hoc operator C maps the target f to a draft f_C using only target weights and a calibration set, with nominal compute ratio $\rho_C = \mathrm{FLOPs}(f_C)/\mathrm{FLOPs}(f)$. A support-selection principle (Proposition 1 in Appendix A.1) says that an exact all-readout reduction must delete units whose contribution is identically zero; for a fixed readout, exactness can also hold through a nongeneric output-inactive or functionally redundant unit (Lemma 4). Thus every compute-reducing post-hoc width operator is approximate; the question is how much the approximation costs in top-1 agreement.

For a Qwen-style block $y = W_d(\sigma(W_g x) \odot W_u x)$, deleting channels S gives

$$y_S(x) = y(x) - \Delta y(x), \qquad \Delta y(x) = \sum_{j \in S} a_j(x)\, W_d[:, j], \tag{1}$$

with $a(x) = \sigma(W_g x) \odot W_u x$. We choose S with the activation-aware score $s_j = \mathbb{E}|a_j|\, \|W_d[:, j]\|_2$ and study uniform and non-uniform per-layer allocation. Lemma 2 shows that an exact all-readout width reduction forces $r = h$ (a signed coordinate permutation), so no nontrivial exact width reduction exists; it motivates the family and is not used to prove the audit failure. The deployed statement is the measured lower bound below.

3.2 The rejection layer: definitions

Let $a(x), b(x)$ be the target top-1 and top-2 tokens, $m(x) = \log P_x(a(x)) - \log P_x(b(x))$ the log-margin, and $\Delta_m(x)$ the draft's perturbation of it.

Lemma 1 (Post-hoc top-2 lower bound)**.** *If $m + \Delta_m < 0$, the draft ranks the target's second token above its first, so the token is rejected:*

$$q_g \;\geq\; q_{\mathrm{lb}} := \Pr\big(m(x) + \Delta_m(x) < 0\big). \tag{2}$$

The bound is deterministic on the measured tokens and computable from the same cache as q_g; it is an in-sample post-hoc lower bound, not a pre-deployment predictor. A nested top-k version gives $q_{\mathrm{lb}}^{(2)} \leq q_{\mathrm{lb}}^{(K)} \leq q_g$. The protocol-independent TV floor $q_s = \mathbb{E}[\mathrm{TV}]$ lower-bounds the rejection of any exact token-level rule, and standard speculative sampling attains that bound. q_s and q_g are not ordered in general—a draft can disagree on the argmax while having small TV, or vice versa—so we report both and do not treat $q_s \leq q_g$ as a theorem. On the measured cells $q_s < q_g$, but that is an empirical property of this draft family. Tree verification is outside this claim because its proposal is a tree-induced distribution. The audit threshold for the rejection layer is the $c = 1$ FLOP break-even $q^\star_{\mathrm{FLOP}}$ from (8); a cell is *audit-passed* when the lower endpoint of a sequence-cluster interval for q_{lb} exceeds $q^\star$.

3.3 The decision-boundary resolution model

At high width the draft is a small perturbation of the target, and the rejection rate is governed by the target's decision boundary rather than by bulk information loss. Let $\Delta(x)$ be the perturbation of the top-1/top-2 logit difference. The draft flips the token if m and $m + \Delta$ have opposite signs, or if it promotes a token outside the target top-2. At 99–99.99% width the second term is small: the measured q is within 0.1–1.5 points of $\Pr[\mathrm{sign}(m) \neq \mathrm{sign}(m + \Delta)]$ (Figure 2b). Deleting channels changes the margin by

$$\Delta(x) = \sum_{j \in S} a_j(x)\,\big(W_d[a(x), j] - W_d[b(x), j]\big), \tag{3}$$

which is token-specific even when the average score $\mathbb{E}|a_j|$ is small. The target margin has two relevant components: an atom of exact bf16 ties $\pi_{\mathrm{tie}} = \Pr(m = 0)$ and a near-boundary mass $B(\varepsilon) = \Pr(0 < |m| < \varepsilon)$. The exact sign-flip event is $E_{\mathrm{sign}} = \{\mathrm{sign}(m) \neq \mathrm{sign}(m + \Delta)\}$. The tie atom gives the certified lower bound

$$q(w) \;\geq\; \pi_{\mathrm{tie}}\,\Pr(\mathrm{flip} \mid m = 0). \tag{4}$$

The near-boundary mass gives a superset of the non-tie sign flips, hence an upper-bound decomposition, not a lower bound:

$$q(w) \;\leq\; \pi_{\mathrm{tie}}\,\Pr(\mathrm{flip} \mid m = 0) \;+\; (1 - \pi_{\mathrm{tie}})\,\Pr\big(0 < |m| < |\Delta(w)|\big) \;+\; \text{promotion}. \tag{5}$$

We use (5) as a heuristic decomposition of where the flips concentrate, not as a certified inequality; the certified statement is the measured q_{lb}. The floor as $w \to 1$ is set by the smallest nonzero deletion, not by the bulk channel information: as long as $|\Delta|$ exceeds the margin on a positive-measure set, the measured q stays above the vanishing $q^\star$. This is the model we test in Section 5; it predicts that the near-identity failure is insensitive to the tested channel rankings but sensitive to width and to the numerical realisation of the boundary.

3.4 The cost layer: definitions

Let $T_T(B, n)$ and $T_D(B, n)$ be target and draft times. A speculative step over γ tokens costs $\gamma T_D(B, 1) + T_T(B, \gamma + 1)$ and emits A tokens, where the target alone needs $AT_T(B, 1)$ passes. Hence

$$\rho_B = \frac{T_D(B, 1)}{T_T(B, 1)}, \qquad \tau_B = \frac{T_T(B, \gamma + 1)}{T_T(B, 1)}, \tag{6}$$

and

$$S_B = \frac{A(1 - q, \gamma)}{\gamma\rho_B + \tau_B}, \qquad A(\alpha, \gamma) = \sum_{i=0}^{\gamma} \alpha^i. \tag{7}$$

Break-even $S_B \geq 1$ is

$$A(1 - q, \gamma) \;\geq\; \gamma\rho_B + \tau_B. \tag{8}$$

For fixed rules A is decreasing in q, so there is a unique threshold $q^\star$ when the right side is below $\gamma + 1$. Define the nominal ratios $\rho_{\mathrm{FLOP}} = F_D/F_T$ and $\rho_{\mathrm{mem}} = M_D/M_T$ and $\rho_{\mathrm{roof}} = \min(\rho_{\mathrm{FLOP}}, \rho_{\mathrm{mem}})$. Under the same-kernel, equal-efficiency assumptions, $q^\star(\rho_B, \tau_B, \gamma) \le q^\star_{\mathrm{roof}}(\rho_{\mathrm{roof}}, \gamma)$; the padded batch-1 implementation is an upper bound on relative cost, so the true implementation lies between the roofline lower bound and the measured upper bound.

For a draft-side kernel advantage c, the relative pass ratio is ρ_B/c and break-even is $A(1-q,\gamma) = \gamma\rho_B/c + \tau_B$. We report the minimum c that reopens a cell. If the draft pass has an MLP share s of kernel time, an FFN-only speedup c_{FFN} maps to a whole-pass advantage

$$c = \frac{1}{(1-s) + s/c_{\mathrm{FFN}}}. \tag{9}$$

The free-FFN ceiling is $c_{\max} = 1/(1-s)$; a cell with $c^\star > c_{\max}$ is closed to every FFN-only kernel under the measured conversion. The stronger shape ceiling is $c_{\mathrm{shape}} = 1/\rho_{\mathrm{FLOP}}$: a width-only kernel cannot beat the roofline of the narrow FFN, and a portable kernel improvement cancels from the relative comparison (Proposition 8). The cost verdicts are ordered $q^\star_{\mathrm{meas}} \le q^\star_{\mathrm{regime}} \le q^\star_{\mathrm{FLOP}}$, with the measured free-FFN threshold reported as a separate diagnostic (Appendix A.5). Under the optimistic frontier, width deletion always leaves a positive acceptance window (Corollary 1), so a frontier no-go is a rejection statement, not a kernel statement. This is what separates the operator-level R/R-free verdicts from the implementation-conditional K/G verdicts.

Deployment coverage. A draft is deployed on a prefix distribution π_{dep}, not on the evaluation set μ. The transfer statement we use is the TV-Lipschitz bound $|q^\pi_{\mathrm{lb}} - q^\mu_{\mathrm{lb}}| \le \mathrm{TV}(\pi, \mu)$ (Proposition 3). Fitting target responses on μ controls only μ; this separates the post-hoc audit failure, which sees the full target weights, from the open distillation question. The exactness statements and cost assumptions are in Appendix A.1.

3.5 Operator taxonomy and scope of the theory

Why the exactness result is a motivation, not the deployed claim. Lemma 2 assumes a shared intermediate basis and equality for all inputs and all block weights. It rules out a general linear projection as an exact replacement for the gated FFN: the Jacobian is $P^\top \mathrm{diag}(d(Pz))P$, and equality for all readouts forces the off-diagonal terms to vanish. The deployed operator does not satisfy the all-readout quantifier; it deletes channels and refits a readout. For a fixed readout, Lemma 4 gives the exact condition $W_{d,D}a_D = (R - W_{d,K})a_K$; output-inactive channels and an exact functional redundancy are the two escapes, both nongeneric algebraic conditions. We do not formalize a generic-weight quantifier for the deployed $h > d$ regime; the earlier full-row-rank sketch does not apply there. Lemma 6 then gives an exact decomposition of the deployed map, so the audit is measured on the composed operator rather than on the idealized pruning step. This is the sense in which the theory motivates and explains the operator without claiming that the operator is exact.

What the support-selection principle covers. The principle says that an exact all-readout reduction must delete units whose contribution is identically zero. It applies to FFN width, MoE active experts, attention heads, and depth, because each is a compute-bearing unit with a readout. The instantiations differ in what "delete" means: an FFN channel has a nonzero activation almost surely, a routed expert may be unselected, a KV group can be removed with a projection, and a trailing block can be dropped only if the residual stream is preserved. Table 4 summarizes the taxonomy.

Table 4: Post-hoc compute-reducing operators and the support-selection principle. Exact reduction is possible only by deleting units that are identically inactive; every nontrivial compute-reducing operator therefore perturbs the function. The lower bound and cost test are operator-agnostic; the last column is the evidence in this paper.

Family	Compute unit	Exact reduction	Perturbation	Evidence
FFN width	intermediate channel	no nontrivial exact reduction; $r = h$	$\Delta y = \sum_{j \in S} a_j W_d[:, j]$	13-width sweep
MoE active experts	selected expert	$k' = k$ or zero gate	deletion + renormalization	35B-A3B scan
Attention head	head	zero-output head only	head-output removal	principle only
Depth	residual block	identity block only	residual-stream change	9B depth scan
Low-rank	rank	full rank	basis mixing	SVD baseline
Quantization	bit width	memory-only	rounding	control

From block perturbation to logits. Because Δ_m is measured from the draft logits, the certificate does not require differentiating through the remaining layers. The first-order interpretation in Appendix A.1 is $\Delta_m = \langle J_a - J_b, \Delta y \rangle + O(\|\Delta y\|^2)$, where J_a, J_b are the Jacobians of the two target logits with respect to the block output. The measured Δ_m is exact for the composed network, so the interpretation is explanatory, not an approximation used in the bound.

Scope of the assumptions. A1 excludes gated ReLU ($d(t) = 0$ for $t < 0$); the theorem covers SiLU/GELU-style smooth gating, and the measured models use SiLU. A2 is a shared-basis assumption; independent gate/up bases add a basis-mixing term and are measured as a baseline, not covered by the all-readout statement. A5 is the same-kernel roofline assumption for the cost layer. The certificate layer only needs the measured target margin and draft logits; it does not need A5. This separation matters when reading the results: a failure of A5 (for example, a much faster target kernel) changes the cost-only layer, not the measured rejection floor.

The deployment-distribution bounds. Propositions 2–5 formalize why a draft trained on a finite target-response dataset can fail on a deployment prefix distribution. The three consequences are Lipschitz continuity of rejection in the prefix distribution, a coverage bound for a draft with small average rejection, and an uncovered-component lower bound for a data-only learner. The last bound does not apply to a post-hoc operator, which sees the full target weights; it is the formal reason we do not claim that distillation fails. It also explains why the screen is a diagnostic, not a certificate: it estimates the deployment margin mass, and the certificate is measured on the deployment tokens when the screen is below threshold.

4 Protocol

Takeaway. The protocol makes the two layers separately falsifiable: a width sweep measures top-1 rejection, and a clean batch-1 cost protocol measures the draft and verification passes on the same GPU. We report greedy top-1 rejection q_g, exact-sampling rejection $q_s = \mathbb{E}[\mathrm{TV}]$, and the post-hoc top-2 lower bound q_{lb}; tree verification is outside the claim. All audits are in-sample on the evaluation tokens.

Terminology. *Rejection* is top-1 disagreement between draft and target; *wide* is 5–50% retained FFN width, *mid* 70–95%, and *high* 98–99.99%. A cell is *point-above* when $q_g > q^\star$ and *audit-passed* when the lower endpoint of the interval for q_{lb} exceeds $q^\star$; *audit failure* means either. (Some appendix tables abbreviate "audit failure" as "no-go".) $q^\star$ is the break-even rejection from (8); $c^\star$ is the minimum draft-kernel advantage that would reopen a cell; ρ_1, τ_1 are the measured per-token and verification ratios at batch 1; $L = \Pr(m < 0.5)$ is the low-margin mass; π_{tie} is the fraction of exact target top-1/top-2 ties. "Raw" means non-target-generated human text (operational definition in Appendix A.30).

Table 5 fixes the terminology for the several uses of "no-go" in the paper. The confirmatory claim is only the certified rejection no-go; the cost-layer verdicts R/G/K and the screen prediction are scoped separately.

Table 5: Terminology for the word "no-go". The terms are not interchangeable: the first row is the confirmatory rejection claim; R/G/K are cost-layer verdicts of decreasing strength; the acceptance-independent row is an implementation statement; the screen row is a triage prediction. Only the first row carries the paper's confirmatory claim.

Term	Definition	Scope / evidence
certified rejection no-go	lower endpoint of q_{lb} exceeds $q^\star$ on measured tokens	confirmatory in the $36+9$ family
roofline-bound (R)	$q > q^\star_{\mathrm{FLOP}}$ under same-kernel $c = 1$	operator-level; batch- and hardware-invariant
regime-bound (G)	$q^\star_{\mathrm{regime}} < q \le q^\star_{\mathrm{FLOP}}$	actual FLOP/memory regime
realized-kernel-bound (K)	$q^\star_{\mathrm{meas}} < q \le q^\star_{\mathrm{regime}}$	a draft-exclusive kernel could reopen
acceptance-independent cost no-go	$C_{\mathrm{meas}} > \gamma + 1$, so $q^\star_{\mathrm{meas}} = 0$	implementation statement, not operator-level
screen predicted no-go	predicted $q > q^\star$ from the margin mass	triage only; direct audit required

Decision order. The audit is sequential. (1) *Rejection first*: compute q_{lb} and compare its lower endpoint with $q^\star_{\mathrm{FLOP}}$; if it is above, the cell is R or R-free and the cost layer is not needed. (2) *Cost second*: if the rejection audit passes, place the cell in the W/K/G taxonomy using the measured and roofline thresholds; only K cells are candidates for a draft-exclusive kernel. (3) *Screen last*: the margin-mass prediction is a triage step; a below-threshold prediction must be followed by a direct audit, and an above-threshold prediction is evidence of a no-go, not a certificate. This order is why the screen cannot override the rejection audit and why the cost layer cannot rescue an R cell.

Models and data. Target and compressed draft are the same model. The main sweep is Qwen3.5-9B; cross-family checks are Qwen3.5-27B/4B, Qwen3-4B-Instruct-2507, Llama-3.1-8B-Instruct, K2-Horizon-3.7B, and a Qwen3.5-35B-A3B MoE. In-domain data are MATH-500 target-generated continuations; raw corpora are WikiText-2 [30], FineWeb-Edu [33], C4, four Wikipedia languages, Chinese web text [34], CodeParrot [16], PG-19, and arXiv (100 chunks each). Deployment-like suites are 100 SWE-bench-Lite prefixes [18] and 80 MT-Bench chat/instruction turns. We score channels by $s_j = \mathbb{E}|a_j| \, \|W_d[:, j]\|_2$ on a 200k-token target-aligned calibration set and sweep $r/h \in \{0.05, 0.1, 0.2, 0.3, 0.5, 0.7, 0.8, 0.85, 0.9, 0.95, 0.98, 0.99, 0.999, 0.9999, 1.0\}$; the fine 0.999/0.9999 points delete 12 and one channel per layer on 9B. MoE reduces the active-expert count k from 8 to $6, 4, 3, 2, 1$; depth drops the last $1, 2, 4, 8, 16$ of 32 layers; head pruning removes KV groups on Llama; combined scans use width+depth and width+head (Appendix A.25).

Rejection protocol. We cache the target top-256 distribution and the draft logits on the evaluation tokens and compute match, q_g, q_s, m, Δ_m, and q_{lb}. All CIs are 95% sequence-cluster bootstrap intervals (2,000 resamples, seed 0). For the fine-grained analysis we additionally evaluate same-corpus oracle importance (activation scores collected on the evaluation corpus), random selection (seed 0), and weight-norm selection (no activation scores).

Cost protocol. The clean measurement pre-clones the 512-token prefix cache outside the timed region and reports GPU kernel time from windows of back-to-back passes between CUDA events; the Python-loop wall clock is launch-bound at batch 1 and is retained only as a reference (Appendix A.6). We measure ρ_1, τ_1, and the zero-FFN counterfactual by replacing the MLP with a zero path, then convert a whole-pass advantage to an FFN-only speedup with (9). We use "padded narrow MLP" for the implementation and "clean kernel-time" for this timing protocol. The reported 5.00–5.28 range at 98–99% is clean kernel-time; the earlier padded full-pass protocol gives 5.25–5.28 on 9B and 5.37–5.66 on 27B and is an upper bound, not the reported cost. The 27B measurement uses one resident model at a time; the 9B hybrid loop uses exact re-prefill as the main number, and the negative-crop loop is an optimistic reference.

Metric status. The lower bound uses only the target top-2 margin and the draft logits on those two tokens, so it is insensitive to top-256 cache truncation and third-token flips and is conservative on small samples. q_g and q_{lb} agree on the $36 + 9$ confirmatory raw cells; in the 55-cell breadth suite the only raw exception is

code at 50% width in the eleven-corpus breadth suite ($q_g = 0.221$, q_{lb} lower endpoint 0.116, $q^\star = 0.140$). Full protocol and statistical plan: Appendix A.30.

4.1 Protocol and statistical plan details

Evaluation sets and cluster structure. Table A.84 in Appendix A.30 lists the evaluation sets, the number of sequences, and the token counts. Every interval is a sequence-cluster bootstrap over the evaluation sequences, with 2,000 resamples and seed 0. The cluster is the sequence, not the token, because adjacent tokens in a sequence are correlated and a token-level interval would be too narrow. For the confirmatory families the interval level is Bonferroni-adjusted within the family; the exploratory audit is descriptive and is not used to certify a cell.

Confirmatory families. The confirmatory family was fixed before the runs. It contains 36 compressed Qwen3.5-9B English raw cells (12 widths × three corpora: WikiText, FineWeb, C4; the 100% identity cells are controls and are excluded). A separate nine-cell cross-family family contains six Llama-3.1-8B raw cells (widths 0.95/0.98/0.99 on WikiText and FineWeb) and three K2-Horizon-3.7B raw WikiText cells (widths 0.95/0.98/0.99 from the 100/100 powered replication, Table A.66); the 30/50 target-generated K2 check is qualitative and is not in the confirmatory family. The in-domain MATH family is reported separately and is not pooled with raw text. This split replaces an earlier 45-test family that double-counted the Qwen cells. The corrected family-level result is in Table A.91; the per-cell confirmatory sweep is Table A.88, the cross-family cells are Table A.74 (six Llama cells) and Table A.66 (three K2 raw cells), and the bad MATH cells are listed in Table A.87.

Calibration. Channel scores are computed on a 200k-token target-aligned calibration set. The calibration size sensitivity at 98% width is at most 0.006 in rejection across 50k–200k tokens (Table A.36); every cell remains far above $q^\star$. The ridge readout is fitted on held-out activations from the same calibration set. We do not tune the channel score or the ridge penalty on the evaluation corpora; the same procedure is used for every corpus and width.

Hardware and timing. The quality measurements use one A100-80GB per run with the target and draft resident. The batch-1 latency measurements use a single-resident protocol: the target and the compressed draft are loaded in separate processes so that a 27B checkpoint fits on one GPU, and each process exits after its measurement. The end-to-end serving loops use the same GPU for the target-only and speculative runs. The reported timings are medians over repeated passes after warm-up. They are service benchmarks, not production throughput claims; the roofline bound is the implementation-independent statement.

What is post-hoc admissible. A repair is admissible if it never sees target-generated responses, uses only target weights, activations, and top-1/top-2 labels, and solves a closed-form or convex problem within a fixed small budget. Ridge refitting and margin-aware calibration satisfy the definition. Gradient updates to gate/up, 0.5M-token adaptation, distillation, QAT, and trained drafts are adaptation controls, not post-hoc repairs. The distinction matters for the claim: the negative result is about what can be done without target responses, not about what can be done with them. Appendix A.19 traces the adaptation budget curve.

4.2 Statistical families and the confirmatory split

Table 6 defines what each family is allowed to support. The family composition is defined in Section 4.1: 36 compressed Qwen raw cells plus a separate nine-cell Llama/K2 family (six Llama cells at widths 0.95/0.98/0.99 on WikiText/FineWeb and three K2 raw WikiText cells at the same widths from the 100/100 replication); the in-domain MATH family is reported separately and is not pooled with raw text. This split replaces an earlier 45-test family that double-counted the Qwen cells.

Appendix Table A.91 reports the family-level result: the 36 Qwen raw cells and the nine Llama/K2 cells are fully certified, with minimum adjusted margins +0.0320 and +0.0330 and two-level bootstrap lower bounds +0.0333 and +0.0344. The MATH family is the in-domain exception (5/12 certified). Appendix Table A.88 gives the 36-cell confirmatory sweep, Table A.74 the six Llama cells, and Table A.66 the three K2

Table 6: Statistical families and what each is allowed to support. The confirmatory families were fixed before the runs; the descriptive and exploratory aggregates overlap with the confirmatory cells and are breadth evidence, not additional family-wise tests. "Bad" means point rejection above $q^\star$ but an adjusted lower endpoint at or below $q^\star$. The 200-cell raw summary is descriptive; the 218-cell audit is exploratory.

Family	Tests	Correction	Role	Main?
Qwen raw36 (Wiki/FineWeb/C4, 12 compressed widths)	36	Bonferroni 0.05/36 (0.05/39 sens.)	Confirmatory; all certified, min. margin +0.0320	Yes
Llama + K2 raw9	9	Bonferroni 0.05/9	Independent cross-family; all certified, min. margin +0.0330	Yes
Qwen MATH (in-domain)	13	Bonferroni 0.05/13	In-domain exception; 5/12 certified, 5 bad	Separate
Raw200 (all raw cells)	200	Descriptive	Breadth; 200/200 point-above, 199/200 certified	No
Audit218 (all compressed cells)	218	Exploratory	Breadth and negative results; 26 bad, all in-domain MATH	No
Breadth / operator / cross-family	varies	Exploratory	Robustness checks in Section 8 and Appendix A.25	No

raw cells; Table A.87 lists the bad MATH cells. The descriptive 200-cell raw summary is 200/200 point-above and 199/200 certified; the exploratory 218-cell audit has 26 point-above-but-adjusted-lower-bound-below cells, all in-domain MATH. We do not use either count as a family-wise test. Every interval is a sequence-cluster bootstrap over evaluation sequences (2,000 resamples, seed 0), and the confirmatory intervals are Bonferroni-adjusted within the family.

5 The rejection layer

Takeaway. Raw-text rejection stays above the $c = 1$ break-even at every compressed width. The near-identity residual is a decision-boundary resolution floor, not a channel-ranking artifact: one deleted channel per layer already flips 1.2–1.4% of top-1 decisions, and an oracle channel ranking removes only 5–10% of that floor. The floor is a property of the target's margin distribution, so it is stable across kernels and batch sizes.

5.1 Does width reduction preserve the target's top-1 decision?

Table 7 reports the Qwen3.5-9B sweep. In-domain rejection falls from 0.58 at 5% width to 0; measured raw English prose stays at 0.83/0.50/0.07 at 5/50/99%. On the main English raw suite the top-2 bound exceeds $q^\star_{\mathrm{FLOP}}$ at every compressed width; at 99% it is 0.038–0.046 against 0.0024 (16–19×), $q_s = 0.058/0.051$ is 21–24×, and the greedy $q_g = 0.065$–0.068 is 27–28×. Target-generated MATH has a significant 50% FLOP-only window on 9B ($q = 0.1297$, cluster interval $[0.1262, 0.1337]$, $q^\star_{\mathrm{FLOP}} = 0.1400$), but $q > q^\star_{\mathrm{mem}} = 0.1220$ closes it on the memory frontier; the 30% point (0.2098 versus 0.2102) is inside the interval and is not a significant window.

Table 7: Qwen3.5-9B width sweep. q_g: greedy top-1 rejection; $q_s = \mathbb{E}[\mathrm{TV}]$: sampling rejection; q_{lb}: post-hoc top-2 lower bound with a 95% sequence-cluster interval. "W/F" denotes WikiText/FineWeb; each W/F entry is the pair of corpus means. "audit-passed" means the lower endpoint exceeds $q^\star$ on measured tokens. 100% is the identity control; the breadth suite is reported in Section 7.

Width	ρ_{FLOP}	$q^\star$	q_g W/F	q_s W/F	q_{lb} W [95% CI]	q_{lb} F [95% CI]	status
0.10	0.452	0.2933	0.778/0.762	0.760/0.761	0.361 [0.355, 0.366]	0.362 [0.354, 0.369]	point+audit
0.50	0.696	0.1400	0.502/0.505	0.516/0.516	0.269 [0.262, 0.276]	0.267 [0.260, 0.275]	point+audit
0.90	0.939	0.0250	0.208/0.209	0.181/0.177	0.134 [0.130, 0.138]	0.125 [0.121, 0.130]	point+audit
0.98	0.988	0.0049	0.097/0.096	0.080/0.076	0.064 [0.061, 0.066]	0.057 [0.054, 0.061]	point+audit
0.99	0.994	0.0024	0.068/0.065	0.058/0.051	0.045 [0.043, 0.048]	0.038 [0.036, 0.040]	point+audit
1.00	1.000	0.0000	0/0	0/0	0 [0, 0]	0 [0, 0]	identity control

The high-width part of the sweep is the informative one. A 1% channel deletion removes very little bulk information, so a 6.8% top-1 change invites a ranking explanation; the next two subsections show that ranking is part of the story at 99% but not at the near-identity floor.

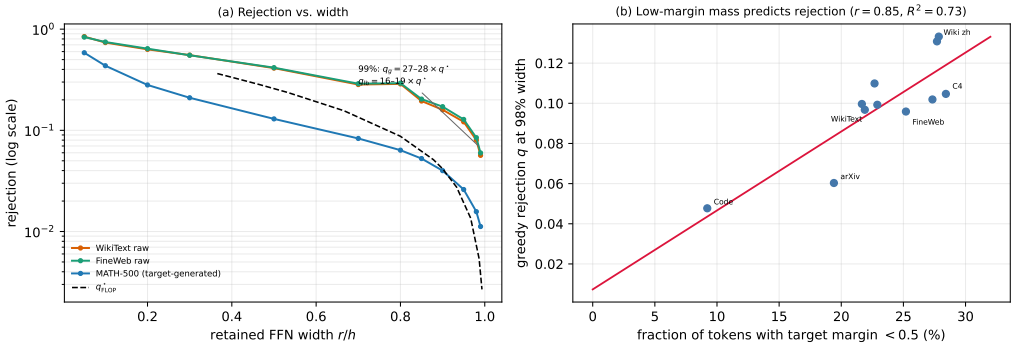


Figure 1: (a) Greedy rejection versus retained FFN width on Qwen3.5-9B for WikiText, FineWeb, and target-generated MATH, with $q^\star_{\mathrm{FLOP}}$ ($\tau = 1$, $\gamma = 4$). At 99% width the raw curves are 27–28× the threshold; the top-2 bound is 16–19×. (b) Low-margin mass $L = \Pr(m < 0.5)$ versus rejection at 98% width for eleven raw corpora, with a leave-one-corpus-out fit. The mechanism predicts the ordering; the screen must still audit directly below threshold.

5.2 The high-width residual is a resolution floor

Figure 2 and Table 9 vary the number of deleted channels per layer from 246 (98% width) to one (99.99% width). At one channel per layer—0.01% of the block—9B rejection is 1.30% and 4B rejection is 1.44%, 500–550× the break-even. Same-corpus oracle selection reduces this floor by only 5–10% (1.23% and 1.30%); at 99% width the oracle is more useful (3.73% versus 6.77% on 9B), but it still leaves 15× the threshold. Random selection is indistinguishable from the default at 99.9% (2.38% versus 2.33%), while weight-norm selection is much worse (8.31%). Ranking therefore matters in the selection-driven part, not at the floor.

The floor follows from the target's decision boundary. An exact tie is $m = 0$: the top-1 and top-2 log-probabilities are bit-equal in the stored bf16 target. In bf16 the margin has an atom of such ties ($\pi_{\mathrm{tie}} = 3.1\%$ on 9B, 2.6% on 4B) and a near-boundary mass $B(0.1) = 3.7\%/3.9\%$; the margin grid is quantized in 1/16 steps, so there is almost no mass between 10^{-3} and 0.06. A clean float32 rebuild removes the exact tie atom but leaves the near-boundary mass (Table 8): one deleted channel still flips 0.31% of tokens, 129× the float32 break-even. The tie component is therefore a bf16-specific part of the floor, not its explanation; the flip is not a tie-order rule, because at width 1.0 the draft is bit-identical and $q = 0$. Deleting one channel per layer leaves the median token untouched ($|\Delta| \approx 0.001$ on 9B) but creates a heavy tail: the 90th percentile of $|\Delta|$ is 0.125, and that tail flips the near-boundary tokens. Figure 2b shows that flips concentrate in the tie and small-margin bins; the measured q is within 0.1–1.5 points of $\Pr[\mathrm{sign}(m) \neq \mathrm{sign}(m + \Delta)]$ (Eq. 5). Under the tested ranking, the tie component alone is 0.60% at one channel per layer, already 246× the break-even. The excess grows as $q^\star$ vanishes faster than the boundary mass.

Table 8: Clean float32 control for the one-channel resolution floor on Qwen3.5-9B WikiText. The target cache and the draft are both rebuilt in float32, so this is a different engine, not a tie-only ablation; the near-boundary mass changes from 3.68% to 5.4%. The exact bf16 tie atom disappears, but the floor remains: the one-channel rejection falls from 1.30% to 0.31% (95% cluster interval $[0.26, 0.38]$), still 129× the float32 $c = 1$ break-even. Exact ties are therefore not necessary for the floor.

Precision	Width	q	π_{tie}	$B(0.1)$	median $\lvert\Delta_m\rvert$	$q/q^\star$
bf16 target	0.9999	1.30% [1.21,1.40]	3.09%	3.68%	0.0010	534×
float32 target	0.9999	0.31% [0.26,0.38]	0.00%	5.4%	0.0072	129×
bf16 target	0.99	6.77%	3.09%	3.68%	0.1255	27.7×
float32 target	0.99	6.65%	0.00%	5.4%	0.1426	27.2×

Table 9: The high-width resolution floor. "Deleted" is the number of FFN channels removed per layer; at 99.99% width this is one channel per layer (0.01% of the block). q is greedy rejection with the activation-importance ranking; q_{oracle} uses the same-corpus importance; $q^\star_{\text{FLOP}}$ is the $c = 1$ break-even. The tie component is the measured part of q on exact bf16 ties. Deleting a single channel per layer still flips 1.2–1.4% of top-1 decisions, 500–550× the break-even, and the same-corpus oracle reduces this floor by only 5–10%. The gap grows as the width approaches one because $q^\star$ vanishes faster than the boundary mass.

Model	Width	Deleted	q	q_{oracle}	tie comp.	$q^\star_{\text{FLOP}}$	$q/q^\star$
Qwen3.5-9B	0.98	246	0.0968	0.0528	0.0142	4.89e-03	20×
Qwen3.5-9B	0.99	123	0.0677	0.0373	0.0130	2.44e-03	28×
Qwen3.5-9B	0.999	12	0.0233	0.0163	0.0085	2.44e-04	96×
Qwen3.5-9B	0.9999	1	0.0130	0.0123	0.0060	2.44e-05	533×
Qwen3-4B	0.99	97	0.0676	0.0324	0.0111	2.68e-03	25×
Qwen3-4B	0.999	10	0.0235	0.0153	0.0082	2.68e-04	88×
Qwen3-4B	0.9999	1	0.0144	0.0130	0.0063	2.67e-05	537×

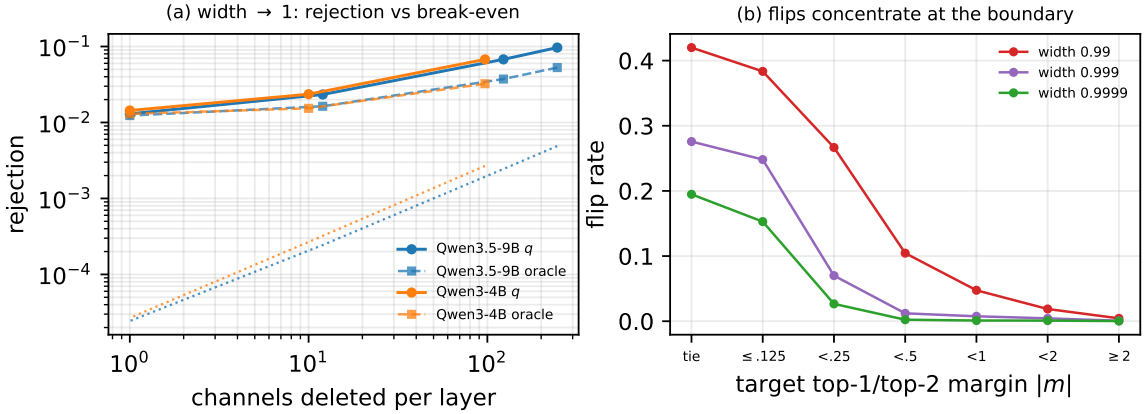


Figure 2: The high-width resolution floor. (a) Rejection versus channels deleted per layer on WikiText, for default and same-corpus oracle selection; the dotted lines are $q^\star_{\text{FLOP}}$. One channel per layer still flips 1.2–1.4% of tokens, 500–550× break-even, and the oracle reduces the floor by only 5–10%. (b) Flip rate in target top-1/top-2 margin bins: flips concentrate at exact ties and small margins, and the affected margin range shrinks as the width approaches one.

5.3 Mechanism: low-margin entropy mass

The negative result is not "measured raw English prose is hard" in a generic sense. It has a specific mechanism: the target's predictive uncertainty puts a mass of tokens near the top-1/top-2 decision boundary, and a sub-logit perturbation flips that mass. This subsection expands the mechanism evidence.

Margin is negatively correlated with entropy. Table A.77 reports the median top-1/top-2 log-margin, the low-margin mass $\Pr(m < 0.5)$, the mean target entropy, and the Spearman correlation between margin and entropy. On WikiText, FineWeb, and C4 the correlation is $-0.80/-0.74/-0.69$; on MATH it is -0.99. The target is confident where it is certain and uncertain where the next-token distribution is broad. This is not a calibration failure of the raw corpora; it is the ordinary behavior of a language model on prose.

The gap is composition, not an abnormal calibration. The raw/MATH gap can be decomposed by reweighting one corpus to the other's entropy histogram. Reweighting MATH to the WikiText entropy histogram raises its low-margin mass from 0.033 to 0.343. Reweighting WikiText to the MATH entropy histogram lowers it from 0.219 to 0.008. At fixed entropy, MATH has equal or higher low-margin mass. The raw/MATH difference is therefore mostly the composition of entropy values, not a different conditional relationship between margin and entropy. This matters for deployment: a screen should be recalibrated when

the deployment entropy distribution shifts, even if the target model is unchanged.

The low-margin quintile carries the rejection. Table A.76 stratifies the 98%-width rejection by target margin quintile. On WikiText the lowest quintile rejects at 0.327 while the highest quintile rejects at essentially zero; on MATH the lowest quintile rejects at 0.078 and the rest is near zero. The mean absolute perturbation $|\Delta_m|$ is not much smaller in the high-margin quintiles (the deletion is not concentrated on low-margin tokens), but the target margin is large enough that the same perturbation does not flip the decision. The binding quantity is the ratio of perturbation to margin, not the perturbation alone. This is why reconstruction-error improvements can leave rejection unchanged: they reduce $|\Delta y|$ but not the low-margin mass.

Why a calibration fix does not close the gap. Margin-aware post-hoc calibration (Appendix A.21) reweights the draft logits to reduce the average margin shift on a calibration set. It moves the mean but not the tail: the low-margin tokens are the ones whose margin is already small, and a global affine logit correction cannot make a negative margin positive without changing the sign on another part of the distribution. The calibration control reduces a global bias but leaves the 98%-width raw rejection above $q^\star$. The same logic applies to the ridge readout and to gradient fitting: a better fit to the block output does not target the decision boundary.

Implication for the screen. The mechanism suggests a simple diagnostic: estimate the low-margin mass $L = \Pr(m < 0.5)$ on a small prefix sample from the deployment distribution, and compare the predicted rejection with the break-even threshold. The screen is not a certificate; it is a way to decide whether a direct measurement is needed. Section 7 gives the calibration, the exceptions, and the two-stage procedure.

5.4 Do post-hoc repairs close the gap?

Appendix Table A.33 lists the admissible repairs at 98% width. Layer-wise ridge refit reduces layer MSE by 35% but moves q by less than ± 0.005. Gradient fitting, top-1 margin calibration, and 0.5M-token adaptation are worse. Same-corpus held-out selection reaches 0.049–0.056 and the evaluation-corpus oracle reaches 0.043–0.053; both leave the cell audit-failed (9–11× $q^\star$). Non-uniform per-layer allocation at a fixed average width reduces q by 14–15% (front/back is best; alternating and random are worse), far short of the $\sim 20\times$ reduction needed; on 9B the best pattern is 0.0826 versus 0.0964 at 98% and 0.0582 versus 0.0683 at 99%. Independent-SVD and shared-rotation bases at matched FLOP are worse. The conclusion is that, within the tested uniform-width activation-ranking family, the audit failure is not a missing channel-ranking or simple-allocation trick. Global subset search, per-layer width DP, and an OBS-style score are tested in Section 5.5.

One combination is not measured: oracle selection followed by a ridge readout refit. It is an upper bound on the selection family, and we flag it as an open cell rather than claim that it closes the gap. The same applies to a structured OBS/SparseGPT variant with a Hessian-weighted channel score and a readout update; the weight-norm, SVD, and rotation baselines in the operator zoo are weaker than that, so our ordering of repairs is a lower bound on the best post-hoc competitor, not a proof that no such competitor exists.

5.5 Is the gap a search artifact?

The main sweep fixes a uniform per-layer width and an activation-aware score. A natural objection is that a joint subset search, an optimized per-layer allocation, or a Hessian-style score could close the gap. We test that objection on the same evaluation cells. The selectors are: a global score-ranked channel subset (a top-R knapsack over all FFN channels under one total-width budget, not an exhaustive subset search); a per-layer width DP with a convex removal penalty; a sensitivity-weighted DP that measures each layer's deletion response on the first 30 evaluation sequences (an oracle diagnostic, not a deployable selector); an OBS-style second-moment score; and a random per-layer allocation on a held-out calibration split. The search uses 200k calibration tokens, a 32-channel DP grid, and 12 random candidates; selection took under one A100 GPU-hour and is not saturated. The main rows use the same 100-sequence target cache and audit; the random check is audited separately on 50 held-out chunks. The search rows are pruning-only; a ridge-readout

spot check at 98% changes the reference by at most 0.004. It is not a full ridge-refit test of every search row (Appendix A.12.1).

Table 10: Global post-hoc selectors on WikiText at fixed average FFN width. All rows use the same calibration text and target cache and the same 100-sequence audit. The search rows are pruning-only (no ridge readout); a 98% ridge spot check moves the reference by at most 0.004, but it covers only the uniform and global OBS rows and is not a full ridge-refit test (Appendix A.12.1). $q_{\mathrm{lb}}^{\mathrm{lo}}$ is the lower endpoint of the 95% sequence-cluster interval (single-cell, not family-wise); $q^\star$ is the $c = 1$ FLOP break-even. The selectors are post-hoc choices on the same evaluation cells, not new confirmatory cells. The global allocation rows are budgeted heuristics, not a proof of global optimality. The lowest point q and the largest certified margin need not coincide: $q_{\mathrm{lb}}^{\mathrm{lo}}$ is the comparison the paper uses; ratios use the unrounded $q^\star$. The nonuniform rows are rejection-only: the $q^\star$ column is the $c = 1$ FLOP threshold at the average width, and no measured-cost (R/G/K) verdict is assigned. The full selector matrix is Table A.38; FineWeb is Table A.37.

Width	Selector	q	q_{lb}	$q_{\mathrm{lb}}^{\mathrm{lo}}$	$q^\star$	$q_{\mathrm{lb}}^{\mathrm{lo}}/q^\star$	Verdict
30%	Uniform, activation-aware	0.6565	0.3319	0.3245	0.2102	1.5×	no-go
	Uniform, OBS second-moment	0.5682	0.3004	0.2937	0.2102	1.4×	no-go
50%	Uniform, activation-aware	0.5017	0.2690	0.2617	0.1400	1.9×	no-go
	Uniform, OBS second-moment	0.4622	0.2529	0.2461	0.1400	1.8×	no-go
98%	Uniform, activation-aware	0.0968	0.0636	0.0608	0.0049	12.4×	no-go
	Uniform, OBS second-moment	0.0863	0.0568	0.0541	0.0049	11.1×	no-go
99%	Uniform, activation-aware	0.0677	0.0455	0.0432	0.0024	17.7×	no-go
	Uniform, OBS second-moment	0.0635	0.0412	0.0390	0.0024	16.0×	no-go

On the tested Qwen3.5-9B WikiText/FineWeb cells, OBS and random lower point q without improving the certified lower bound; the default activation-aware uniform selector retains a larger certified margin than both. The OBS score lowers point q at 98% width to 0.0863 on WikiText and 0.0920 on FineWeb, a reduction of about 10.8% on WikiText, but it also lowers the certified lower bound: the lower-bound ratios are 11.1× and 10.5× $q^\star$ at 98%, and 16.0× and 13.3× at 99%. The global subset and DP allocations are worse than uniform OBS at every tested width; at 30% the global/DP OBS variants improve on the activation-aware uniform point rejection, while the activation-aware variants do not, and neither improves the certified margin or beats uniform OBS. The score mass is not a sufficient statistic for per-layer sensitivity, and the sensitivity-weighted DP is closer to uniform but still does not beat it. A best-of-12 random allocation, selected on 30 sequences and audited on 50 held-out chunks, reaches $q = 0.0805/0.0725$ at 98% and $q = 0.0573/0.0507$ at 99%; it lowers the point estimate further, but its lower-bound ratios are 9.6–11.1× and 12.3–14.6× $q^\star$ and do not improve on the default (Appendix A.12.1). At 90/50/30% width the default activation-aware selector again has a larger certified margin than OBS and random, with lower-bound ratios 5.2×, 1.9×, and 1.5× $q^\star$; the OBS selector narrows the 30% ratio to 1.4×. At 30% the OBS point estimate is 0.5682 ($2.7 \times q^\star$) while its lower endpoint is 1.4×; the default margin is 1.5×. The low-width cells are the closest to reopening, but none opens, and q remains above the point threshold as well. These are single-cell lower endpoints, not family-wise bounds. The 90% and FineWeb cells are in Appendix A.12.1 (Tables A.38 and A.37).

A cross-family check on Llama-3.1-8B points the same way. The OBS score is worse than the activation-aware score at 98%, and the global OBS allocation lowers point q at 99% but its lower bound remains 12–13× $q^\star$, so it does not reopen a cell either (Appendix A.12.1).

This is a best-of-budget heuristic, not a proof of global optimality; the exact combinatorial optimum remains open, and the 30% cell at 1.4× is the first place a larger search budget should be spent. It does not change the confirmatory $36 + 9$ family, and it does not support a claim about all post-hoc compressors. It does support a narrower statement: on the measured raw cells, the tested global selectors do not reopen a rejection window; a score change at uniform width lowers the point estimate, but the default activation-aware selector retains a larger certified margin than OBS and random, and global allocation does not help.

6 The measured cost layer

Takeaway. The cost result has a different logical status from the rejection result. Under the optimistic $c = 1$ roofline, width deletion always leaves a positive break-even threshold $q^\star_{\text{roof}} > 0$ (Corollary 1); the frontier no-go is therefore a rejection statement, not a cost statement (Theorem 1). At 98–99% width the measured raw rejection is 13–20× the optimistic FLOP threshold, so every target is *roofline-bound* (R), a verdict that is invariant to batch size, hardware, and any kernel in the same family. The measured clean kernel-time cost is also above 5 in the padded implementation, which makes the cell a point-estimate implementation diagnostic as well; that is an additional diagnostic, not the load-bearing reason and not a production-kernel claim. The only cells where the cost layer is load-bearing are *realized-kernel-bound* (K) or *regime-bound* (G). The 50% raw cells are R-free: even a free FFN cannot supply enough acceptance. The Llama MATH 50–70% cells are K, with a 0.4–2.3% shape margin inside the timing spread; the Llama 90% cell is R.

Box A: the same-kernel roofline. Fixing $c = 1$ gives the operator-level statement. It uses only the draft/target FLOP or byte ratios and the break-even rule (8); it does not need a clock. This is the box that contains the rejection layer: on measured raw English prose $q_{\text{lb}} > q^\star_{\text{FLOP}}$ at every compressed width. The bound is hardware-invariant because ρ_{FLOP} and ρ_{mem} are model ratios, not hardware ratios; the hardware only chooses which term binds.

Box B: the measured batch-1 kernel. The cost-only statement uses the measured ρ_1 and τ_1 on one A100 with a padded narrow-MLP implementation. It is an upper bound on relative cost and is reported with the zero-FFN counterfactual as a diagnostic, not as a ceiling. The two boxes agree on every raw cell, but they are falsifiable by different evidence: a faster draft kernel would move Box B and leave Box A unchanged, while a different evaluation distribution would move Box A and leave the kernel protocol unchanged.

6.1 The measured batch-1 cost

The clean kernel-time decomposition is in Table 11 and Appendix Table A.64. At 98–99% the realized $4\rho_1 + \tau_1$ is 5.004–5.28 ≥ 5 on every tested target; replacing the MLP with a zero path gives 3.1–4.1, so a hypothetical free FFN would put the cost below 5 everywhere; this is a counterfactual about a fully removed MLP, not a width-only kernel. The difference between the two rows is the entire practical question: the operator changes the MLP, and the MLP is only $s \approx 0.23$–0.52 of the measured draft pass. The remaining $1 - s$ is shared with the target, so the compressed draft cannot become much cheaper than the target merely by shrinking the FFN.

Two protocol details matter. First, the numbers are GPU kernel times from windows of back-to-back passes with the prefix cache pre-cloned outside the timed region; the Python-loop wall clock is launch-bound at batch 1 and overstates the residual non-MLP cost. Second, the narrow MLP is implemented as a padded dense GEMM. This is an upper bound on relative cost, not a lower bound: a production kernel could recover part of the gap. We therefore report the roofline and the measured kernel as two separate objects, and we phrase the cost conclusion as an implementation-conditional audit failure.

6.2 Applying the cost taxonomy

The three roofline lower bounds on cost induce the ordered thresholds $q^\star_{\text{meas}} \leq q^\star_{\text{regime}} \leq q^\star_{\text{FLOP}}$ (Appendix A.5); the measured free-FFN threshold $q^\star_{\text{free}}$ is a separate diagnostic because it uses a different accounting. The interval containing the measured q is the verdict: W (window), K (realized-kernel-bound), G (regime-bound), R (roofline-bound), or R-free (even a free draft fails). Table A.11 applies the taxonomy to the six targets at 98% width: all six are R. Table A.13 gives the in-domain cells where the cost layer is load-bearing.

The binding question for a K cell is how much draft-side kernel advantage is needed. Let s be the MLP share of the measured draft kernel time ($s = 1 - \rho_{\text{zero}}$). A free FFN gives at most $c_{\max} = 1/(1 - s)$ whole-pass advantage, and the required FFN-only speedup is

$$c^\star_{\text{FFN}} = \frac{s}{1/c^\star - (1 - s)}, \tag{10}$$

Table 11: Whole-pass versus FFN-only kernel conversion at 98% width. s is the MLP share of the measured draft kernel time ($s = 1 - \rho_{\text{zero}}$); $c_{\max} = 1/(1-s)$ is the largest whole-pass advantage a *free* FFN can deliver; $c^\star$ is the whole-pass advantage needed for break-even at the measured q; $c^\star_{\text{FFN}}$ is the implied FFN-only speedup, $s/(1/c^\star - (1-s))$. A $c^\star$ above $c_{\max}$ is structurally closed (no FFN-only kernel under the measured conversion can help). At 50% width $c^\star \approx 3.0$–$6.1 > c_{\max}$ on every measured target; at 98% the targets need 1.7–15.8× FFN-only speedup, inside the published 1.2–2.7× fused-MLP band only for Llama and 27B.

Model	s	$c_{\max}$	$c^\star$	$c^\star_{\text{FFN}}$	cost	zero-FFN cost
Qwen3.5-9B	0.304	1.44	1.30	4.04	5.033	3.83
Qwen3-4B	0.317	1.46	1.42	15.84	5.276	4.01
Qwen3.5-4B	0.229	1.30	1.26	8.87	5.004	4.09
Llama-3.1-8B	0.520	2.08	1.26	1.67	5.060	3.12
K2-3.7B	0.342	1.52	1.33	3.70	5.266	3.90
Qwen3.5-27B	0.393	1.65	1.25	2.06	5.066	3.50

where $c^\star$ is the minimum whole-pass advantage that reopens the cell. At 50% width the raw cells are R-free: even a free FFN fails because the measured rejection is above $q^\star_{\text{free}}$. At 98–99% $c^\star$ is below $c_{\max}$ and the targets need 1.7–15.8× FFN-only speedup (Llama 1.67×, 27B 2.06×, 9B 4.04×, Qwen3.5-4B 8.87×, Qwen3-4B 15.8×; K2 is R). Published fused-MLP kernels report 1.2–2.7× over naive dense baselines, but they are portable: applying one to both target and draft moves the relative ratio toward one (Proposition 8). Only a draft-exclusive kernel counts. The shape ceiling at 98% is $c_{\text{shape}} = 1.011$–1.015, while the required advantage is 1.25–1.42, so the gap is not a kernel-tuning gap.

The 98–99% band needs a further caveat. A 1.2–1.4× whole-pass advantage would require a draft-specific kernel that is faster than the target's kernel on the *whole* pass, while the draft and target compute are nearly identical ($\rho_{\text{FLOP}} \approx 0.99$). A fused or low-precision kernel that accelerates the whole pass would also accelerate the target, so it does not create a relative advantage. We therefore report the band as an arithmetic conditional, not as a deployment window. The contrast is the useful part: the 50% cells are R-free by arithmetic; the 98–99% cells are R by the rejection floor, with the realized padded kernel as an independent point-estimate diagnostic.

6.3 Regimes and what would reopen a cell

Table 12 summarizes the width bands. The 5–90% range is a rejection audit-failure that is robust to a several-fold kernel gain; 95–99% is also closed by the rejection floor, and the $c^\star$ row is conditional on a kernel that is not draft-specific. At $B = 128$ the measured cost is 5.41, so batching does not open the compute-bound corner in the measured range (Appendix Table A.21). The 27B in-domain window behaves consistently: its clean kernel-time cost is 4.23/4.53/4.84 at 50/70/90% width, so the implied break-even rejection is 0.083/0.050/0.016, below the measured MATH $q = 0.088/0.056/0.026$; the 50–90% cells are K at the clean cost, and the 98% point is R. The Llama-3.1-8B low-width MATH window is the only K case: at 50–70% width the measured padded kernel closes it, with a 0.4–2.3% shape margin inside the 9% timing spread; at 90% it is R (Table A.13). In our reference KV-cache loop, end-to-end greedy throughput is 0.19–0.65× target-only (Appendix Table A.65); the 9B identity control reaches only 0.607×, so this is a harness-inclusive reference, not an operator cost claim.

The cost layer is therefore narrower than the rejection layer but not weaker: it says that under the measured implementation, even a free FFN would not save the mid-width cells, and the high-width cells that a free FFN would reopen are not reachable by a width-only kernel when the rest of the pass is shared. A reader who distrusts the wall clock can still use the rejection layer; a reader who distrusts the certificate can still use the measured cost. The two agree on every raw cell in the audit.

7 The deployment screen

Takeaway. The mechanism yields a cheap triage rule. Estimate the low-margin mass $L = \Pr(m < 0.5)$ on a small prefix from the deployment distribution, predict rejection with a per-target fit $q = a_w + b_w L$, and measure the certificate directly when the prediction is below threshold. The rule is asymmetric: a prediction

Table 12: Decision map. $c^\star$ is the minimum draft-kernel throughput advantage over the target that reopens the cost-only window at the measured rejection; the two columns give $\tau = 1$ (best case) and the measured τ_1. The 5–90% range is a rejection audit failure; 95–99% is also closed by the resolution floor, and the $c^\star$ row is an arithmetic conditional on a whole-pass draft-specific kernel, not a realizable window (the draft shares 98–99% of the pass with the target). At $B = 128$ the measured cost is 5.41, so batching does not open the compute-bound corner. In the clean kernel-time decomposition the realized cost is 5.00–5.28 > 5 at 98–99% on every tested target. Those cells are roofline-bound by 13–20×, so the measured cost is an additional acceptance-independent implementation failure, not the load-bearing reason. A free-FFN counterfactual would put the cost below 5 on all six targets, but it is not a width-only kernel: the shape ceiling is 1.011–1.015, below the required 1.25–2.72. All six targets are R; K2 is not R-free once the raw K2 cells are used. *At 10% width even a free draft is below break-even because $A(1-q,\gamma) < \tau_1$.

Width band	Audit	Batch-1 cost	$c^\star$ ($\tau = 1$)	$c^\star$ (τ_1)	Action
5–50%	fail	fail	3.0–6.1	$\geq 4.6^*$	don't use; robust to a several-fold kernel gain
70–95%	fail	fail	1.4–2.2	1.6–2.9	don't use
98–99%	fail	fail	1.2–1.3	1.3–1.4	closed under post-hoc family; $c^\star$ conditional

above threshold is evidence of an audit failure; a prediction below threshold is not evidence of a window. On the evaluated raw corpora the screen has leave-one-corpus-out MAE 0.0138 at 98% width and no observed false negatives, but its coefficients are per-target and its false-negative upper bound is not zero.

7.1 The diagnostic

The screen is the deployable part of the mechanism. It estimates whether a new deployment distribution is likely to be audit-failed before the expensive certificate measurement. It is a diagnostic, not a guarantee. The quantity is the target's own low-margin mass: no draft logits, no channel scores, and no response data are needed, only one target forward pass on a small prefix. The mechanism of Section 5.3 predicts that this quantity orders deployments, because the flip rate is driven by the mass of tokens whose margin is comparable to the deletion perturbation.

7.2 Fit, leave-one-corpus-out validation, and calibration

We fit $q = a_w + b_w L$ on ten raw corpora and predict the held-out eleventh (leave-one-corpus-out). Appendix Table A.79 reports the per-width LOCO errors. The MAE is 0.0466 at 50% width and 0.0138 at 98%; every held-out raw corpus is predicted above $q^\star$ at every width, including code at 50% width (predicted 0.231, measured 0.221, $q^\star = 0.140$). The two-feature variant adds the mean absolute margin perturbation $\mathbb{E}|\Delta_m|$ and lowers the MAE to 0.0270 at 50% and 0.0114 at 98%. Refitting the same screen on the standard-sampling metric $q_s = \mathbb{E}[\mathrm{TV}]$ instead of q_g gives $q_s = 0.0208 + 0.244L$ with LOCO MAE 0.0088 (max 0.0184); every raw corpus is still predicted above $q^\star = 0.0049$. The screen is therefore not specific to the greedy protocol, but the main-text coefficients are for q_g.

Appendix Table A.82 gives the pooled coefficients. At 98% width the fit is $q = 0.0073 + 0.393L$; the coefficient is small because the deletion is near-identity at high width. The threshold sensitivity in the same table varies the low-margin cutoff over 0.1, 0.25, 0.5, and 1.0: the predicted rejection at 98% width stays above $q^\star = 0.0049$ for WikiText, FineWeb, and MATH at every cutoff. The 0.5 cutoff is therefore not tuned to the evaluation set.

Appendix Table A.81 reports the calibration of actual on predicted rejection: slope 0.989, $R^2 = 0.964$, and 92.7% coverage of the 95% residual interval. The decision errors are zero on the 55 raw LOCO cells. The external suites (MT-Bench, SWE-bench, MATH) have a larger bias (+0.0333) and slope (1.404), driven by MT-Bench at 50% width, which is under-predicted (0.214 predicted versus 0.397 measured) but still above $q^\star$. The in-domain MATH cell at 50% is the only predicted-break-even external cell, and the measurement agrees (0.130 versus $q^\star = 0.140$). That is the in-domain window, not a measured raw-text failure.

7.3 Boundary cases and false-negative accounting

Appendix Table A.83 is a synthetic corpus of templated JSON records, repetitive code, and key-value logs. The target is highly confident (median margin 8.75, low-margin mass 4.3%), yet rejection remains above $q^\star$ at every width because the deletion perturbation is large (mean $|\Delta_m| = 1.17$–3.41). The mass-only screen under-predicts 50% width (0.123 versus measured 0.152 and $q^\star = 0.140$); the two-feature screen with $\mathbb{E}|\Delta_m|$ corrects the decision (0.355). This is a boundary case, not a window. It shows that the mass-only screen is a lower bound in the high-perturbation regime and that the two-feature variant is the safer default when the deployment distribution is unusually regular.

We say "no observed false negatives on the evaluated raw corpora," not "no false negatives." The zero counts are 0/55 cells and 0/11 corpora. The one-sided 95% upper bound from the rule of three is 5.3% per cell and 23.9% per corpus. The external suites add 0/14 cells with no decision error. These bounds are wide enough that a deployment with a different distribution should not treat a single zero as a guarantee.

The coefficients are also per-target. Applying the 9B fit to WikiText on four other targets preserves the binary raw decision (Appendix Table A.8) but over-predicts the raw K2 cell by 1.2× (conservative); the earlier 30/50 target-generated K2 check is under-predicted and is not the raw transfer cell. A new target requires a refit on a small prefix from that target's own deployment distribution; the false-negative guarantee is scoped to the target on which the screen was fitted.

7.4 Screen-to-audit algorithm

The operational recipe is deliberately small. It uses a prefix sample X and the deployment threshold $q^\star$ implied by (ρ_1, τ_1, γ).

A 1,024-token prefix simulation over 200 draws predicts above $q^\star$ in 100% of draws for every measured raw/deployment cell at every width. That is a statement about these corpora, not a distribution-free guarantee. The asymmetric rule is the operational content: measure when the screen is below threshold, and do not report a window from the screen alone.

7.5 Practitioner decision guide

Figure 3 turns the audit into a decision procedure. The first branch is scope: if target responses are available, a trained or adapted draft is a different object outside this audit, and we do not evaluate or rank it. If the goal is memory rather than FLOPs, weight-only quantization is the orthogonal axis. Only when the practitioner specifically considers a post-hoc FFN-width draft does the screen apply, and then only as a triage step before a direct certificate measurement. On measured raw English prose at 95–99% width the answer is already settled by the rejection floor; in-domain traffic needs its own prefix audit because the window, when it exists, is cost-specific and in-model-generated.

8 Breadth and negative results

Takeaway. The rejection layer is not specific to one model family or to FFN width. Every raw cell in the cross-family, depth, head, and MoE checks is audit-failed at compressed widths, and the deployment-like suites are not windows either. The negative results also rule out the obvious repairs: a better channel ranking, a readout refit, a non-uniform allocation, or a small amount of adaptation does not move the binding variable. Adaptation, quantization, and other draft families remain different objects, and we do not rank them here.

8.1 Cross-family replication

Appendix Tables A.72, A.74, A.73, A.41, and A.46 give the full sweeps for Qwen3.5-27B, Llama-3.1-8B, Qwen3-4B, Qwen3.5-4B, and K2-Horizon-3.7B. All raw-text cells are certified at every compressed width. The families differ in the size of the in-domain window: Llama-3.1-8B has a narrow MATH window at high width, Qwen3-4B has a narrow one at 50%, and Qwen3.5-27B has a 50–90% 35B-generated fixed-continuation window. The Llama window is cost-closed in the padded narrow-MLP implementation under the clean protocol (5.067/5.066/5.154 at 50/70/90% width), so it is not a counterexample to the measured cost layer.

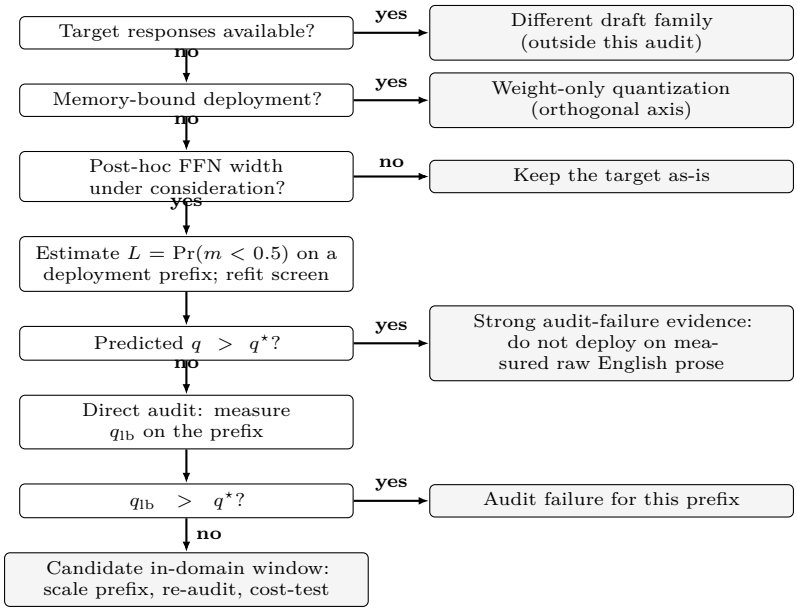


Figure 3: Scope and decision guide. The first branch marks a different draft family as outside this audit; it is not a comparative recommendation. The audit is only relevant after the first three branches. A screen prediction above threshold is evidence of an audit failure; a prediction below threshold triggers a direct certificate measurement, not deployment. The measured raw-text answer at 95–99% width is already settled by the rejection floor; in-domain traffic requires its own prefix audit.

The raw-text floor is stable across families even when the in-domain window moves, which is the cross-family content of the claim.

K2 is a standard dense Transformer without hybrid linear attention. Its raw WikiText rejection is 0.079 at 98% and 0.063 at 99% in the 100/100 powered replication (Table A.66); the earlier 30/50 target-generated check gives 0.31 at 99% and is qualitative, not the raw confirmatory cell. The raw K2 cells are still R, but they are not higher than the Qwen3.5 family. Llama is the strongest dense replication: at 98% width WikiText rejection is 0.081 against $q^\star = 0.0060$, and at 99% it is 0.060 against 0.0030. Qwen3-4B is useful because the target is a different family and tokenizer; raw rejection at 99% is 0.068/0.070 against 0.0027.

8.2 Other operators: width, depth, head, MoE

The audit failure is not specific to FFN width. Appendix Table A.35 compares channel selection with weight-norm ranking, independent SVD of each FFN matrix, and a shared rotation before the nonlinearity at matched nominal FLOP. Activation-aware width is the best of the family on raw text, and even it is audit-failed. Weight-norm ranking is worse at every width, which is the expected consequence of ignoring the activation magnitude. Independent SVD is worse still: it mixes coordinates and does not preserve the target's top-1 decisions. The shared rotation gives $q = 1$ at every rank in the unfused implementation; this is a bound on the unfused rotation, not on a fully optimized SliceGPT pipeline, and we do not use it to argue that rotation-based pruning is useless.

Appendix Table A.69 reduces the active-expert count of Qwen3.5-35B-A3B from 8 to 1. Every compressed cell is certified; at $k = 1$ raw rejection is 0.79/0.74 against $q^\star = 0.137$. The cost saving is modest because the active routed-expert set is about one third of active FLOPs, so the FLOP ratio moves less than the expert count suggests. Appendix Table A.71 drops the last 1–16 of 32 layers; even dropping one layer gives raw rejection 0.32/0.29 against $q^\star = 0.011$. Appendix Table A.59 physically removes KV groups on Llama; 25% head removal leaves WikiText rejection at 0.77 against $q^\star \approx 0.009$. Appendix Table A.26 composes width reduction with real packed INT4: the FLOP-axis audit failure remains at every raw width, and the memory-axis certificate is not claimed because the lower bound does not clear $q^\star_{\mathrm{mem}}$ at 50%.

8.3 Deployment-like suites

Appendix Table A.24 evaluates 100 SWE-bench-Lite issue-to-patch prefixes (23,412 scored tokens). Rejection is 0.305/0.153/0.095/0.067 at 50/90/98/99% width, all certified. The agentic prefix is not an easier distribution than raw prose: it is closer to code and issue text, and its low-margin mass is not small. Appendix Table A.32 evaluates 80 MT-Bench chat/instruction turns; rejection is 0.397/0.147/0.098/0.063/0.043 at 50/90/95/98/99%, all above $q^\star$. The chat suite is also not a window; at 50% width it is worse than WikiText.

These suites are worth separating from "raw" in the operational sense. They are not target-generated, but they are also not pretraining-like prose. The fact that they are audit-failed at every compressed width strengthens the distribution robustness of the rejection layer; it does not convert them into a deployment claim.

8.4 Adaptation controls: a shallow slope

Appendix Table A.53 traces the adaptation budget. A 0.5M-token down-projection adaptation gives raw rejection 0.448/0.201, better than the pure post-hoc repair but still above the raw break-even. Distillation to 2B/4B reaches in-domain rejection near 0.009, but its raw rejection is 1.25–2.07× the off-the-shelf draft. Adaptation buys in-domain fit at the cost of raw rejection; it does not remove the low-margin bottleneck. This is why we do not claim that distillation fails. It is a different objective, and this audit does not rank it against the post-hoc operator.

Appendix Table A.55 gives the target-native MTP head as a cheap trained control. The MTP draft has nominal $\rho \approx 0.126$ (it shares the target backbone and LM head), raw rejection 0.304/0.241, and low-margin-quintile rejection 0.591/0.533. It is much cheaper than any post-hoc width point and its nominal rejection can clear break-even. It shares the target backbone, so its effective cost is not the nominal FLOP ratio; we report it as a low-cost control for the low-margin bottleneck, not as a deployment recipe and not as a ranking against other draft families.

8.5 The complementary capacity scan

Appendix Table A.49 reports the local exponent of greedy rejection versus draft capacity for the earlier off-the-shelf direction. In-domain acceptance improves with capacity, but the raw-text rejection floor falls slowly: the local exponent is shallow, and the capacity multiple needed to halve q is large. The raw-text floor is about 0.13 across corpora and targets, consistent with the margin-mass mechanism. Appendix Table A.17 checks the geometric acceptance-length approximation used in the cost model; the empirical acceptance length is within 1.00–1.12 of the geometric value on raw text, so the cost model is conservative.

8.6 What the negative results do not show

They do not show that structured pruning is useless for perplexity, that quantization fails, that trained drafts are bad, or that every engine will see the same wall clock. They show that the post-hoc structured operator, without target-response training, does not clear the same-kernel certificate on raw text, and that the measured batch-1 cost does not clear break-even for the tested targets. Adaptation, quantization, and other draft families are outside the operator; this audit does not rank them.

9 Reproduction Protocol

This section gives the protocol details needed to reproduce the main tables without referring to a code release. The full model versions and the artifact map are in Appendix A.22; this is the short version.

Models and tokenizers. The main target is Qwen3.5-9B. The replications use Qwen3.5-27B, Qwen3.5-4B, Qwen3-4B-Instruct-2507, Llama-3.1-8B-Instruct, K2-Horizon-3.7B, and Qwen3.5-35B-A3B. Target and compressed draft are the same checkpoint; the draft is produced by deleting FFN channels and refitting the readout. Channel scores are computed once per model on a 200k-token target-aligned calibration set, and the same scores are used for every evaluation corpus at a given width.

Data. WikiText-2, FineWeb-Edu, C4, four Wikipedia languages, FineWeb-2 Chinese, CodeParrot, PG-19, arXiv, 100 SWE-bench-Lite issue-to-patch prefixes, 80 MT-Bench first turns, and MATH-500 target-generated continuations. Each raw corpus uses 100 chunks; token and cluster counts are in Table A.84. The target cache stores the top-256 target distribution and the draft logits for the evaluation tokens.

Quality pipeline. For each width: load the target, collect activations on the calibration set, score channels by $s_j = \mathbb{E}|a_j| \, \|W_d[:, j]\|_2$, delete the lowest-scoring channels, refit the readout by residual ridge on held-out activations, cache the target top-256 distribution and the draft logits on the evaluation tokens, and compute q_g, q_s, and q_{lb}. Intervals are 95% sequence-cluster bootstrap over evaluation sequences (2,000 resamples, seed 0). The confirmatory families are fixed before the runs; the Bonferroni level is 0.05 divided by the family size.

Latency pipeline. The batch-1 measurement loads the target and the draft in separate processes so that a 27B checkpoint fits on one A100-80GB. With a 512-token prefix in the KV cache, it times one target token pass, a $\gamma + 1$ target verification pass, one draft token pass, and γ sequential draft passes. The prefix cache is pre-cloned outside the timed region. We report two protocols: a target-resident wall-clock loop with per-pass CUDA events, and a kernel-time window that queues a batch of pre-cloned passes between events and sums CUDA kernel time. At batch 1 the wall clock is launch-bound, so the decomposition uses kernel time; both are saved in the artifact. The zero-FFN control replaces the MLP forward with a zero path and measures the same draft pass. End-to-end loops use exact KV rollback on the dense models and a negative crop (or exact re-prefill) on the 9B hybrid cache.

Hardware and scope. Quality measurements use one A100-80GB per run; the 27B batch-1 measurement and the dense zero-FFN control each use one A100-80GB. The new float32 floor control and the Llama MATH cost measurement each use one A100-80GB. The statistical and mechanism analyses are CPU-only. The production-kernel question remains open; our cost numbers are a padded-kernel upper bound, not a lower bound. The code is not attached to this submission; the protocol above, the model versions, and the reported statistics are sufficient to reproduce the audit on public checkpoints. The full sweep, replications, and controls are on the order of a few hundred A100-hours.

Minimal reproduction checklist.

1. Qwen3.5-9B (or a smaller target) with 100 WikiText-2 and FineWeb-Edu chunks at 512 continuation tokens.
2. Calibration set (200k tokens); score channels; sweep widths 0.5/0.9/0.98/0.99.
3. Top-256 target cache; compute q_g, q_{lb}, the exact tie mask, and the tie component.
4. Compare with $q^\star_{\mathrm{FLOP}}$ at $\tau = 1$ and $q^\star(\rho_1, \tau_1, \gamma)$; report the end-to-end loop as the cost check.
5. For the screen: fit $q = a_w + b_w L$ on ten corpora, predict the eleventh, and refit per target before using it on a new model.

Latency protocol caveat. The first batch-1 protocol called `copy.deepcopy` on the KV cache inside the timed pass; that copy is not part of a serving pass and biases ρ and τ toward one. A clean protocol should pre-clone the cache outside the timed region (or crop it back after the pass) and should run on an isolated GPU. We report the cost layer as implementation-conditional and use the roofline and end-to-end loops as the primary cost evidence.

10 Discussion

Takeaway. The result has two layers with different scopes. The rejection layer is a same-kernel, $c = 1$ statement: on measured raw English prose $q_{\mathrm{lb}} > q^\star_{\mathrm{FLOP}}$ at every compressed width, and the near-identity failure is a decision-boundary resolution effect that channel ranking cannot remove. The cost layer is nested

inside it: under the optimistic roofline, width deletion always leaves a positive break-even threshold $q^{\star}_{\text{roof}} > 0$, so the frontier no-go is a rejection statement. At 98–99% the raw cells are roofline-bound by 13–20×; the measured batch-1 cost is also above break-even in the padded implementation, an implementation-conditional diagnostic. At 50% the cells are R-free. The cost-load-bearing cells we identify are the Llama MATH 50–70% points, with a 0.4–2.3% shape margin inside the timing spread, and the 27B MATH 50–90% points, with a 5–60% margin. A best-of-budget global search on the tested Qwen3.5-9B WikiText/FineWeb cells over channel subsets, per-layer widths, and an OBS-style score does not reopen a raw cell. OBS and random lower point q without improving the certified lower bound; the default retains a larger certified margin than both. For the default, OBS, and random selectors the single-cell lower endpoints are 1.4–18× $q^{\star}$ (Section 5.5).

10.1 A nested result, two audiences

The rejection layer carries the conclusion: at same-kernel efficiency, the compressed draft rejects the target's top-1 token more often than the roofline allows. The cost layer is a practitioner-facing check nested inside that conclusion: with this kernel and this serving regime, the draft is not cheaper. It is not a second independent conclusion. Keeping the layers separate prevents either an overclaim ("compression never works") or an underclaim ("we measured one kernel"). That separation is the methodological contribution, and it is why we report $c^{\star}$ and the batch scan instead of a single speedup.

10.2 Why the rejection floor is the more robust result

The rejection floor is measured on tokens, not on a clock. It depends on the target margin and the draft logits, both of which are stable across kernels and batch sizes. The cost result depends on the engine, the batch, the prefix length, and the implementation. A reader who distrusts the wall-clock can still use the certificate; a reader who distrusts the certificate can still use the measured cost. The two agree on every raw cell in our audit, which is the empirical reason we believe the conclusion is not an artifact of one measurement.

The resolution-floor part is more portable than the specific numbers. The lesson is that top-1 agreement under a compression operator is not a smooth function of reconstruction error: as the operator approaches identity, the rejection rate stops falling with the perturbation size and is set by the target's near-boundary mass. This applies to any structured compressor whose perturbation is small relative to the margin distribution, including operators other than FFN-width deletion. The exact bf16 tie atom is one precision-specific component; a clean float32 rebuild removes it but leaves a 0.31% floor (129× the float32 break-even), so the mechanism is not a tie-order artifact. We measure the FFN case because it is the natural self-draft baseline; we do not claim that every compressor has the same floor.

The in-domain result does not change the operational conclusion. On the measured 9B and 27B targets the MATH window is closed by the memory frontier or by the clean cost. The Llama-3.1-8B low-width MATH window has a FLOP window at 50–70% width and is cost-closed in the padded narrow-MLP implementation under the clean protocol (5.067/5.066/5.154); the shape margin is 0.4–2.3%, inside the timing spread, so a draft-exclusive kernel could reopen those two points. The 90% point is roofline-bound and is closed by rejection. The existence of these windows shows that the audit failure is distribution-conditional, not that a deployment recommendation follows. A practitioner with in-domain traffic still needs the prefix audit of Section 7.

10.3 The role of the screen

The screen is a triage tool. A deployment team can run one target forward pass on a small prefix sample, estimate the low-margin mass, and decide whether a certificate measurement is needed. The screen is calibrated on eleven raw corpora and two deployment suites, but its false-negative upper bound is not zero. The asymmetric rule—measure when below threshold—is the operational content. The screen is also a diagnostic for the mechanism: if a new deployment distribution has a low-margin mass similar to the measured corpora, the mechanism predicts a similar floor. Its coefficients are per-target, and the template/tool-call control is a characterized boundary case rather than a success.

Table 13: Open experiments and the claim layer each would move. None is required for the raw-text audit failure at $c = 1$; they matter for the cost layer, the in-domain window, or the cross-family scope.

Experiment	Why it matters	Layer moved
Production W4/W4A8 kernel	Could move ρ_1 toward ρ_{mem}; quantization is a memory-only axis.	Cost
9B hybrid exact-rollback production loop	The exact re-prefill loop is an upper bound; a production hybrid cache would give a realized number.	Cost
Tree verification on target top-2 branches	The current claim excludes tree proposals; a restricted tree could inherit the top-2 bound.	Rejection
Exact combinatorial global optimum	The tested global subset/DP/random search is a best-of-budget heuristic; an exact integer optimum remains open.	Rejection

10.4 What would change the conclusion

The conclusion has two layers, so there are two kinds of change. A result that moves the certificate layer must change measured top-1 agreement on the evaluation distribution; a result that moves the cost layer must change the realized kernel or the serving regime. We list the conditions separately and say which layer each one moves. Table 13 is the short list of open experiments; the paragraphs below give the reasoning.

A faster draft kernel. The 98–99% arithmetic $c^\star \approx 1.2$–1.4 is not a realizable window under the post-hoc operator family. It would require a draft-specific kernel that is faster than the target's kernel by that factor on the whole draft pass; at 98–99% width the draft and target compute are nearly identical ($\rho_{\text{FLOP}} \approx 0.99$) and share a kernel family, and the only draft-specific component is the FFN. A fused or W4A8 kernel that accelerates the whole pass would also accelerate the target, so it does not create a relative advantage. Our Triton fused MLP is 0.42–0.72× cuBLAS, and compiled kernels are slower than the padded GEMM. The mid-width cells need 3–6× and are robust to this change. A different operator family—quantization, retrieval, or exact identity—is the real escape, not a faster kernel on the same operator.

A better verification kernel. The cost is $\gamma\rho/c + \tau$. Lowering τ_1 relaxes the cost condition. On the dense Qwen3-4B the clean free-FFN cost is $4.01 < 5$ and $q = 0.10$ is just below the implied $q_0^\star = 0.11$, so the realized MLP is the binding cost and a production MLP kernel is the lever. On a target with a very efficient verification kernel, more cells become cost-feasible; the certificate layer is unchanged. This is the sense in which the batch-1 cost result is implementation-conditional and should not be read as a theorem about every engine.

A different deployment distribution. In-domain text with high target margin has a window. On Qwen3.5-9B, MATH shows a significant 50% FLOP-only window under the FLOP frontier with $\tau = 1$; it is closed by the memory frontier or the measured cost. The 30% point is not significant (the gap to $q^\star_{\text{FLOP}}$ is 0.0004, inside the cluster interval). On Qwen3.5-27B the 35B-generated fixed-continuation MATH window is 50–90%; the clean kernel-time cost is 4.23–4.84, and the implied break-even rejection (0.083/0.050/0.016) is below the measured 0.088/0.056/0.026, so rejection closes the window at the clean cost; the 98% point is also a cost no-go ($5.07 > 5$). A deployment dominated by such text would need its own audit; the raw-text no-go does not apply to it. The screen is designed for this case: estimate the low-margin mass on the deployment prefix, and measure the certificate if the screen is below threshold.

Quantization. Weight-only quantization changes precision, not the FLOP ratio. It is a memory-only axis: a production W4 kernel or a W4A8 tensor-core path could move ρ_1 toward ρ_{mem} and reopen the cost-only window for a quantized draft. Our unoptimized packed kernels are not faster than bf16 at batch 1, but the literature reports 1.2–2.7× speedups with optimized kernels. This does not contradict the FLOP-axis audit; it changes the axis. We do not claim a measured end-to-end quantization win under the present harness.

A compute-bound serving regime. At large batch the verification pass becomes more expensive and the cost condition changes. Our measured scan to $B = 128$ does not open a window: the cost stays above 5 (5.41 at 98% width). A different engine or a much larger batch could change ρ_B and τ_B; the roofline bound is the implementation-independent statement, and the measured batch-1 cost is the service benchmark. We do not claim a wall-clock theorem for every batch size.

A retrieval or prompt-lookup baseline. Retrieval and prompt-lookup drafts are the $\rho \to 0$ limit; they have no compressed target weights to certify. They are a practical alternative when the deployment text has long verbatim spans. We do not include a measured anchor for them in this report, so we do not make a comparative recommendation; the statement is about cost structure, not a controlled comparison.

A different compression operator. The certificate and the cost test apply to any draft; what changes is the perturbation. A low-rank or rotation-based operator with a smaller Δ_m on low-margin tokens could in principle do better. Our measured rotation and SVD baselines are worse, but they are not an exhaustive search. The support-selection principle says that an exact compute-reducing operator must delete inactive units; an approximate operator can be better, and the question is empirical. The margin-mass diagnostic is the right screening tool for such an operator.

11 Limitations

Audit and protocol. q_{lb} is computed on the evaluation tokens; the only formal transfer statement is the TV-Lipschitz bound of Proposition 3. The 36 Qwen raw cells share one target, tokenizer, and evaluation format, so they are cells rather than independent replications; the nine Llama/K2 cells are the cross-family check. We do not estimate a practical TV radius for a new deployment; the screen and the mandatory direct audit are the operational substitute, and the screen's false-negative bound is empirical, not distribution-free. The cost numbers use one A100, batch 1, and a padded narrow-MLP implementation; they are an upper bound on relative cost, not a wall-clock theorem for every engine. The 9B hybrid cache does not support exact rollback, so the negative-crop loop is optimistic and the exact re-prefill loop is an upper bound; we do not report an end-to-end throughput claim for that model.

Resolution floor. The exact-tie atom is a bf16 property of the evaluated target. A clean float32 rebuild removes the tie atom but leaves a 0.31% one-channel floor (129× the float32 break-even), so the floor is a near-boundary mass phenomenon, not a bf16 tie-order artifact. The deployed target is bf16; we do not claim the exact percentages for an arbitrary rounding engine. The floor is insensitive to the tested channel rankings; it is not a claim that every perturbation is equally harmful.

Cost layer. The clean measurement reports both a Python-loop wall clock and a kernel-time window. The wall clock is launch-bound at batch 1 and overstates the residual non-MLP cost; realized costs are quoted from the kernel-time protocol. The four cost thresholds are nested (Appendix A.5); the 98–99% raw cells are roofline-bound by 13–20×, so the operator-level verdict does not depend on the kernel, batch, or hardware. The measured cost is at or above 5, an additional acceptance-independent implementation diagnostic (point estimate). A portable production MLP kernel moves the relative ratio toward one (Proposition 8); only a draft-exclusive kernel can move a K cell. We therefore do not claim a zero-FFN ceiling.

Screen. No observed false negatives on eleven raw corpora and two deployment suites; the one-sided 95% upper bound is 5.3% per cell and 23.9% per corpus. The coefficients are per-target: cross-model transfer preserves the binary raw decision but over-predicts the raw K2 cell by 1.2× (conservative); the earlier 30/50 target-generated K2 check is under-predicted. A new target still requires a refit. The template control gives a mass-only false negative at 50%; below-threshold predictions must be measured. The screen is calibrated for FFN-width deletion; we do not claim that the same coefficients transfer to depth, head, or MoE operators without a per-operator calibration.

Scope. All sweeps use a uniform r/h per layer except the non-uniform control; the main sweep is Qwen3.5-9B; the selected-width cross-family check covers those widths; the agentic suite is a static prefix. "Raw" is operational; contamination cannot be established. The global stress test is Qwen3.5-9B only; it is not replicated cross-family. Other draft families, tree verification, retrieval, and compute-bound serving are different objects, and we do not run a matched-latency comparison against them. A self-draft also keeps a

second FFN copy; width reduction shrinks it by r/h, so the memory saving is partial and is not the object of this paper.

12 Conclusion

We audited the natural no-training self-draft baseline: delete FFN channels from the target and use the compressed model as its own draft. On the evaluated raw English token population, under the stated operator and audit, the baseline fails the rejection layer before the cost layer matters. At 99% width the top-1 rejection is 27–28× the $c = 1$ break-even; at one deleted channel per layer it is still 500–550×, and an oracle channel ranking reduces that floor by only 5–10%. The mechanism is a decision-boundary resolution floor built from exact bf16 ties and near-boundary mass, not a channel-ranking defect; a clean float32 control removes the tie atom but leaves a 0.31% floor. The cost layer is nested inside the rejection layer: the 98–99% raw cells are roofline-bound by 13–20×, the measured clean kernel-time cost is also at or above break-even in the padded implementation (a point estimate, not a production-kernel claim), and the 50% cells are closed even to a free FFN. The cost-load-bearing cells we identify are the Llama MATH 50–70% points, where the shape margin is inside the timing spread, and the 27B MATH 50–90% points, whose margins are 5–60%. Confirmatory evidence is the 36 Qwen raw cells plus 9 independent Llama/K2 cells; the 200-cell raw count is descriptive. The practical output is a margin-mass screen with a mandatory direct audit, and a decision map that separates the rejection layer from the measured-regime cost layer. The result is about this natural but non-strongest post-hoc baseline and the best-of-budget global search tested in Section 5.5; it is not evidence that speculative decoding, other draft families, or tree verification fail in production.

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Table A.1: Appendix index. Section numbers are automatic; this table is for navigation.

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A.1 Proofs

Throughout the appendix, "certificate" means the in-sample measured lower bound q_{lb} on the evaluation tokens, not a pre-deployment guarantee.

Table A.2: Notation. Batch-dependent quantities carry a subscript B; τ_1 is the $B = 1$ verification ratio and $\tau(\gamma)$ makes the draft length explicit.

Symbol	Meaning
P_x, Q_x	target and draft next-token distributions at prefix x
q_g, q_s	greedy top-1 rejection; standard-sampling rejection $\mathbb{E}[\mathrm{TV}(P_x, Q_x)]$
$q_{\mathrm{lb}}^{(K)}$	nested top-K certificate $\Pr(E_K)$; $q_{\mathrm{lb}}^{(2)}$ is the main bound
m, Δ_m	target top-1/top-2 log-margin; draft-induced perturbation of that margin
$\rho_{\mathrm{FLOP}}, \rho_{\mathrm{mem}}$	nominal draft/target matmul-FLOP and byte ratios
ρ_B, τ_B	measured per-token draft/target pass ratio and verification ratio at batch B
$\tau_1, \tau(\gamma)$	$B = 1$ verification ratio; same ratio with explicit draft length
$q^\star_{\mathrm{FLOP}}$	break-even rejection under the FLOP roofline ($\tau = 1$)
$q^\star_{\mathrm{mem}}$	break-even rejection under the memory roofline
$q^\star_{\mathrm{meas}}$	break-even rejection at the measured clean kernel-time cost
$q^\star_{\mathrm{free}}$	break-even rejection for a free-FFN counterfactual (diagnostic)
$C_{\mathrm{meas}}, C_{\mathrm{roof}}$	measured cost $\gamma\rho_1 + \tau_1$; roofline cost $\gamma\rho_{\mathrm{roof}} + 1$
$\alpha, \alpha^\star$	per-token acceptance $1 - q$; break-even acceptance $1 - q^\star$
$\phi_{\mathrm{MLP}}, \phi_E$	FFN share of model FLOPs; routed-expert share of MoE active FLOPs
$c, c^\star$	draft-kernel effective throughput; minimum advantage that reopens a cell
c_{shape}, s	width-only shape ceiling $1/\rho_{\mathrm{FLOP}}$; measured MLP time share
W/K/G/R	cost verdicts: window, realized-kernel, regime, roofline

Lemma 2 (Exact all-readout width reduction forces $r = h$)**.** *Consider the gated FFN $y = W_d(\sigma(W_g x) \odot W_u x)$ with a shared intermediate basis $F_P(x) = W_d P^\top(\sigma(PW_g x) \odot (PW_u x))$, $PP^\top = I_r$. If F_P equals y for all x and all block weights, then P is a signed coordinate permutation and $r = h$; in particular, no nontrivial exact width reduction exists.*

Lemma 3 (Perturbation of deleting channels)**.** *For deleted channels S and $a(x) = \sigma(W_g x) \odot W_u x$, $y_S(x) = y(x) - \Delta y(x)$ with $\Delta y(x) = \sum_{j \in S} a_j(x) W_d[:, j]$.*

Proposition 1 (Support selection)**.** *Consider a module whose output is a sum over compute-bearing units $u \in U$, $y = \sum_{u \in U} A_u \phi_u(x; \theta_u)$, where the readout A_u is linear and independent across units. If an operator removes a set R and is exact for all inputs and all readouts, then every removed unit satisfies $\phi_u(x) = 0$ on the input domain. For a fixed readout, exactness can also hold when $A_u \phi_u(x) = 0$ (the unit is output-inactive), the fixed-weight exception of Lemma 4. Consequently: (i) FFN width: exact reduction under all block weights is a signed coordinate selection, and $r = h$ (Lemma 2); (ii) MoE: reducing the active-expert count is exact only if the marginal expert has zero gate (Lemma 5); (iii) attention head: exact deletion requires the head output to be zero for every input; (iv) depth: exact layer deletion requires the residual block to be the identity.*

Every nontrivial compute-reducing post-hoc operator is therefore approximate: deleting a unit that the target already does not use saves compute but does not change the function.

Lemma 4 (Fixed-weight exactness and redundancy)**.** *Let $a(x) = \sigma(W_g x) \odot W_u x$, $y(x) = W_d a(x)$, $K \subseteq [h]$ retained, $D = [h] \setminus K$. For a fixed candidate readout R (chosen by the operator, possibly refitted), the compressed block with retained activations a_K is exact on an open set U if and only if*

$$W_{d,D}\, a_D(x) \;=\; (R - W_{d,K})\, a_K(x) \qquad \text{for all } x \in U. \tag{11}$$

The "if and only if" is for a given R; the existence of some R is a weaker condition and is characterized by whether $W_{d,D} a_D$ lies in the span of a_K on U. For pure deletion ($R = W_{d,K}$) the condition reduces to $W_{d,D} a_D(x) = 0$ on U. Two fixed-weight escapes exist: an output-inactive channel ($W_d[:, j] = 0$), or a functional redundancy $a_D(x) = M a_K(x)$ with a refitted readout. Both are nongeneric; for generic weights and the SiLU activation used here, $a_D = 0$ on an open set forces the deleted gate/up rows to vanish. The all-readout lemma forbids the redundancy escape.

Lemma 5 (MoE perturbation)**.** *For a top-k MoE block, active-expert reduction to k' has the exact perturbation*

$$\Delta y(x) = \sum_{e \in S(x) \setminus S'(x)} g_e(x) E_e(x) + \sum_{e \in S'(x)} \big(g'_e(x) - g_e(x)\big) E_e(x), \tag{12}$$

with nominal ratio

$$\rho_C = 1 - \phi_E (1 - k'/k), \tag{13}$$

where ϕ_E is the routed-expert share of active FLOPs. Expert deletion at fixed k is memory-only.

Proposition 2 (Shift and mixture)**.** *For any π, μ, Q, $|q^\pi_s(Q) - q^\mu_s(Q)| \le \mathrm{TV}(\pi, \mu)$ and likewise for q_g. If $\pi = (1 - \epsilon)\mu + \epsilon\nu$, then $q^\pi_s = (1 - \epsilon) q^\mu_s + \epsilon q^\nu_s$.*

Proposition 3 (Certificate transfer)**.** *For any two prefix distributions π, μ and any draft Q,*

$$\big|q^\pi_{\mathrm{lb}}(Q) - q^\mu_{\mathrm{lb}}(Q)\big| \;\le\; \mathrm{TV}(\pi, \mu), \qquad q^\pi_{\mathrm{lb}}(Q) := \Pr_{x \sim \pi}\big(m(x) + \Delta^Q_m(x) < 0\big). \tag{14}$$

Proof. The event $\{m + \Delta^Q_m < 0\}$ is a 0–1 loss, bounded in $[0, 1]$. The difference of its expectations under π and μ is at most the total variation distance. □

Proposition 4 (Coverage requirement)**.** *If $q^\pi_s(Q) \le q^\star$, then $\pi(\{x : \mathrm{TV}(P_x, Q_x) \ge \delta\}) \le q^\star/\delta$ for every $\delta > 0$: the draft must be δ-close on at least $1 - q^\star/\delta$ of the deployment mass. For q_g, let $\ell(x) = \mathbf{1}\{\arg\max_v Q_x(v) \ne \arg\max_v P_x(v)\}$. The same argument with this bounded 0–1 loss says that the draft must agree with the target's top-1 on at least $1 - q^\star$ of the deployment mass. Equivalently, for every set A with $\pi(A) > q^\star$, the draft cannot be wrong on every $x \in A$. There is no δ-tube version because ℓ is already 0–1.*

Proposition 5 (Uncovered-region lower bound)**.** *Let $\pi = (1 - \epsilon)\mu + \epsilon\nu$ with μ, ν mutually singular, and let two targets agree on $\mathrm{supp}(\mu)$ but have disjoint top-1 sets on $A \subseteq \mathrm{supp}(\nu)$. A learner that sees only μ cannot distinguish them, so its worst-case rejection on A is at least*

$$\frac{\epsilon\, \nu(A)}{2}. \tag{15}$$

If this exceeds $q^\star$, no data-only learner trained on μ guarantees break-even on π; the q_g version replaces TV *by the bounded 0–1 indicator. The construction is realizable: take two copies of a base model that agree on the continuation distribution of μ and are instruction-tuned on disjoint domains in ν; it is a worst-case separation, not a claim about a particular model pair.*

Proof of Lemma 2. Because the equality holds for every W_d, the two intermediate vectors must agree:

$$P^\top\big(\sigma(PW_gx)\odot PW_ux\big)=\sigma(W_gx)\odot W_ux \tag{16}$$

for all x, W_g, W_u. The left side lies in the column space of $P^\top$, which has dimension at most r. The right side can be any vector in $\mathbb{R}^h$: take $x=e_1$, let the j-th row of W_g have first entry α_j and the j-th row of W_u have first entry $\beta_j=v_j/\sigma(\alpha_j)$ for a fixed α_j with $\sigma(\alpha_j)\neq 0$. Hence $r=h$. With $r=h$, $PP^\top=I_h$ makes P orthogonal, and (16) becomes $F_P(z)=P^\top(\sigma(Pz)\odot Pz)=\sigma(z)\odot z$ for all $z\in\mathbb{R}^h$ (choose $W_g=W_u$ with $W_gx=z$ and $W_d=I_h$).

The Jacobian is

$$J_P(z)=P^\top\operatorname{diag}\big(\sigma'(Pz)\odot Pz+\sigma(Pz)\big)\,P.$$

If $F_P=G$ for all z, then $J_P(z)$ must be diagonal for all z. The entries $d_j(w)=\sigma'(w_j)w_j+\sigma(w_j)$ of the inner matrix have open image as w varies, and $w=Pz$ covers $\mathbb{R}^h$. For an off-diagonal entry (k,l), $k\neq l$, we have $\sum_j P_{j,k}d_j(Pz)P_{j,l}=0$ for all such d; hence $P_{j,k}P_{j,l}=0$ for every j. To see the last step: d_j depends only on w_j, and the w_j vary independently over an open set, so $(d_1,\dots,d_h)$ ranges over an open rectangle; a linear form that vanishes on an open rectangle has all coefficients zero. Thus each row of P has at most one nonzero entry. With P orthogonal, the nonzero entries are ±1 and distinct rows select distinct coordinates. A negative entry is impossible: at $z=te_i$ with $t>0$, it gives $-\sigma(-t)t<0$ while the target's i-th coordinate is $\sigma(t)t>0$. Hence P is a positive coordinate selection for SiLU. For a general C^1 activation whose $d(t)=\sigma'(t)t+\sigma(t)$ has open image, the same off-diagonal argument gives coordinate selection up to signs; the sign is fixed only when the activation is one-signed on the relevant ray. This argument does not require $d\geq h$; it only uses $d\geq 1$.

Proof of Lemma 4. Subtracting the retained contribution from $y(x)$ gives $y(x)-W_{d,K}a_K(x)=W_{d,D}a_D(x)$, so $Ra_K(x)=y(x)$ is equivalent to (11). For $R=W_{d,K}$ the right side is zero. In the regime $|D|\leq d$, if $W_{d,D}$ has full row rank and $a_D(x)$ spans $\mathbb{R}^{|D|}$ on U, then $W_{d,D}a_D(x)=0$ for all $x\in U$ implies $a_D(x)=0$ on U. Componentwise, $a_j(x)=\sigma(w_{g,j}\cdot x)(w_{u,j}\cdot x)$. For SiLU, $\sigma(t)=t\,\mathrm{sigmoid}(t)$ vanishes only at $t=0$. If $w_{u,j}\neq 0$, the linear functional $w_{u,j}\cdot x$ is nonzero on the complement of a hyperplane, an open dense set; on that set $a_j=0$ forces $w_{g,j}\cdot x=0$, hence $w_{g,j}=0$. Thus a channel that is not output-inactive and has a nonzero up row can vanish on an open set only in the degenerate case $w_{g,j}=0$; for generic weights this does not occur. Thus, in the $|D|\leq d$ regime, generic fixed weights force coordinate deletion of inactive channels. For $|D|>d$ we keep the exact identity and do not claim the generic conclusion; the deployed 50% width deletion can be in that regime. The redundancy escape (ii) is a nongeneric algebraic relation; it is exactly what the all-readout quantifier in Lemma 2 rules out, because a generic readout A_D would expose the missing term $A_Da_D(x)$.

Fixed-weight exactness: what we do and do not claim. Lemma 4 is the exact statement: for a fixed readout R, equality on an open set is equivalent to $W_{d,D}a_D(x)=(R-W_{d,K})a_K(x)$; for pure deletion this is $W_{d,D}a_D(x)=0$. A deleted channel can be output-inactive ($W_d[:,j]=0$), or the deleted activations can satisfy an exact linear relation $a_D=Ma_K$ that a refitted readout absorbs. These are nongeneric algebraic conditions, but we do not formalize a generic-weight quantifier here. A full-row-rank sketch would assume W_g and W_d full rank and reduced the block to $a(z)=\sigma(z)\odot z$; that reduction needs assumptions the deployed FFNs do not satisfy. In a typical FFN $W_g\in\mathbb{R}^{h\times d}$ with $h>d$, so W_g cannot have full row rank; if one restricts to $d\geq h$ and a full-row-rank W_g, the reduction still has to track the kernel component of W_u, which the sketch did not do. We therefore state the exact identity (Lemma 4) and treat the nongeneric escapes as a remark, not a theorem. Lemma 2 does not have this gap: it quantifies over all W_g, W_u, and $x=e_1$ makes the intermediate vector arbitrary.

Lemma 6 (Refitted-readout decomposition)**.** *Let K be the retained channels, D the deleted channels, and let the deployed draft use an arbitrary readout R, i.e. $y_R(x) = Ra_K(x)$. Then the perturbation is exactly*

$$\Delta y(x) := y(x) - y_R(x) = W_{d,D}a_D(x) + (W_{d,K} - R)a_K(x). \tag{17}$$

For the residual-ridge readout used in Appendix A.11,

$$R = W_{d,K} + W_{d,D}\Sigma_{DK}(\Sigma_K + \lambda I)^{-1}, \qquad \Sigma_K = \mathbb{E}[a_K a_K^\top], \quad \Sigma_{DK} = \mathbb{E}[a_D a_K^\top], \tag{18}$$

so

$$\Delta y(x) = W_{d,D}\Big[a_D(x) - \Sigma_{DK}(\Sigma_K + \lambda I)^{-1}a_K(x)\Big]. \tag{19}$$

The first term is the deleted-channel perturbation; the second is the readout correction. With the empirical moments used by the fit, (19) holds exactly on every input, so the lower bound is a statement about the deployed (refitted) draft, not only the pruning step.

Proof of Lemma 6. Equation (17) is algebra: $y = W_{d,K}a_K + W_{d,D}a_D$ and $y_R = Ra_K$. For the residual-ridge objective $\min_R \mathbb{E}\|y - Ra_K\|^2 + \lambda\|R - W_{d,K}\|_F^2$, set $R = W_{d,K} + D$. The first-order condition is

$$\mathbb{E}\big[(W_{d,D}a_D - Da_K)a_K^\top\big] - \lambda D = 0 \quad \Longrightarrow \quad D\big(\Sigma_K + \lambda I\big) = W_{d,D}\Sigma_{DK},$$

where $\Sigma_K = \mathbb{E}[a_K a_K^\top]$ and $\Sigma_{DK} = \mathbb{E}[a_D a_K^\top]$. Hence $D = W_{d,D}\Sigma_{DK}(\Sigma_K + \lambda I)^{-1}$, which is (18); substituting into (17) gives (19). Replacing population moments by their calibration-sample counterparts makes the same identity exact for the fitted readout on every input. The $\lambda = 0$ and $\lambda \to \infty$ limits follow by inspection.

For interpretation, let J_v be the gradient of the logit of token v with respect to the block output. A first-order expansion gives $\Delta_m = \langle J_a - J_b, \Delta y\rangle + O(\|\Delta y\|^2)$. This is why a small Δy can still flip a token whose target margin m is small; the certificate uses the measured Δ_m, not this expansion.

Lemma 7 (Nested top-k lower bound)**.** *With target top-K tokens $v_1, \dots, v_K$ and $E_K(x) = \{\exists j \in \{2, \dots, K\} : z_Q(v_j) > z_Q(v_1)\}$, $E_2 \subseteq \cdots \subseteq E_V = \{$the draft rejects $v_1\}$, so $q_{\mathrm{lb}}^{(2)} \le q_{\mathrm{lb}}^{(K)} \le q_g$ and $q_{\mathrm{lb}}^{(V)} = q_g$.*

Proof of Lemma 7. The events are nested because E_K is a disjunction over a growing set. If E_V holds, some token has draft logit above $z_Q(v_1)$, so the draft's argmax is not v_1 and the token is rejected. If the draft rejects v_1, its argmax $v^* \neq v_1$ has $z_Q(v^*) > z_Q(v_1)$ and v^* appears somewhere in the full vocabulary, so E_V holds. Hence E_V is exactly the rejection event and $q_{\mathrm{lb}}^{(V)} = q_g$. Each E_K is a subset of the rejection event, giving the bounds.

Proof of Proposition 1. Write the module output as $y = \sum_{u \in U} A_u \phi_u(x; \theta_u)$ with independent linear readouts A_u. If the operator removes R and is exact, then $\sum_{u \in R} A_u \phi_u(x; \theta_u) = 0$ for all inputs and all readouts. Fix x and set all readouts except A_u to zero; varying A_u gives $A_u \phi_u(x; \theta_u) = 0$ for all A_u, hence $\phi_u(x; \theta_u) = 0$. The FFN, head, and residual-layer instantiations identify A_u with the corresponding output-projection block or residual path. The MoE instantiation is Lemma 5: the gate $g_e(x)$ is the readout weight, and it depends on the router rather than being free, so the exactness condition is that the gate is zero on the input domain.

Proof of Lemma 5. Equation (12) follows by subtracting the two sums and splitting $S(x) = (S(x) \setminus S'(x)) \cup (S(x) \cap S'(x))$; the surviving terms differ only in their renormalized gates. If $S'(x) = S(x)$, the first sum is empty and $g'_e = g_e$, so $\Delta y = 0$. If k is reduced, $S'(x) \subset S(x)$ and the marginal expert $e^\star \in S(x) \setminus S'(x)$ has gate $g_{e^\star}(x) > 0$ whenever it is selected; exactness for all expert weights requires its output to vanish or its gate to be zero for every x. Deleting an expert with fixed k is the special case where $S'(x) = S(x)$ except on inputs that select the deleted expert; exactness again requires that expert never to be selected (or to have zero output).

Lemma 8 (Protocol-independent rejection floor)**.** *Any exact SD rule that uses the draft distribution Q has per-position rejection probability at least $\mathrm{TV}(P, Q)$; standard speculative sampling attains the bound, so $q_s = \mathbb{E}[\mathrm{TV}(P, Q)]$ lower-bounds every exact token-level rule.*

Proof of Lemma 8. Let r be the rejection probability and let $\mu(w)$ be the probability that a draft token w is accepted and emitted. Exactness requires $\mu(w) \le Q(w)$ and $\mu(w) \le P(w)$, so $\sum_w \mu(w) \le \sum_w \min(P(w), Q(w)) = 1 - \mathrm{TV}(P, Q)$. Hence $r = 1 - \sum_w \mu(w) \ge \mathrm{TV}(P, Q)$. Residual sampling with acceptance $\min(1, P/Q)$ uses $\mu = \min(P, Q)$ and attains the bound.

Proposition 6 (Draft-agnostic lower bound)**.** *For any draft Q, post-hoc or distilled, with margin perturbation Δ_m^Q, $q_g(Q) \ge \Pr(m(x) + \Delta_m^Q(x) < 0)$; the TV floor is also draft-agnostic. What changes for a distilled draft is the origin of Δ_m^Q: a training outcome rather than a structural deletion term.*

Proof of Proposition 6. Apply Lemma 1 to the arbitrary draft Q: if $m(x) + \Delta_m^Q(x) < 0$, the draft ranks the target's second token above its first, so the token is rejected. The sampling floor is Lemma 8.

Proof of Proposition 2. Both q_s and q_g are expectations of a loss bounded in $[0, 1]$: q_s uses $\mathrm{TV}(P_x, Q_x) \in [0, 1]$ and q_g uses the 0–1 disagreement. A bounded loss is 1-Lipschitz in total variation, so $|q^\pi - q^\mu| \le \mathrm{TV}(\pi, \mu)$. The mixture identity is linearity of expectation: $\mathbb{E}_{(1-\epsilon)\mu+\epsilon\nu}[\ell] = (1-\epsilon)\mathbb{E}_\mu[\ell] + \epsilon\mathbb{E}_\nu[\ell]$.

Proof of Proposition 4. Apply Markov's inequality to the nonnegative random variable $Z(x) = \mathrm{TV}(P_x, Q_x)$: $\pi(Z \ge \delta) \le \mathbb{E}_\pi[Z]/\delta = q_s^\pi(Q)/\delta \le q^\star/\delta$.

Proof of Proposition 5. Because μ and ν are mutually singular, the response distribution on μ is identical under P^0 and P^1; a learner that sees only responses from μ therefore outputs the same (randomized) draft distribution under both. Fix a draft Q (or condition on the learner's randomness). On A, the top-1 sets of P_x^0 and P_x^1 are disjoint, so Q_x's argmax can equal at most one of them. Thus the average over the two targets of the 0–1 rejection on A is at least $1/2$. Taking the worst case over the two targets and integrating over A gives $\epsilon\nu(A)/2$. If this exceeds $q^\star$, no such learner can guarantee break-even on π.

Scope of A1 and independent bases. A1 excludes gated ReLU: for $\sigma(t) = \mathrm{ReLU}(t)$, $d(t) = t\,\mathbf{1}\{t > 0\}$ is identically zero on $(-\infty, 0]$, so the open-rectangle argument does not apply. The measured models use SiLU, which satisfies A1; we do not claim Lemma 2 for gated ReLU. Lemma 2 also assumes a shared basis P for gate and up. For independent bases $P_g \ne P_u$, a general projection adds a basis-mixing term; a second-order expansion at $z = 0$ forces agreement only on the active coordinates, and we do not prove the all-readout statement for the independent case. The independent-SVD baseline in Appendix A.12 is worse than shared-basis channel deletion, so this is a measured boundary, not a claimed theorem.

On approximate non-coordinate operators. Lemma 2 is an exact-equality statement. We do not claim a quantitative perturbation lower bound for non-coordinate operators: a local continuity statement would introduce an unestimated constant and would not support any numerical claim. Instead, the operator zoo in Appendix A.12 measures independent SVD and shared-rotation baselines end-to-end at matched nominal FLOP. This is the formal sense in which other linear compute-reducing operators add a basis-mixing term that channel deletion does not have, but the evidence for their ordering is empirical.

Support dominance (why width deletion is the minimal compute-reducing operator). Lemma 2 says that an exact width reduction is coordinate selection. A non-coordinate operator (a rotation, a joint SVD basis, or a layer removal) is not supported only on a deleted channel set: it also changes retained coordinates, adding a basis-mixing term that channel deletion does not have. Channel deletion is therefore the minimal-support compute-reducing post-hoc operator at a fixed retained width. This is a structural statement, not a proof that its rejection is always smaller; the measured operator zoo in Appendix A.12 confirms the ordering for weight-norm, independent SVD, and shared rotation at matched nominal FLOP. Weight-only quantization is a memory-only axis and is not covered by this argument.

A.2 Claim–Evidence–Condition Map

Table A.3: Claim–evidence–condition map. The paper's no-go is not a single theorem; it is a chain from an exactness statement, to a measured post-hoc lower bound, to a lower bound on relative cost. "Strength" separates theorem-backed statements from measured or fitted ones. Each row states the condition under which the claim is used.

Claim	Strength	Evidence	Condition / scope
Exact all-readout FFN width reduction is coordinate selection.	theorem	Lemma 2 (algebraic).	Equality for all inputs and all block weights; fixed-target redundancy is Lemma 4.
Fixed-weight exactness is characterized by an exact identity.	theorem	Lemma 4 (algebraic).	Pure deletion: $W_{d,D}a_D = 0$. Output-inactive channels or a functional redundancy are nongeneric escapes; we do not formalize a generic-weight quantifier.
Deleted channels perturb the block output, and the refitted readout has an exact decomposition.	identity	Lemmas 3 and 6 (exact identities).	Pure deletion is the first term; residual ridge adds the readout-correction term. The deployed certificate is measured on the composed map.
Top-2/top-K rejection lower bound.	theorem + measured	Lemmas 1 and 7; measured draft logits.	Post-hoc empirical bound; requires evaluation-token logits; not a pre-deployment predictor.
TV floor for exact token-level protocols.	theorem + measured	Lemma 8 (standard).	Exact SD rule using the draft distribution; tree verification is outside.
The near-identity resolution floor is a measured decision-boundary effect, not a tie-order or channel-ranking artifact.	measured	E-highwidth-floor, E-float32-floor; Tables 9 and 8.	Qwen3.5-9B and Qwen3-4B, WikiText raw, one deleted channel per layer (99.99% width), activation-aware ranking, bf16 target plus a float32 control. It is a mechanism diagnostic, not a deployable configuration.
Relative cost is at least the roofline ratio, uniformly in batch size and hardware.	theorem	Assumptions 1–3; Proposition 7 and Theorem 1.	Same-kernel fair comparison, equal efficiency, $\tau_B \geq 1$, nominal FLOP/byte ratios. The bound is hardware-invariant because it uses model ratios, not hardware peaks.
Cost verdicts are ordered: W, K, G, R, with R-free as a stronger sub-label.	theorem + measured	The four thresholds of Appendix A.5; Tables A.11, A.12, A.13.	Measured clean kernel-time cost for W/K; nominal FLOP and byte ratios for G/R; measured time share for the free-FFN counterfactual. The cost layer is load-bearing only for K and G; raw 98–99
Raw-text rejection no-go.	measured, confirmatory	Measured $q_g, q_s, q_{\mathrm{lb}}$ vs. $q^{\star}_{\mathrm{roof}}$; 200/200 raw cells point-above, 199/200 certified.	Nominal ϕ_{MLP} (sensitivity in Appendix A.5); measured distributions. Confirmatory families are 36 Qwen3.5-9B raw cells at $\alpha = 0.05/36$ and 9 Llama/K2 raw cells at $\alpha = 0.05/9$; the joint union bound is at most 0.10. The 200/218/55/99/135 counts are

A.3 Full Width Sweep

Table A.4: Full width sweep on Qwen3.5-9B with 95% sequence-cluster bootstrap intervals. Generated OOD is target-generated from the same 9B model.

Width	MATH-500	WikiText raw	FineWeb raw	WikiText gen	FineWeb gen
0.05	0.5846 [0.577,0.592]	0.8319 [0.824,0.840]	0.8161 [0.807,0.825]	0.7445 [0.714,0.772]	0.6495 [0.617,0.679]
0.10	0.4350 [0.428,0.442]	0.7782 [0.769,0.787]	0.7616 [0.750,0.773]	0.6079 [0.577,0.636]	0.4131 [0.383,0.442]
0.20	0.2808 [0.275,0.287]	0.7033 [0.693,0.714]	0.6914 [0.679,0.704]	0.3716 [0.346,0.397]	0.2183 [0.193,0.245]
0.30	0.2098 [0.205,0.215]	0.6565 [0.643,0.670]	0.6284 [0.614,0.643]	0.2258 [0.203,0.250]	0.1440 [0.122,0.169]
0.50	0.1297 [0.126,0.134]	0.5017 [0.489,0.514]	0.5047 [0.490,0.520]	0.1530 [0.135,0.171]	0.1120 [0.094,0.133]
0.70	0.0831 [0.081,0.086]	0.3618 [0.353,0.371]	0.3643 [0.352,0.377]	0.1145 [0.102,0.127]	0.0827 [0.069,0.098]
0.80	0.0637 [0.062,0.066]	0.2876 [0.280,0.295]	0.2926 [0.282,0.303]	0.0963 [0.087,0.106]	0.0691 [0.058,0.081]
0.85	0.0526 [0.051,0.055]	0.2513 [0.245,0.258]	0.2518 [0.242,0.262]	0.0862 [0.078,0.095]	0.0614 [0.051,0.073]
0.90	0.0399 [0.039,0.042]	0.2080 [0.202,0.214]	0.2085 [0.200,0.217]	0.0750 [0.069,0.082]	0.0505 [0.043,0.059]
0.95	0.0260 [0.025,0.027]	0.1502 [0.145,0.155]	0.1551 [0.148,0.163]	0.0596 [0.055,0.065]	0.0366 [0.031,0.042]
0.98	0.0157 [0.015,0.016]	0.0968 [0.093,0.101]	0.0959 [0.090,0.102]	0.0415 [0.039,0.045]	0.0233 [0.020,0.027]
0.99	0.0112 [0.011,0.012]	0.0677 [0.064,0.071]	0.0649 [0.061,0.069]	0.0334 [0.031,0.036]	0.0169 [0.014,0.019]
1.00	0.0000 [0.000,0.000]	0.0000 [0.000,0.000]	0.0000 [0.000,0.000]	0.0000 [0.000,0.000]	0.0000 [0.000,0.000]

Fine-grained high-width control. Table 9 and Figure 2 vary the number of deleted channels per layer from 246 (98% width) to one (99.99% width). At one channel per layer the default ranking gives $q = 1.30\%$ on 9B and 1.44% on 4B; same-corpus oracle selection gives 1.23%/1.30%, random selection is indistinguishable from the default at 99.9% (2.38% versus 2.33% on 9B), and weight-norm selection is worse (8.31%). The floor is therefore insensitive to ranking, while the selection-driven part at 99% is not (oracle 3.73% versus default 6.77%).

An exact tie is $m = 0$ in the stored bf16 target. The tie atom is not a float16 storage artifact: a float32 rebuild of the Qwen3-4B WikiText cache gives the same stored tie statistics at 98/99% width ($\pi_{\text{tie}} = 0.02552$; tie component 0.01305/0.01115). More importantly, a clean float32 rebuild of both the Qwen3.5-9B target cache and the draft (Table 8) removes the exact tie atom but leaves a 0.31% one-channel floor, 129× the float32 break-even, with 5.4% near-boundary mass. The flip is not a tie-order rule: at width 1.0 the draft is bit-identical and $q = 0$. The floor is therefore a near-boundary resolution phenomenon; the exact bf16 tie atom is one precision-specific component. We do not claim the exact percentages for an arbitrary rounding engine.

Table A.5: The top-2 certificate $q_{\text{lb}} = \Pr(m + \Delta_m < 0)$ of Lemma 1. A cell is certified no-go (✓) when the lower 95% confidence bound of q_{lb} exceeds $q^\star_{\text{FLOP}}$; the table prints the point q_{lb}, and the checkmarks use the lower bound; because $\tau \geq 1.32$ in the measured model the true threshold is lower, so the certificate is conservative. The certificate is computed on the same tokens as q. On both raw OOD sets every compressed width is certified; on MATH widths ≥ 0.85 are certified.

Width	$q^\star_{\text{FLOP}}$	q_{lb} MATH	q_{lb} Wiki	q_{lb} FineWeb	cert. MATH	cert. OOD
0.05	0.3168	0.2415	0.3879	0.3871	–	✓
0.10	0.2933	0.1917	0.3607	0.3617	–	✓
0.20	0.2498	0.1417	0.3396	0.3362	–	✓
0.30	0.2102	0.1189	0.3319	0.3159	–	✓
0.50	0.1400	0.0884	0.2690	0.2674	–	✓
0.70	0.0791	0.0629	0.2124	0.2071	–	✓
0.80	0.0513	0.0504	0.1789	0.1719	–	✓
0.85	0.0379	0.0424	0.1588	0.1493	✓	✓
0.90	0.0250	0.0331	0.1337	0.1254	✓	✓
0.95	0.0123	0.0217	0.0985	0.0930	✓	✓
0.98	0.0049	0.0126	0.0636	0.0574	✓	✓
0.99	0.0024	0.0087	0.0455	0.0381	✓	✓
1.00	0.0000	0.0000	0.0000	0.0000	–	–

High-width tie floor and allocation controls. Table A.6 gives every measured 98/99% tie cell; Table A.7 gives the per-layer allocation control at fixed average width; Table A.8 gives the cross-model transfer of the

screen; Table A.9 gives the in-domain window significance tests. These are the supporting tables for the main-text claims.

Figure A.1 decomposes the high-width residual into the exact-tie component and the remaining near-boundary mass; Figure A.2 gives the corresponding full width curve.

Table A.6: Full bf16 tie-floor table: every measured cell at 98 and 99% width. The tie component is a measured component of the draft rejection, not a lower bound for arbitrary drafts; it is the part of the measured rejection attributable to exact top-1/top-2 ties. K2 raw cells are in Table A.66; K2 is omitted here because the 30/50 target-generated check does not have a raw tie mask.

Model	Corpus	Width	π_{tie}	q_{tie}	tie comp.	non-tie comp.	q	$q^\star$
Qwen3.5-9B	WikiText	0.98	0.031	0.460	0.0142	0.0826	0.0968	0.0049
Qwen3.5-9B	WikiText	0.99	0.031	0.420	0.0130	0.0547	0.0677	0.0024
Qwen3.5-9B	FineWeb	0.98	0.038	0.447	0.0170	0.0789	0.0959	0.0049
Qwen3.5-9B	FineWeb	0.99	0.038	0.393	0.0149	0.0500	0.0649	0.0024
Qwen3.5-9B	MATH	0.98	0.005	0.467	0.0023	0.0134	0.0157	0.0049
Qwen3.5-9B	MATH	0.99	0.005	0.427	0.0021	0.0091	0.0112	0.0024
Qwen3.5-4B	WikiText	0.98	0.032	0.467	0.0148	0.0734	0.0883	0.0049
Qwen3.5-4B	WikiText	0.99	0.032	0.418	0.0133	0.0489	0.0622	0.0022
Qwen3.5-4B	MATH	0.98	0.005	0.445	0.0024	0.0164	0.0188	0.0049
Qwen3.5-4B	MATH	0.99	0.005	0.414	0.0022	0.0106	0.0128	0.0022
Qwen3-4B	WikiText	0.98	0.026	0.511	0.0131	0.0874	0.1005	0.0054
Qwen3-4B	WikiText	0.99	0.026	0.437	0.0111	0.0564	0.0676	0.0027
Qwen3-4B	FineWeb	0.98	0.030	0.477	0.0142	0.0896	0.1038	0.0054
Qwen3-4B	FineWeb	0.99	0.030	0.387	0.0115	0.0582	0.0697	0.0027
Qwen3-4B	MATH	0.98	0.007	0.471	0.0035	0.0189	0.0224	0.0054
Qwen3-4B	MATH	0.99	0.007	0.468	0.0035	0.0120	0.0155	0.0027
Llama-3.1-8B	WikiText	0.98	0.032	0.417	0.0135	0.0673	0.0807	0.0060
Llama-3.1-8B	WikiText	0.99	0.032	0.415	0.0134	0.0463	0.0597	0.0030
Llama-3.1-8B	FineWeb	0.98	0.039	0.440	0.0173	0.0594	0.0767	0.0060
Llama-3.1-8B	FineWeb	0.99	0.039	0.443	0.0174	0.0408	0.0582	0.0030
Llama-3.1-8B	MATH	0.98	0.009	0.405	0.0039	0.0151	0.0189	0.0060
Llama-3.1-8B	MATH	0.99	0.009	0.402	0.0038	0.0104	0.0142	0.0030

Table A.7: Non-uniform per-layer width allocation at a fixed average width (Qwen3.5-9B, WikiText; the Qwen3-4B rows are the earlier cross-model control). "Front/back" gives the first and last three layers +0.04 width and removes the same mass from the middle; "alternating" alternates $+0.04/-0.04$; "random" draws per-layer widths with the same mean. All patterns keep the parameter-weighted average at 0.98 or 0.99. On 9B the best tested simple pattern reduces q by 14–15%, and $q/q^\star$ is still 17–29×; break-even requires a $\sim 20\times$ reduction. The optimized per-layer DP in Section 5.5 is also worse than uniform OBS. Non-uniform allocation is therefore not a missing post-hoc repair.

Average	Pattern	q	q_{lb}	$q/q^\star$	Outcome
0.98	uniform (9B)	0.0964	0.0639	19.7×	no-go
0.98	front/back	0.0826	0.0577	16.9×	no-go
0.98	alternating	0.1087	0.0704	22.2×	no-go
0.98	random	0.1036	0.0692	21.1×	no-go
0.99	uniform (9B)	0.0683	0.0453	28.5×	no-go
0.99	front/back	0.0582	0.0394	24.3×	no-go
0.98	uniform (Qwen3-4B)	0.1001	0.0693	18.5×	no-go
0.98	front/back (Qwen3-4B)	0.0883	0.0656	16.4×	no-go

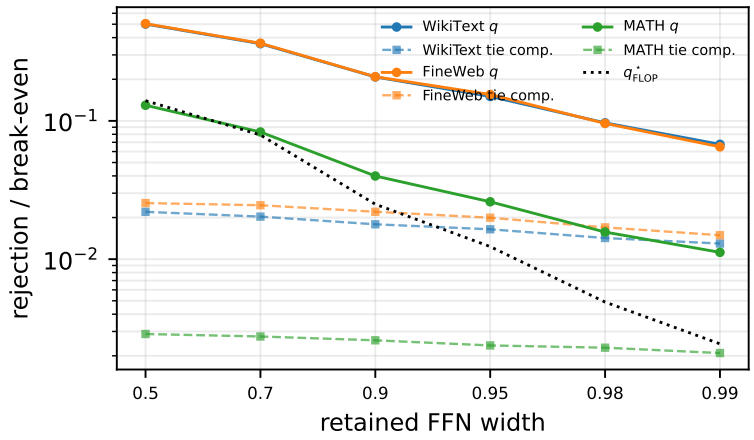


Figure A.1: The tie floor. Solid: greedy rejection q; dashed: the component of q attributable to exact bf16 top-1/top-2 ties; dotted: $q^{\star}_{\mathrm{FLOP}}$ at $c = 1$, $\tau = 1$, $\gamma = 4$. The tie component crosses $q^{\star}$ between 90% and 95% width on raw text and is the dominant residual at 98–99%; MATH has an order of magnitude less tie mass.

Table A.8: Cross-model transfer of the margin-mass screen. The coefficients $q = a_w + b_w L$ are fitted on the eleven Qwen3.5-9B raw corpora at 98% width ($a_w = 0.0073$, $b_w = 0.3932$) and applied without refitting to WikiText on four other targets. The binary raw-text decision survives (all predictions are above $q^{\star} = 0.0049$). On the raw K2 cells the 9B fit is conservative, over-predicting by 1.2× (0.098 versus 0.079); the earlier 30/50 target-generated K2 check is under-predicted and is not the raw transfer cell. The screen still requires a per-target refit; its false-negative guarantee is scoped to the target on which it was fitted.

Target	$L(0.5)$	q_{pred} (9B fit)	q measured	error	decision
Qwen3.5-9B	0.219	0.0934	0.0968	-0.0034	no-go
Qwen3.5-4B	0.230	0.0977	0.0883	+0.0094	no-go
Qwen3-4B	0.204	0.0875	0.1005	-0.0130	no-go
Llama-3.1-8B	0.248	0.1048	0.0807	+0.0241	no-go
K2-3.7B	0.231	0.0981	0.0791	+0.0190	no-go

Table A.9: In-domain MATH window significance. "FLOP sig." means the 95% cluster interval for q lies entirely below $q^{\star}_{\mathrm{FLOP}}$; "mem sig." is the same test against $q^{\star}_{\mathrm{mem}}$. The 30% 9B point is not a significant FLOP window (the gap to $q^{\star}_{\mathrm{FLOP}}$ is 0.0004, far inside the interval); the 50% 9B point is, but the memory frontier closes it. The 27B FLOP window is 50–90%; its clean kernel-time cost 4.23/4.53/4.84 closes it by rejection at the implied break-even; the older padded full-pass 5.37–5.66 is an upper bound and is not the reported cost.

Target	Width	q	95% CI	$q^{\star}_{\mathrm{FLOP}}$	$q^{\star}_{\mathrm{mem}}$	FLOP sig.	mem sig.
Qwen3.5-9B	0.30	0.2098	[0.2050, 0.2151]	0.2102	0.1812	no	no
Qwen3.5-9B	0.50	0.1297	[0.1262, 0.1337]	0.1400	0.1220	yes	no
Qwen3.5-27B	0.50	0.0881	[0.0855, 0.0909]	0.1561	–	yes	–
Qwen3.5-27B	0.70	0.0563	[0.0546, 0.0583]	0.0875	–	yes	–
Qwen3.5-27B	0.90	0.0257	[0.0248, 0.0268]	0.0275	–	yes	–
Qwen3.5-27B	0.95	0.0166	[0.0159, 0.0174]	0.0135	–	no	–

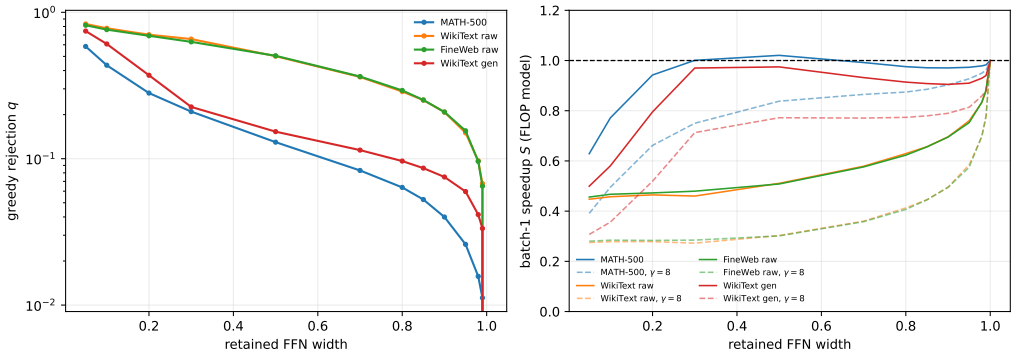


Figure A.2: Left: greedy rejection q_g versus retained FFN width. Right: speedup S under the optimistic FLOP roofline ($\rho = \rho_{\text{FLOP}}$, $\tau = 1$), $\gamma = 4$ (solid) and $\gamma = 8$ (dashed); this is an upper bound, not a measured wall-clock. The measured batch-1 cost is above the FLOP-model cost (clean kernel-time decomposition: Table A.64; the earlier mixed protocol is Table A.20), so the plotted S is an upper bound. The plotted quantity is q_g; the certificate q_{lb} is a lower bound on q_g (Table 7).

Table A.10: Extreme high-width check. The 99.5% and 99.9% points are new; $q^\star$ uses $\tau = 1$, $\gamma = 4$. The excess ratio grows rather than converging to a window.

Width	q Wiki	q FineWeb	$q^\star$	$q/q^\star$ (Wiki)
0.990	0.0677	0.0649	0.00244	27.7×
0.995	0.0502	0.0474	0.00122	41.2×
0.999	0.0233	0.0210	0.00024	95.7×

The calibration corpus contains 745 unique ids, including all 500 MATH evaluation prompts; WikiText and FineWeb evaluation ids are disjoint. This overlap can only make the in-domain MATH result look better than a leak-free calibration would, because the channel ranking has seen the evaluation prompts; raw text is unaffected. Appendix A.4 repeats the key widths with a MATH-excluded calibration file as a control.

We report per-cell 95% cluster-bootstrap intervals and, for the main sweep, a Bonferroni-adjusted check in Appendix A.30. The certified cells are far from the threshold (e.g., 0.041 vs. 0.0026 for the raw K2 WikiText at 99% width), so a Bonferroni correction over the 39 tests (13 widths × three raw corpora) would not change the qualitative no-go. We use "certificate" below as shorthand for the empirical post-hoc lower bound; it requires the draft logits on the evaluation tokens. The target cache is top-256 plus a residual bucket. The residual mass outside the top-256 is below 0.1% on MATH and 1.7–1.8% on raw text prose. A top-1024 re-cache changes q by at most 0.0009 across 20 cells and increases raw-text $\mathbb{E}[\text{TV}]$ by about 0.004; the main tables therefore use top-256 and slightly understate TV on raw text. The certificate q_{lb} is more robust to cache truncation: it uses only the target top-2 margin and the draft logits on those two tokens, both of which are present in a top-256 cache, so a top-1024 cache cannot change q_{lb} except through ties or numerical noise. The top-1024 check therefore targets q and $\mathbb{E}[\text{TV}]$ (third-token flips and residual mass), not the certificate. The per-sequence match vectors used for the cluster bootstrap are stored with the project results.

Relation to a margin-tail assumption. A margin-tail assumption and a TV power law would derive a conditional capacity exponent. Those assumptions are not used here. The certificate (2) is computed directly from the measured margin perturbation, so it does not require estimating a TV exponent or checking an $\varepsilon_D \leq \varepsilon_0$ condition. The off-the-shelf capacity scan is retained only as the complementary direction (Appendix A.18).

A.4 MATH-Excluded Calibration Control

To check that the MATH calibration overlap is not driving the result, we recompute channel importance after removing all 500 MATH evaluation ids and repeat the sweep at widths 0.50, 0.90, and 0.98 on MATH, WikiText, and FineWeb. The raw-text rejection changes by at most 0.0015 and the in-domain MATH rejection by at most 0.0006 at the three widths; MATH remains above the corresponding FLOP threshold at 90–98% width and below it only in the same 50% window. Thus the overlap does not explain the raw-text no-go, and removing it does not open a profitable in-domain window. The full numbers are in the retained result files.

A.5 The cost layer: roofline, batch, kernel conversion, and taxonomy

What this appendix establishes. The rejection layer of Section 3 fixes the kernel ratio $c = 1$ and asks whether top-1 agreement is good enough. The cost layer asks a different question: at the measured rejection q, is the draft pass cheap enough for break-even? We give a complete derivation in four steps. First, the accounting identity and its monotonicity. Second, a roofline bound that holds for every batch size and every hardware, and which shows that width deletion is *never* an acceptance-independent cost no-go under the optimistic frontier. Third, a fair-comparison (Amdahl) bound: a kernel improvement that is available to both target and draft cannot increase the relative advantage, so the only relevant advantage is the operator's shape reduction. Fourth, a nested taxonomy of the measured thresholds that classifies every cell as window, realized-kernel-bound, regime-bound, roofline-bound, or free-draft-bound. The taxonomy separates the load-bearing rejection statement from the implementation-conditional cost statement. The measured inputs are the clean kernel-time decomposition of Appendix A.6 and the width sweeps of Appendix A.3.

A.5.1 Accounting identities

Let $T_T(B, n)$ and $T_D(B, n)$ be the target and draft times to score n positions for B sequences. A speculative step proposes γ tokens and verifies $\gamma + 1$ positions, so it costs $\gamma T_D(B, 1) + T_T(B, \gamma + 1)$ and emits A tokens; the target alone needs A one-token passes. Define

$$\rho_B = \frac{T_D(B, 1)}{T_T(B, 1)}, \qquad \tau_B(\gamma) = \frac{T_T(B, \gamma + 1)}{T_T(B, 1)}, \qquad C_B(\gamma) = \gamma\rho_B + \tau_B(\gamma). \tag{20}$$

With per-token acceptance $\alpha = 1 - q$ and the geometric acceptance length $A(\alpha, \gamma) = \sum_{i=0}^{\gamma} \alpha^i$, the speedup is

$$S_B = \frac{A(1 - q, \gamma)}{\gamma\rho_B + \tau_B(\gamma)} = \frac{A(1 - q, \gamma)}{C_B(\gamma)}, \tag{21}$$

and break-even $S_B \geq 1$ is

$$A(1 - q, \gamma) \;\geq\; \gamma\rho_B + \tau_B(\gamma). \tag{22}$$

For fixed γ, A is strictly decreasing in q, so there is a unique threshold $q^\star(C, \gamma)$ with $A(1 - q^\star, \gamma) = C$ whenever $1 \leq C < \gamma + 1$; if $C \geq \gamma + 1$ no acceptance rate can break even and we set $q^\star = 0$. The cell is a no-go iff $q > q^\star(C, \gamma)$.

Lemma 9 (Monotonicity of the break-even threshold)**.** *For $\gamma \geq 1$ and $\alpha \in (0, 1]$, $A(\alpha, \gamma)$ is strictly increasing in α and strictly increasing in γ. Hence $q^\star(C, \gamma)$ is strictly decreasing in C on $1 \leq C < \gamma + 1$, and strictly decreasing in γ on the same region.*

Proof. $\partial A/\partial\alpha = \sum_{i=1}^{\gamma} i\alpha^{i-1} > 0$ and $A(\alpha, \gamma + 1) = A(\alpha, \gamma) + \alpha^{\gamma+1} > A(\alpha, \gamma)$. The threshold is defined by $A(1 - q^\star, \gamma) = C$; differentiating implicitly, $\partial q^\star/\partial C = -1/[A'(1 - q^\star, \gamma)] < 0$. For γ, the left side increases with γ at fixed q, so the matching $q^\star$ decreases. □

The monotonicity is used twice below: a lower bound on cost gives a higher (more permissive) threshold, and a larger γ gives a lower threshold. It also means the audit is conservative when we replace an unknown cost by a lower bound.

A.5.2 The roofline bound: hardware and batch invariance

Assumption 1 (Same-kernel fair comparison)**.** *Target and draft use the same kernel family with the same efficiency factor $\eta \in (0, 1]$, so their realized pass times are $T_T(B, n) = t_T(B, n)/\eta$ and $T_D(B, n) = t_D(B, n)/\eta$, where t_T, t_D are the roofline times. Equivalently, the realized ratio equals the roofline ratio, $T_D/T_T = t_D/t_T$, and a kernel improvement available to the draft is also available to the target. Section A.5.4 shows that this is not an extra restriction: under the alternative, a portable improvement cancels from the relative comparison.*

Assumption 2 (Verification floor)**.** *Verifying $\gamma + 1$ positions costs at least one target token pass, $\tau_B \geq 1$.*

Assumption 3 (Nominal operator ratios)**.** *$F_D/F_T = \rho_{\text{FLOP}}$ and $M_D/M_T = \rho_{\text{mem}}$ are the operator's nominal FLOP and byte ratios, independent of kernel and batch size.*

For a width-w deletion, let ϕ_{FLOP} and ϕ_{mem} be the FFN shares of target FLOPs and bytes. Then

$$\rho_{\text{FLOP}} = 1 - \phi_{\text{FLOP}}(1 - w), \qquad \rho_{\text{mem}} = 1 - \phi_{\text{mem}}(1 - w), \qquad \rho_{\text{roof}} = \min(\rho_{\text{FLOP}}, \rho_{\text{mem}}). \tag{23}$$

The attention, embedding, and KV terms are unchanged by FFN deletion; only the FFN terms scale with w. The same construction applies to depth, head, MoE, and composed operators with their own ϕ's (Section A.5.8).

Proposition 7 (Batch-uniform roofline bound)**.** *Under Assumptions 1–3, for every batch size B and every hardware with peak compute P and bandwidth W,*

$$\rho_B \;\geq\; \rho_{\text{roof}}, \qquad C_B(\gamma) \;\geq\; \gamma\rho_{\text{roof}} + 1 \;=:\; C_{\text{roof}}(\gamma). \tag{24}$$

Proof. Let $F_T(B), M_T(B)$ be the target FLOPs and bytes for the pass. By Assumption 3, $F_D = \rho_{\text{FLOP}} F_T$ and $M_D = \rho_{\text{mem}} M_T$. The roofline times are $t_T = \max(F_T/P, M_T/W)$ and $t_D = \max(\rho_{\text{FLOP}} F_T/P, \rho_{\text{mem}} M_T/W)$. If $F_T/P \geq M_T/W$ (target compute-bound), then $t_D/t_T = \max(\rho_{\text{FLOP}}, \rho_{\text{mem}} M_T W/(F_T P)) \geq \rho_{\text{FLOP}} \geq \rho_{\text{roof}}$. If $M_T/W \geq F_T/P$ (target memory-bound), then $t_D/t_T = \max(\rho_{\text{FLOP}} F_T P/(M_T W), \rho_{\text{mem}}) \geq \rho_{\text{mem}} \geq \rho_{\text{roof}}$. In both cases $\rho_B = T_D/T_T = t_D/t_T \geq \rho_{\text{roof}}$, where the first equality is Assumption 1. Assumption 2 gives $\tau_B \geq 1$, hence $C_B \geq \gamma\rho_{\text{roof}} + 1$. □

The bound is deliberately optimistic: it uses the smaller of the FLOP and byte ratios and the smallest possible verification ratio. It is also *hardware-invariant*: ρ_{FLOP} and ρ_{mem} are ratios of model FLOPs and bytes, not of hardware peaks. The hardware enters only through which term of the roofline binds, not through the value of the lower bound. The batch size enters only through the same choice. This is why the rejection statement below does not need a clock.

Theorem 1 (The frontier no-go is a rejection statement)**.** *Let $q^\star_{\text{roof}}(\gamma)$ be defined by $A(1 - q^\star_{\text{roof}}, \gamma) = C_{\text{roof}}(\gamma)$. If $q > q^\star_{\text{roof}}(\gamma)$, then for every B, every kernel family satisfying Assumption 1, and every hardware,*

$$A(1 - q, \gamma) \;<\; C_B(\gamma), \tag{25}$$

so the cell is a no-go at every batch size, on every hardware, and for every kernel in the family.

Proof. By Lemma 9, $q > q^\star_{\text{roof}}$ implies $A(1 - q, \gamma) < A(1 - q^\star_{\text{roof}}, \gamma) = C_{\text{roof}}(\gamma)$. Proposition 7 gives $C_{\text{roof}}(\gamma) \leq C_B(\gamma)$. □

Corollary 1 (Cost optimism)**.** *For every $w < 1$ with $\phi_{\text{FLOP}} > 0$ and $\phi_{\text{mem}} > 0$, $\rho_{\text{roof}} < 1$, hence $C_{\text{roof}}(\gamma) < \gamma + 1$ and $q^\star_{\text{roof}}(\gamma) > 0$. Consequently, under the optimistic frontier, width deletion is* never *an acceptance-independent cost no-go: there is always a positive rejection threshold below which some draft could break even. Every frontier no-go is a rejection statement.*

Proof. $\rho_{\text{FLOP}} = 1 - \phi_{\text{FLOP}}(1 - w) < 1$ and $\rho_{\text{mem}} < 1$, so their minimum is < 1. Hence $C_{\text{roof}} = \gamma\rho_{\text{roof}} + 1 < \gamma + 1$, and $A(1, \gamma) = \gamma + 1 > C_{\text{roof}}$, so the continuous strictly decreasing function $A(1 - q, \gamma)$ crosses C_{roof} at a positive $q^\star_{\text{roof}}$. □

Corollary 1 is the logical separation the paper needs. The measured implementation can fail the cost test (and does, at 98–99%), but that failure is an implementation statement: the optimistic frontier always leaves a positive break-even threshold $q^\star_{\text{roof}} > 0$. This does not imply that the measured operator attains such an acceptance rate; it says only that the frontier cannot produce an acceptance-independent cost no-go. The raw-text no-go is carried by $q > q^\star_{\text{roof}}$, not by the padded kernel.

A.5.3 Batch size, throughput, and the crossover

The roofline bound is batch-uniform, but the *measured* cost is not. It is useful to see where the batch scan sits. Let f be the target FLOPs per token per sequence, m_w the weight bytes, and m_k the KV bytes per token per sequence. For B sequences,

$$F_T(B) = Bf, \qquad M_T(B) = m_w + Bm_k, \qquad t_T(B) = \max\left(\frac{Bf}{P}, \frac{m_w + Bm_k}{W}\right). \tag{26}$$

The target is memory-bound when $Bf/P \le (m_w + Bm_k)/W$. Solving for B gives the crossover

$$B^\star = \frac{(P/W)m_w}{f - (P/W)m_k} \tag{27}$$

when $f > (P/W)m_k$; otherwise the workload is memory-bound for every B. For the six targets on an A100-80GB ($P = 312$ TFLOP/s bf16, $W = 2.04$ TB/s, $P/W \approx 153$ FLOP/byte), the formula gives $B^\star \approx 150$–170 at a 512-token prefix. The measured scan reaches $B = 128$, i.e. about 0.75–0.85 of the crossover, which is consistent with the observed rise of τ_B from 1.29 to 1.41 while ρ_B stays near one (Table A.21). Beyond $B^\star$ the target is compute-bound; the roofline ratio then tends to ρ_{FLOP}, which is still at least ρ_{roof}.

The same accounting gives the throughput statement. With B sequences and one token per sequence per step, the emitted-token rate with speculation is $BA/(\gamma T_D(B,1) + T_T(B,\gamma+1))$ and the target-only rate is $B/T_T(B,1)$; their ratio is exactly S_B of (21). Thus the latency speedup and the steady-state throughput speedup coincide for a fixed batch, and the roofline bound applies to both. Continuous batching changes the schedule but not the per-step ratio. The measured $B = 128$ cost is 5.41, so batching does not open the compute-bound corner in the measured range.

A.5.4 Kernel efficiency, portability, and the shape ceiling

The paper reports a minimum draft-kernel advantage $c^\star$. It is important to say what c means. Decompose the target pass into a non-FFN part N and an FFN part F, with measured time share $s = F/(N+F)$. A width-w draft with an FFN-only kernel speedup c_{FFN} has

$$T_D = N + F\,\frac{g(w)}{c_{\mathrm{FFN}}}, \qquad \rho(c_{\mathrm{FFN}}) = (1-s) + s\,\frac{g(w)}{c_{\mathrm{FFN}}}, \tag{28}$$

where $g(w)$ is the shape factor of the narrow FFN relative to the full FFN ($g(1) = 1$, and $g(w) \ge w$ under the roofline because both FLOPs and bytes scale with w). The free-FFN ceiling is $c_{\max} = 1/(1-s)$; the required FFN-only speedup to reach a whole-pass advantage c is

$$c^\star_{\mathrm{FFN}} = \frac{s}{1/c^\star - (1-s)}, \tag{29}$$

which is (9) of the main text.

The key point is that *a portable kernel improvement does not help.* Suppose a kernel transformation multiplies the FFN time of *both* target and draft by $1/c_K$. Then

$$\rho_{\mathrm{port}}(c_K) = \frac{(1-s) + s\,g(w)/c_K}{(1-s) + s/c_K}. \tag{30}$$

At $c_K = 1$, $\rho_{\mathrm{port}} = (1-s) + sg(w)$. As $c_K \to \infty$, $\rho_{\mathrm{port}} \to 1$.

Proposition 8 (Portability/Amdahl bound)**.** *For $0 < g(w) \le 1$ and $s > 0$, $\rho_{\mathrm{port}}(c_K)$ is nondecreasing in c_K on $[1,\infty)$. Hence the relative advantage is maximized at $c_K = 1$; any kernel improvement available to both passes moves the ratio toward one and cannot reopen a cell. A reopening requires a* draft-exclusive *advantage, i.e. a kernel that is faster on the narrow FFN for a reason that does not transfer to the target.*

Proof. Differentiate (30):

$$\frac{\partial \rho_{\mathrm{port}}}{\partial c_K} = \frac{s\,(1-s)\,(1-g(w))}{c_K^2\,((1-s) + s/c_K)^2} \;\ge\; 0.$$

The numerator is nonnegative because $g(w) \le 1$. The limit $c_K \to \infty$ is one. □

Proposition 8 justifies the same-kernel normalization of Assumption 1. More importantly, it turns the "production kernel" objection into a precise requirement: a production fused-MLP kernel is portable, so applying it to the draft but not to the target is not a fair comparison. Only a draft-exclusive kernel counts, and a draft-exclusive kernel cannot be justified by width deletion alone when w is near one.

Corollary 2 (Shape ceiling)**.** *Under the roofline with $g(w) = w$, the best whole-pass advantage a width-only draft can achieve is*

$$c_{\text{shape}}(w) = \frac{1}{\rho_{\text{FLOP}}} = \frac{1}{1 - \phi_{\text{FLOP}}(1-w)}. \tag{31}$$

The largest whole-pass advantage a width-only kernel can deliver is $c_{\text{shape}}(w)$; a larger advantage would require $\rho < \rho_{\text{FLOP}}$, which contradicts the roofline. Equivalently, the best width-only draft has cost $C_{\text{shape}} = \gamma\rho_{\text{FLOP}} + 1$, so it reopens the cell iff $C_{\text{shape}} \le A(1-q)$, i.e. $q \le q^\star_{\text{FLOP}}$. Consequently, if $q > q^\star_{\text{FLOP}}$, no width-only kernel, however efficient, reopens the cell.

Proof. With $g = w$ and $c_{\text{FFN}} \to \infty$, $\rho \to 1 - \phi_{\text{FLOP}}(1-w) = \rho_{\text{FLOP}}$; the advantage is $1/\rho_{\text{FLOP}}$. A larger advantage would require $\rho < \rho_{\text{FLOP}}$, which contradicts the roofline. The best width-only cost is therefore $\gamma\rho_{\text{FLOP}} + 1$, and $q^\star_{\text{FLOP}}$ is defined as its break-even rejection. □

The measured decomposition uses the time share s rather than the nominal ϕ_{FLOP}; the corresponding realized shape ceiling is $1/[(1-s) + sw]$. At 98% width the nominal ceiling is 1.011–1.015 and the realized ceiling is 1.005–1.010, while the required advantage is 1.25–1.42. The gap is not a kernel-tuning gap; it is the arithmetic of deleting 2% of the FFN. This is the quantitative form of "the high-width cost failure is not a production-kernel opportunity."

A.5.5 The cost taxonomy: ordered thresholds

The four lower bounds on cost induce four thresholds. The first three are ordered; the fourth is the measured free-FFN diagnostic. Listed from the strongest (highest threshold) to the weakest:

$$C_{\text{meas}} = \gamma\rho_{\text{meas}} + \tau_{\text{meas}}, \qquad q^\star_{\text{meas}} = q^\star(C_{\text{meas}}, \gamma), \tag{32}$$

$$C_{\text{regime}} = \gamma\rho_{\text{regime}} + 1, \qquad q^\star_{\text{regime}} = q^\star(C_{\text{regime}}, \gamma), \tag{33}$$

$$C_{\text{FLOP}} = \gamma\rho_{\text{FLOP}} + 1, \qquad q^\star_{\text{FLOP}} = q^\star(C_{\text{FLOP}}, \gamma), \tag{34}$$

$$C_{\text{free}} = \gamma(1-s) + \tau_{\text{meas}}, \qquad q^\star_{\text{free}} = q^\star(C_{\text{free}}, \gamma). \tag{35}$$

Here ρ_{regime} is the roofline term that binds in the deployment regime (FLOP or memory), and C_{meas} is the measured clean kernel-time cost. Because $q^\star$ decreases with cost, the first three thresholds are ordered exactly:

$$q^\star_{\text{meas}} \;\le\; q^\star_{\text{regime}} \;\le\; q^\star_{\text{FLOP}}. \tag{36}$$

The free-FFN threshold uses the measured time share and a different accounting (a fully removed MLP, not a width-scaled one), so we report it as a diagnostic rather than as a fourth ordered interval; it normally lies above $q^\star_{\text{FLOP}}$ but can sit a fraction of a point below it when s and ϕ_{FLOP} differ. The ordered verdict is:

Interval	Verdict	What would reopen it
$q \le q^\star_{\text{meas}}$	W: window	nothing; the cell already passes
$q^\star_{\text{meas}} < q \le q^\star_{\text{regime}}$	K: realized-kernel-bound	a better draft kernel
$q^\star_{\text{regime}} < q \le q^\star_{\text{FLOP}}$	G: regime-bound	a less binding roofline term
$q > q^\star_{\text{FLOP}}$	R: roofline-bound	only a non-width/free-FFN kernel

The taxonomy is exhaustive and ordered: every cell falls in exactly one interval. Within R we add the sub-label R-free when $q > q^\star_{\text{free}}$: the measured free-FFN counterfactual also fails, so even a free MLP would not save the cell. R and R-free are *operator-level* statements under Assumption 1; K and G are *implementation/regime* statements; W is the only pass. A measured cost above $\gamma + 1$ is an *acceptance-independent cost no-go*: no acceptance rate can break even. Some tables abbreviate this as "cost-only no-go"; the intended meaning is acceptance-independent, not operator-level. In the taxonomy it is the boundary $q^\star_{\text{meas}} = 0$, not a separate verdict.

Table A.11 applies the taxonomy to the six targets at 98% width. Five are R and K2 is R: the measured q exceeds the optimistic FLOP threshold by factors of 13–20×, so the raw no-go is a rejection statement that survives every kernel, batch, and hardware in the family. The measured cost is above $\gamma + 1 = 5$ on every target, so $q^\star_{\text{meas}} = 0$: even perfect acceptance cannot save the padded implementation. That is an additional, acceptance-independent implementation diagnostic (point estimate), not the load-bearing reason.

Table A.12 repeats the calculation across the 9B width sweep. Every width is R; at 10–95% it is R-free, because even a free draft cannot supply enough acceptance. The cost layer therefore never supplies the primary no-go on raw text. It confirms the rejection layer and quantifies how far the measured kernel is from the frontier.

Table A.13 isolates the cells where the cost layer *is* load-bearing. The 9B MATH 50% point is G: the FLOP frontier has a window, but the memory frontier closes it ($q = 0.1297$ versus $q^\star_{\text{mem}} = 0.1220$). The Llama MATH 50–70% points are K: both frontiers pass, but the measured padded kernel does not; the required shape advantage exceeds the roofline advantage by only 0.4–2.3%, inside the 9% timing spread, so the closure is a point estimate. The Llama MATH 90% point is R: $q = 0.0430$ is above the optimistic FLOP threshold $q^\star_{\text{FLOP}} = 0.0310$, so rejection closes it and the measured cost is redundant. This is the precise version of the paper's statement that the Llama window is cost-closed only at its low-width end.

Table A.11: Cost-layer thresholds at 98% width on WikiText. Each threshold is the break-even rejection $q^\star$ for a lower bound on relative cost: the measured implementation ($q^\star_{\text{meas}}$), the batch-1 memory roofline ($q^\star_{\text{mem}}$), the optimistic FLOP roofline ($q^\star_{\text{FLOP}}$), and a free-FFN counterfactual ($q^\star_{\text{free}}$, measured time share). $q^\star$ decreases with cost, so the thresholds are ordered: $q^\star_{\text{meas}} \le q^\star_{\text{mem}} \le q^\star_{\text{FLOP}}$; the free-FFN column is a measured diagnostic, and R-free is a sub-label of R. The interval containing the measured q is the verdict: W (window), K (realized-kernel-bound), G (regime-bound), R (roofline-bound), R-free (even a free draft fails). All six targets are R: the raw no-go survives the *optimistic* FLOP frontier, so it is a rejection statement, not a kernel statement.

Model	q	$q^\star_{\text{meas}}$	$q^\star_{\text{mem}}$	$q^\star_{\text{FLOP}}$	$q^\star_{\text{free}}$	cost	verdict
Qwen3.5-9B	0.0968	0.0000	0.0043	0.0049	0.1340	5.033	R
Qwen3-4B	0.1005	0.0000	0.0054	0.0054	0.1102	5.276	R
Qwen3.5-4B	0.0883	0.0000	0.0043	0.0043	0.1004	5.004	R
Llama-3.1-8B	0.0807	0.0000	0.0056	0.0060	0.2383	5.060	R
K2-3.7B	0.0791	0.0000	0.0045	0.0052	0.1241	5.266	R
Qwen3.5-27B	0.0802	0.0000	0.0051	0.0054	0.1788	5.066	R

Table A.12: The cost taxonomy across the Qwen3.5-9B raw-text width sweep. The measured q is the WikiText/FineWeb mean. It is above the *optimistic* FLOP threshold at every width, so every cell is R (roofline-bound); at 10–95% it is also above the free-FFN threshold (R-free). The cost layer therefore never supplies the primary no-go on raw text; it confirms it. The thresholds use the nominal FLOP ratio and $\tau = 1$; the free-FFN column uses the measured 9B MLP time share ($s = 0.304$).

Width	q	$q^\star_{\text{mem}}$	$q^\star_{\text{FLOP}}$	$q^\star_{\text{free}}$	verdict
0.10	0.7699	0.2489	0.2933	0.1341	R-free
0.20	0.6974	0.2137	0.2498	0.1341	R-free
0.30	0.6424	0.1810	0.2102	0.1341	R-free
0.50	0.5032	0.1219	0.1400	0.1341	R-free
0.70	0.3630	0.0694	0.0791	0.1341	R-free
0.90	0.2083	0.0221	0.0250	0.1341	R-free
0.95	0.1527	0.0109	0.0123	0.1341	R-free
0.98	0.0963	0.0043	0.0049	0.1341	R
0.99	0.0663	0.0022	0.0024	0.1341	R

Table A.13: In-domain MATH cells, where the cost layer is load-bearing. The 9B 50% point is a FLOP-only window ($q < q^\star_{\text{FLOP}}$) closed by the memory frontier ($q > q^\star_{\text{mem}}$), hence G. The Llama 50–70% points are K: both frontiers pass, but the measured padded kernel does not; the required shape advantage exceeds the roofline advantage by only 0.4–2.3%, inside the 9% timing spread. The Llama 90% point is R: $q = 0.0430$ is above the optimistic FLOP threshold $q^\star_{\text{FLOP}} = 0.0310$, so rejection closes it and the measured cost is redundant. The 27B 50–90% points are K: their clean cost 4.23/4.53/4.84 gives break-even 0.083/0.050/0.016, slightly below the measured 0.088/0.056/0.026. The 27B 90% point has a 95% interval $[0.0248, 0.0268]$ that straddles $q^\star_{\text{mem}} = 0.0261$, so its K/G label is a point estimate; its margin over $q^\star_{\text{meas}}$ is 60%.

Cell	width	q	$q^\star_{\text{mem}}$	$q^\star_{\text{FLOP}}$	cost	verdict
9B MATH	0.50	0.1297	0.1219	0.1400	4.497	G
27B MATH	0.50	0.0880	0.1474	0.1561	4.230	K
27B MATH	0.70	0.0560	0.0830	0.0875	4.530	K
27B MATH	0.90	0.0260	0.0261	0.0275	4.840	K
Llama MATH	0.50	0.1567	0.1656	0.1797	5.067	K
Llama MATH	0.70	0.0871	0.0924	0.0996	5.066	K
Llama MATH	0.90	0.0430	0.0289	0.0310	5.154	R

A.5.6 Gamma, prefill, tree verification, and sampling

Gamma. By Lemma 9, $q^\star(C, \gamma)$ decreases with γ. The roofline cost $C_{\text{roof}}(\gamma) = \gamma\rho_{\text{roof}} + 1$ increases linearly, so the no-go set is an upper interval in γ: if a cell is R at some γ_0, it is R for every $\gamma \geq \gamma_0$. The measured scan in Table A.14 confirms that the raw cells are above the FLOP threshold at $\gamma = 1, 2, 4, 8, 16$; the choice $\gamma = 4$ is not doing the work. At high width the measured cost is above $\gamma + 1$ for every γ because $\rho_1 \approx 1$ and $\tau_1 > 1$, so the implementation failure is also γ-independent.

Table A.14: Gamma scan. The first three columns are $q^\star_{\text{FLOP}}$ at selected widths; the last three are the measured batch-1 cost $\gamma\rho_1 + \tau_1(\gamma)$. Entries in bold exceed $\gamma + 1$ and are cost-only no-goes. $\tau_1(\gamma)$ is measured on the target and is nearly flat at batch 1 (1.29–1.30) because verification is memory-bound; ρ_1 is the measured per-token draft ratio. The raw-OOD rejection is above the FLOP entry at every width for every γ.

γ	$q^\star_{\text{FLOP}}(0.1)$	$q^\star_{\text{FLOP}}(0.5)$	$q^\star_{\text{FLOP}}(0.98)$	$C_1(0.1)$	$C_1(0.5)$	$C_1(0.98)$
1	0.548	0.304	0.012	**2.26**	**2.25**	**2.27**
2	0.426	0.219	0.008	**3.23**	**3.21**	**3.25**
4	0.293	0.140	0.005	**5.18**	**5.13**	**5.22**
8	0.180	0.081	0.003	**9.07**	8.98	**9.16**
16	0.102	0.044	0.001	16.86	16.68	**17.03**

Prefill and short generations. Let P_D be the draft prefill time and N the number of generated tokens. The target prefill is common and cancels. The finite-generation speedup is

$$S_N = \frac{NAT_T}{AP_D + N(\gamma T_D + T_T)} = \frac{S_B}{1 + AP_D/(N(\gamma T_D + T_T))} \;\leq\; S_B. \tag{37}$$

Prefill can only reduce the speedup; it cannot turn a steady-state no-go into a window. For very short generations the draft prefill can dominate, which makes the draft worse, not better. This is why excluding prefill from the main comparison is conservative for the no-go.

Tree verification. A tree with k candidate positions costs at least k draft passes and one target verification pass over k positions, so the cost lower bound is $C \geq k\rho_{\text{roof}} + 1$ with $\tau_k = T_T(B, k)/T_T(B, 1) \geq 1$. The cost taxonomy applies with γ replaced by k. The rejection side is outside the claim because the tree induces a different proposal distribution; the cost side is not. This is the precise scope of the "tree verification is outside" statement.

Sampling. Standard speculative sampling replaces the greedy acceptance $1 - q_g$ by $1 - q_s = 1 - \mathbb{E}[\text{TV}]$. The cost side is unchanged; only the acceptance term in (22) changes. Lemma 8 says that q_s lower-bounds

the rejection of any exact token-level protocol; it does not imply the general inequality $q_s \leq q_g$. On the measured cells $q_s < q_g$, so the sampling audit is the more conservative of the two there, and the measured q_s is above the roofline threshold on every raw cell. That empirical ordering is why the sampling layer does not need a separate cost calculation.

A.5.7 Measured kernel attempts and sensitivities

Measured kernel attempts. We measured the two obvious ways to raise the draft-kernel efficiency. `torch.compile` in Inductor max-autotune mode gives $\rho_1 = 2.81$ at 98% width (2.81× the padded dense GEMM), and a Triton fused gated-MLP kernel is 0.72× the dense GEMM at 98% width and 0.42× at 50% width. Neither reaches $c = 1$, let alone the $c = 1.3$–1.4 high-width row. Published fused gated-MLP, Marlin, and W4A16 kernels report roughly 1.2–2.7× over naive dense baselines, so the $c = 1.3$–1.4 band is plausible on modern hardware if the kernel is applied only to the draft. Proposition 8 explains why that asymmetric application is not the fair comparison: the same kernel is available to the target, and applying it to both moves the ratio toward one. The zero-MLP decomposition is the complementary statement, but it is not a ceiling: in the clean kernel-time protocol the realized cost is 5.004–$5.28 \geq 5$ at 98–99% on every tested target, while the zero-FFN cost is 3.1–$4.1 < 5$. A hypothetical free FFN would put the cost below 5 on all six targets (Table A.64), but it is a fully removed MLP, not a width-only kernel; the shape ceiling is 1.011–1.015. The 98–99% no-go is roofline-bound (R), with the realized kernel as an additional independent failure. The earlier $5.10 > 5$ wall-clock number was taken with a launch-bound loop (see the protocol note in Appendix A.6) and is superseded.

Table A.15: Break-even kernel-efficiency ratio. If the draft kernel has effective throughput c times the target's, the cost is $\gamma\rho_{\text{FLOP}}/c + \tau$. $c^\star$ is the minimum c that would reopen the cell at the measured raw-text rejection (an arithmetic conditional, not a deployable window); "–" means no finite c helps: even a free draft (cost τ_1) cannot break even because $A(1-q,\gamma) < \tau_1$. The last three columns are $q^\star(c)$ at $c = 2, 3, 5$ with the measured $\tau_1 = 1.323$. The 0.5+W4 row uses the composed operator's measured rejection; q is the mean of the WikiText and FineWeb measurements.

Width	ρ_{FLOP}	q	$c^\star$ ($\tau = 1$)	$c^\star$ (τ_1)	$q^\star(c{=}2)$	$q^\star(c{=}3)$	$q^\star(c{=}5)$
0.10	0.452	0.7699	6.1	–	0.4193	0.5035	0.5863
0.20	0.513	0.6974	4.8	19.2	0.3896	0.4791	0.5682
0.30	0.574	0.6424	4.2	10.2	0.3619	0.4560	0.5508
0.50	0.696	0.5032	3.0	4.6	0.3113	0.4135	0.5182
0.70	0.817	0.3630	2.2	2.9	0.2661	0.3748	0.4880
0.90	0.939	0.2083	1.6	1.9	0.2251	0.3395	0.4599
0.95	0.970	0.1527	1.4	1.6	0.2154	0.3311	0.4531
0.98	0.988	0.0963	1.3	1.4	0.2096	0.3261	0.4491
0.99	0.994	0.0663	1.2	1.3	0.2078	0.3245	0.4478
0.5 + W4	0.696	0.5255	3.2	5.2	0.3113	0.4135	0.5182

Table A.16: Selected-width kernel sensitivity, companion to Table A.15 (which gives $c^\star$ at all widths). The $c^\star$ columns repeat the 50/90/98/99% rows for convenience; the $q^\star(c)$ columns use the measured $\tau_1 = 1.323$. The 98–99% rows need 1.2–1.3 at $\tau = 1$ and 1.3–1.4 at the measured τ; the 50% and 90% rows need 3.0 and 1.6. The mid-width no-go survives a several-fold kernel advantage.

Width	q	$c^\star$ ($\tau = 1$)	$c^\star$ (τ_1)	$q^\star(1.2)$	$q^\star(1.3)$	$q^\star(1.4)$	$q^\star(1.5)$
0.50	0.5032	3.0	4.6	0.1592	0.1847	0.2078	0.2288
0.90	0.2083	1.6	1.9	0.0579	0.0858	0.1110	0.1340
0.98	0.0963	1.3	1.4	0.0400	0.0682	0.0938	0.1172
0.99	0.0663	1.2	1.3	0.0378	0.0661	0.0917	0.1151

Table A.16 expands the calculation at $c = 1.2, 1.3, 1.4, 1.5$. The 98–99% cells sit just inside the measured-τ boundary: $c^\star = 1.3$–1.4, while the same cells need only 1.2–1.3 under $\tau = 1$. The 50% cell needs 3.0–4.6, and the 90% cell needs 1.6–1.9. The difference between the $\tau = 1$ and measured-τ columns is the reason the main text says 1.2–1.4, not a single number. The entries are a conditional band, not a deployable window; our own

fused-MLP Triton kernel and `torch.compile` path are both slower than the padded dense GEMM on this GPU.

Acceptance geometry. Equation (21) uses the geometric acceptance length $A(1-q,\gamma) = \sum_{i=0}^{\gamma}(1-q)^i$, which assumes a constant conditional acceptance rate. We test this directly on the cached match sequence: for every start position we count the length of the consecutive matching run, truncated at γ, and add the bonus token. This empirical A_{emp} preserves within-sequence correlation. Table A.17 reports $A_{\text{emp}}/A_{\text{geo}}$ on raw WikiText: the ratio is 1.00–1.12, so the geometric model is slightly conservative on OOD at large γ. On MATH the ratio is 0.99 at high width and larger at low width; the geometric model is conservative there too. The per-position conditional rates are almost constant ($\alpha_1 \approx 0.90$, $\alpha_i \approx 0.90$ for $i \leq 4$ at 98% width), so the i.i.d.-position approximation is not the source of the negative result. The cost test uses the geometric A as a lower bound on the empirical yield; using A_{emp} would make the no-go stronger, not weaker.

Table A.17: Empirical acceptance length versus the geometric approximation $A(1-q,\gamma) = \sum_{i\leq\gamma}(1-q)^i$. The empirical value is the average number of consecutive matching tokens from a uniformly sampled start, truncated at γ, plus one; it preserves the within-sequence correlation. Entries are $A_{\text{emp}}/A_{\text{geo}}$ for raw WikiText. Ratios above one mean the geometric model is conservative for the cost test; the largest deviation is below 8%.

Width	$\gamma = 1$	$\gamma = 2$	$\gamma = 4$	$\gamma = 8$	$\gamma = 16$
0.10	1.000	1.017	1.032	1.036	1.036
0.30	1.000	1.023	1.061	1.088	1.098
0.50	1.000	1.016	1.052	1.095	1.116
0.70	1.000	1.008	1.029	1.067	1.100
0.90	1.000	1.002	1.006	1.020	1.045
0.98	1.000	1.000	1.001	1.005	1.020
0.99	1.000	1.000	1.000	1.002	1.009

MLP FLOP-share sensitivity. Table A.18 varies the nominal MLP FLOP share by ±10% and ±20%. The certified raw-text cells at 50% and 98% width are robust across the whole range; the 10% width loses the lower-bound certificate only at $\phi_{\text{MLP}} = 0.731$ (+20%), where the point estimate $q = 0.778$ is still far above $q^\star = 0.385$. For all measured widths $C_{\text{FLOP}} < \gamma + 1$, so $q^\star_{\text{FLOP}}$ is finite. At 100% width $C_{\text{FLOP}} = \gamma + 1$ and $q^\star = 0$; the draft is the target.

Table A.18: Sensitivity to the MLP FLOP share ϕ_{MLP}. Nominal is 0.609; rows vary it by ±10% and ±20%. Entries are $q^\star_{\text{FLOP}}$; the last three columns mark whether the lower confidence bound of q_{lb} still certifies raw WikiText at widths 0.10, 0.50, and 0.98. The 10% width loses the lower-bound certificate only in the most extreme +20% row; its point estimate remains above threshold. High widths are robust.

ϕ_{MLP}	$q^\star(0.10)$	$q^\star(0.50)$	$q^\star(0.98)$	cert 0.10	cert 0.50	cert 0.98
0.487	0.2179	0.1086	0.0039	✓	✓	✓
0.548	0.2540	0.1241	0.0044	✓	✓	✓
0.609	0.2935	0.1401	0.0049	✓	✓	✓
0.670	0.3368	0.1567	0.0054	✓	✓	✓
0.731	0.3849	0.1738	0.0059	–	✓	✓

Required versus measured acceptance. Table A.19 and Figure A.3 give the same comparison in acceptance space. At 99% width the roofline requires 0.998 while the draft accepts 0.934; at 50% width it requires 0.860 while the draft accepts 0.497. The gap is the quantitative form of "small perturbation, not small rejection." The figure uses the optimistic FLOP roofline ($\tau = 1$, $\gamma = 4$); the measured C_{meas} column is the padded batch-1 implementation and is an upper bound, as in Section A.5.5.

Table A.19: Required versus measured acceptance on Qwen3.5-9B English raw text (mean of WikiText and FineWeb). $\alpha^{\star}_{\text{FLOP}}$ uses the optimistic FLOP roofline ($\tau = 1$, $\gamma = 4$); $\alpha_{\text{acc}} = 1 - q_g$ is measured. C_{meas} is the measured batch-1 cost $4\rho_1 + \tau_1$; "–" means the measured cost exceeds $\gamma + 1$, so no acceptance can break even. This table is the earlier mixed KV-cache/single-resident protocol and is shown for continuity; the clean kernel-time Table A.64 supersedes it for the cost verdict. In this earlier table only the 30% row has a finite measured $\alpha^{\star}_{\text{meas}} = 0.990$ (cost $4.90 < 5$); every other row is cost-infeasible.

Width	ρ_{FLOP}	$q^{\star}_{\text{FLOP}}$	$\alpha^{\star}_{\text{FLOP}}$	α_{acc}	gap	C_{meas}
0.05	0.422	0.3168	0.6832	0.1760	0.5072	5.61
0.10	0.452	0.2933	0.7067	0.2301	0.4766	5.19
0.20	0.513	0.2498	0.7502	0.3026	0.4476	5.28
0.30	0.574	0.2102	0.7898	0.3576	0.4322	4.90
0.50	0.696	0.1400	0.8600	0.4968	0.3632	5.17
0.70	0.817	0.0791	0.9209	0.6370	0.2840	5.29
0.80	0.878	0.0513	0.9487	0.7099	0.2388	5.11
0.85	0.909	0.0379	0.9621	0.7484	0.2136	5.13
0.90	0.939	0.0250	0.9750	0.7917	0.1833	5.26
0.95	0.970	0.0123	0.9877	0.8473	0.1404	5.14
0.98	0.988	0.0049	0.9951	0.9037	0.0914	5.28
0.99	0.994	0.0024	0.9976	0.9337	0.0639	5.25

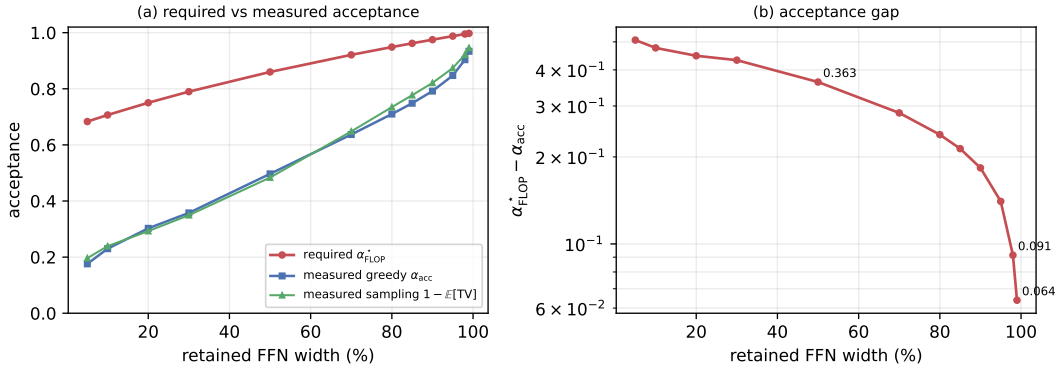


Figure A.3: (a) Required acceptance $\alpha^{\star}_{\text{FLOP}} = 1 - q^{\star}_{\text{FLOP}}$ under the optimistic FLOP roofline and measured greedy/sampling acceptance on Qwen3.5-9B English raw text (mean of WikiText and FineWeb). (b) The acceptance gap. At 99% width the roofline requires 0.998 while the draft accepts 0.934; at 50% width it requires 0.860 while the draft accepts 0.497. This is the quantitative form of "small perturbation, not small rejection."

A.5.8 Operator instantiations

The same accounting covers the operator family. For depth deletion of a fraction d of layers, $\rho_{\text{FLOP}} = 1 - \phi_{\text{depth}} d$ and $\rho_{\text{mem}} = 1 - \phi^{\text{bytes}}_{\text{depth}} d$, where the shares include the deleted attention and FFN. For head pruning, ϕ is the attention-head share. For MoE active-expert reduction from k to k', $\rho_{\text{FLOP}} = 1 - \phi_E(1 - k'/k)$ (Lemma 5). For weight-only quantization, the FLOP ratio is one and the byte ratio is the quantized weight ratio; this is a memory-only axis, and the taxonomy's regime term is the relevant one. For a composed operator, ρ_{FLOP} and ρ_{mem} are the composed shares; the roofline and taxonomy apply without change. In every case the operator-level statement is R or R-free; only the implementation/regime terms are conditional.

A.5.9 Reading the measured numbers

Three protocol facts matter for interpretation. First, the clean kernel-time protocol pre-clones the prefix cache outside the timed region and sums CUDA kernel time over a window of back-to-back passes; the Python-loop wall clock is launch-bound at batch 1 and overstates the residual non-MLP cost (Appendix A.6). Second, the

narrow MLP is a padded dense GEMM, so the measured ρ_1 is an upper bound on the *implementation* cost, not a lower bound on the operator cost. An upper bound above 5 does not certify that the true cost is above 5; it says the measured implementation is at or above 5. The certified no-go is the rejection layer. Third, the measured cost is a median over repeats; the target-pass spread is up to 9% of the pass. A cell whose margin over 5 or over $A(1-q)$ is smaller than that spread is a *point-estimate* closure, not a confidence-bound closure. The Llama MATH 50–70% and 27B MATH 50–90% cells are the point-estimate cells (the 27B 90% K/G label is itself interval-ambiguous); the 98–99% cells are R with a factor 13–20× margin, so their verdict does not depend on the timing spread.

The measured cost can be moved by a better kernel, but the movement is bounded by Proposition 8. A portable kernel moves ρ toward one, not toward ρ_{FLOP}; only a draft-exclusive kernel can move it below the shape ceiling. The fused-MLP, Marlin, and W4A16 kernels in the literature report 1.2–2.7× over naive dense baselines and are portable. Applying one to the draft and not to the target is an asymmetric deployment choice, not a property of width deletion. This is why the paper reports $c^\star$ as an arithmetic conditional and does not treat the 98–99% band as a production-kernel opportunity.

What would change the cost conclusion. A cell classified R would need a draft-exclusive whole-pass advantage above the required $c^\star$ (1.25–1.42 at 98–99%), which is already above the width-only shape ceiling 1.011–1.015; the escape is not a faster MLP but a different operator (quantization on the memory axis, or tree/top-k with a different proposal). A cell classified K would need a draft-exclusive kernel large enough to close the gap to $q^\star_{\mathrm{meas}}$; the Llama MATH 50–70% and 27B MATH 50–90% cells are the measured candidates; the Llama margin is inside the timing spread, and the 27B margins are larger. A cell classified G would need a different regime (a less memory-bound pass) or a different hardware where the FLOP term binds; the 9B MATH 50% point is the measured example. A cell classified W is already a window and needs no cost change. The taxonomy therefore says exactly which evidence would overturn which verdict, which is the sense in which the cost layer is falsifiable rather than a wall-clock theorem.

A.6 Measured Batch-1 Cost

Table A.20: Measured batch-1 cost components with KV caches (512-token prefix, 20 repeats, one A100-80GB). Times are milliseconds. $\rho_1 = T_D^{(1)}/T_T^{(1)}$, $\tau_1 = T_T^{(\gamma+1)}/T_T^{(1)}$, $\gamma = 4$. $S_{\max} = 5/(4\rho_1 + \tau_1)$ is the speedup at perfect acceptance. A row with $4\rho_1 + \tau_1 > 5$ is a point-estimate, acceptance-independent cost no-go at batch 1. This is the earlier KV-cache protocol; the clean kernel-time Table A.64 supersedes it for the cost verdict. The padded narrow MLP is an upper bound on ρ_1; the FLOP frontier in the main text is the corresponding lower bound on cost.

Width	$T_T^{(1)}$	$T_D^{(1)}$	ρ_1	$T_T^{(5)}$	τ_1	$4\rho_1 + \tau_1$	$S_{\max}$	q Wiki
0.10	45.2	43.9	0.973	58.8	1.303	5.193	0.963	0.7782
0.30	48.2	44.2	0.916	59.7	1.239	4.905	1.019	0.6565
0.50	42.8	41.2	0.961	56.6	1.322	5.166	0.968	0.5017
0.70	41.2	40.5	0.985	55.4	1.346	5.285	0.946	0.3618
0.90	43.5	42.7	0.982	58.2	1.337	5.265	0.950	0.2080
0.98	41.6	40.9	0.983	55.9	1.344	5.277	0.948	0.0968
0.99	44.0	43.0	0.977	59.3	1.346	5.254	0.952	0.0677

Table A.20 reports absolute timings and the cost components. The important number is not the exact millisecond but the ratio: at 10% width the draft still costs essentially one target token pass ($\rho_1 \approx 0.97$–1.00), while verification costs 1.30–1.35 target passes. The measured implementation pads the narrow MLP to a multiple of 128 channels and uses ordinary dense GEMMs; it is an upper bound on ρ_1, not a fused-kernel lower bound. The FLOP frontier in the main text is the corresponding lower bound.

Clean timing protocol. The earlier protocol cloned the 512-token KV cache inside every timed call, which is not part of a serving pass. The clean measurement pre-clones the prefix cache outside the timed region and reports two quantities: a target-resident wall-clock loop with per-pass CUDA events, and a kernel-time

window that queues a batch of pre-cloned passes between two events and sums CUDA kernel time. At batch 1 the wall-clock loop is launch-bound: on Llama-3.1-8B the kernel time is 14.4 ms of a 20.2 ms pass, and a zero-MLP path reduces kernel time from 14.4 to 6.5 ms but wall time only from 20.2 to 17.5 ms. Wall-clock ratios therefore understate the benefit of a narrower draft and overstate the residual zero-FFN cost. The decomposition in Table A.64 uses the kernel-time protocol; the realized wall-clock costs (5.1–5.5 at 98–99%) are retained in the artifact. This protocol change is why the 9B zero-FFN cost moves from the earlier 5.10 to 3.83.

Width-monotonicity caveat. The padded narrow MLP is an upper bound and is not width-monotone: at low width, padding and launch overhead keep ρ_1 nearly flat, so the measured cost should be read cell-by-cell. The zero-MLP decomposition is computed at the same cell; $c_{\max}$ is not a smooth function of width, and we do not extrapolate it. The rejection layer does not use ρ_1 and is independent of this caveat.

Table A.21: Batch-size scan at widths 0.50 and 0.98 ($\gamma = 4$, 512-token prefix, 8 repeats, prefill through the backbone to avoid materializing $[B, \text{prefix}, V]$ logits). The scan now reaches $B = 128$, near but below the predicted crossover $B^\star \approx 150$–170. $\rho_B = T_D(B,1)/T_T(B,1)$, $\tau_B = T_T(B,\gamma+1)/T_T(B,1)$. This is the padded batch-scan protocol, not the clean kernel-time decomposition; the $B = 1$ rows are not the reported clean costs. The measured cost remains above 5 at every measured batch; larger batches raise τ_B while ρ_B stays near one for this padded narrow MLP, so the compute-bound regime does not open a window.

Width	B	ρ_B	τ_B	$\gamma\rho_B + \tau_B$	regime
0.50	1	0.988	1.291	5.244	memory-bound
0.50	16	0.998	1.293	5.284	memory-bound
0.50	32	0.996	1.307	5.290	transition
0.50	64	0.948	1.331	5.125	compute-heavy
0.50	128	0.966	1.412	5.275	near $B^\star$
0.98	1	0.991	1.288	5.253	memory-bound
0.98	16	0.996	1.287	5.270	memory-bound
0.98	32	0.995	1.302	5.282	transition
0.98	64	0.999	1.332	5.330	compute-heavy
0.98	128	1.000	1.412	5.413	near $B^\star$

Batch-size scan. Table A.21 repeats the three-component measurement at $B = 1, 16, 32, 64, 128$ for widths 0.50 and 0.98. The padded narrow MLP stays memory-bound: ρ_B remains 0.99–1.00 and τ_B remains 1.27–1.31, so $\gamma\rho_B + \tau_B > 5$ at every measured batch. This is consistent with the FLOP bound and with the roofline picture: larger batches do not make this implementation cheaper per token, and a compute-bound implementation is bounded below by C_{FLOP}.

The zero-path decomposition in Table A.64 gives the corresponding kernel-time shares. At 98% width replacing the MLP with a zero path removes 30% of the pass on 9B, 32% on Qwen3-4B, 23% on Qwen3.5-4B, 52% on Llama, 34% on K2, and 39% on 27B. The residual non-MLP path (linear/full attention, embeddings, LM head, normalization) is therefore the majority of a batch-1 pass on the hybrid and 4B targets. This is why FFN-only width reduction cannot approach the nominal FLOP ratio at batch 1, and why the result should be stated as "FFN width reduction alone does not make this draft cheap enough," not as a claim that all compression is futile. At larger batch the MLP share grows with compute, but verification also grows; the roofline inequality $\rho_B^{\text{roof}} \geq \rho_{\text{roof}}$ covers that trade-off.

Gamma-resolved latency. The decomposition run also measures $\tau(\gamma)$ and $\gamma\rho_1$ for $\gamma = 1, 2, 4, 8, 16$ at widths 0.1, 0.5, and 0.98. The measured cost-only condition $\gamma\rho_1 + \tau(\gamma) > \gamma + 1$ holds at 50–99% width for $\gamma = 4$ and for most of the larger-γ cells; at 30% it is marginal. Where the cost condition is not binding, the rejection term is: the measured q is above the corresponding $q^\star$ in Table A.14. The full JSON is retained with the project.

27B batch-1 measurement. Table A.22 reports the same three-component measurement for Qwen3.5-27B, which we measure here. We load one model at a time (target, then one compressed draft per width) so that the 27B checkpoint fits on one 80GB A100. The target one-token pass is 82.3 ms and the four-token

Table A.22: Qwen3.5-27B batch-1 cost (new measurement). $T_T^{(1)}$ is a one-token target pass and $T_T^{(5)}$ a four-token verification pass with a 512-token prefix in cache; $\rho_1 = T_D^{(1)}/T_T^{(1)}$, $\tau_1 = T_T^{(5)}/T_T^{(1)}$. This is the padded full-pass upper bound, not the clean kernel-time protocol. Cost $4\rho_1 + \tau_1 > 5$ at every width here. The clean protocol gives 4.23/4.53/4.84 at 50/70/90% and 5.07 at 98%: the 50–90% cells are K at the clean point estimate (q above the measured-cost break-even), and 98% is R.

Width	$T_T^{(1)}$ (ms)	$T_T^{(5)}$ (ms)	ρ_1	τ_1	$4\rho_1 + \tau_1$	$S_{\max}$
0.50	82.3	111.3	1.004	1.352	5.368	0.931
0.90	82.3	111.3	1.076	1.352	5.655	0.884
0.98	82.3	111.3	1.019	1.352	5.429	0.921
0.99	82.3	111.3	1.027	1.352	5.459	0.916

verification pass is 111.3 ms ($\tau_1 = 1.352$). The padded narrow MLP gives $\rho_1 = 1.004$–1.076 in that padded full-pass protocol, a 5.37–5.66 upper bound. The clean kernel-time decode protocol gives $\rho_1 = 0.790$–0.998 and a realized cost of 4.23–5.07 at widths 0.50/0.70/0.90/0.98: 98% is a cost no-go ($5.07 > 5$), while the 50–90% points are K at the clean cost (the implied break-even is 0.083/0.050/0.016, below the measured 0.088/0.056/0.026). The 27B zero-MLP cost is 3.50, below 5; a free FFN would reopen a *cost-only* window here too, but not the audit, so it is not a ceiling. The result closes the 27B gap.

Table A.23 reports the trained-draft control.

Table A.23: Trained-draft control. The target-native MTP head (nominal $\rho \approx 0.126$; it shares the target backbone and LM head) is much cheaper than any post-hoc width point and its rejection can clear the nominal break-even, while the post-hoc family cannot; its low-margin-quintile rejection is still high. This is a low-cost control for the low-margin bottleneck; it is not a recommendation and not a matched-latency comparison against a stronger trained draft. The 27B batch-1 measurement is in Table A.22.

Trained draft control (nominal FLOP ratio)					
Draft	ρ	Wiki q	FineWeb q	Wiki low-quintile q	FineWeb low-quintile q
post-hoc width 0.50	0.696	0.502	0.505	0.731	0.717
post-hoc width 0.98	0.988	0.097	0.096	0.327	0.292
target-native MTP	0.126	0.304	0.241	0.591	0.533

Pretrained-draft latency. The pretrained-draft comparison uses the same prefix, batch size, GPU, and warm-up protocol. On this A100, the measured batch-1 per-token ratios are 0.73 (0.8B), 0.74 (2B), and 1.01 (4B) against the 9B target. The nominal FLOP ratios are 0.095, 0.237, and 0.530. The 4B draft is therefore as expensive as the target at batch 1 even though its nominal FLOP ratio is one half; memory boundness compresses the cost differences for both pretrained and compressed drafts. This is why the matched-FLOP comparison, not the measured batch-1 ratio alone, is the controlled comparison.

Verdicts and uncertainty. The measured cost $C_{\text{meas}} = \gamma\rho_1 + \tau_1$ is a median over repeats and an upper bound on the *implementation* cost, because the padded narrow MLP is slower than a fused kernel. An upper bound above 5 therefore does not certify that the true cost is above 5; it certifies the point estimate for this implementation. We convert it to the threshold $q^\star_{\text{meas}}$ and place each cell in the taxonomy of Appendix A.5. At 98–99% the measured cost exceeds $\gamma + 1 = 5$, so $q^\star_{\text{meas}} = 0$: even perfect acceptance cannot break even in this implementation. We call this an acceptance-independent cost no-go; older tables abbreviate it "cost-only no-go," meaning acceptance-independent, not operator-level. It is an acceptance-independent implementation diagnostic (point estimate), and it is additional to the operator-level verdict, which is R in Table A.11. The 98–99% cells are R by a factor 13–20×, so their verdict is not sensitive to the timing spread. The point-estimate K closures are the Llama MATH 50–70% cells, where the margin over 5 is 1.3% and the target-pass spread is up to 9%, and the 27B MATH 50–90% cells, whose margins are 5–60%; Table A.13 marks them K, not R. The 27B 90% point has clean cost 4.84 and measured MATH $q = 0.026$, above the measured-cost break-even $q^\star_{\text{meas}} = 0.0163$ but below the optimistic FLOP threshold $q^\star_{\text{FLOP}} = 0.0275$; it is K, with a 60% margin. It is not R, and the cost layer is the binding constraint at the clean point estimate.

A.7 Agentic Deployment Prefix: SWE-bench-Lite

Table A.24: Agentic deployment prefix: 100 SWE-bench-Lite issue-to-patch examples (23,412 scored tokens). Prefix is an agent instruction plus the GitHub issue and hints (up to 1024 tokens); continuation is the human patch (up to 512 tokens). All cells are certified no-go; the agentic distribution is not easier than raw prose.

width	ρ_{FLOP}	q_g	$q_{\mathrm{lb}}^{(2)}$	95% CI	$q^{\star}_{\mathrm{FLOP}}$	status
0.50	0.696	0.305	0.170	[0.162, 0.179]	0.140	certified
0.90	0.939	0.153	0.097	[0.090, 0.103]	0.025	certified
0.98	0.988	0.095	0.061	[0.057, 0.066]	0.0049	certified
0.99	0.994	0.067	0.043	[0.040, 0.046]	0.0024	certified

The main raw suite is static single-turn prose. To test a more agentic prefix distribution we build 100 examples from the public SWE-bench-Lite test split [18]. The prefix is an agent-style instruction plus the GitHub issue and its hints (truncated to 1024 tokens); the scored continuation is the human patch (truncated to 512 tokens). The target cache is the same Qwen3.5-9B bf16 model, and the draft is the same activation-aware width deletion. This is a static single-turn proxy, not a full tool-call loop: the public Lite split does not ship repository snapshots, so we cannot replay real tool traces. It still shifts the prefix distribution toward code, issue reports, and patch syntax, which is the part of the deployment distribution that the margin mechanism is sensitive to.

Table A.24 gives the result. Rejection is 0.305 at 50% width and falls to 0.067 at 99%, comparable to WikiText/FineWeb and higher than clean code at several widths. The top-2 lower bound is above $q^{\star}_{\mathrm{FLOP}}$ in every cell, including the 50% cell that is only point-above in the code corpus. The agentic prefix therefore does not open a profitable window; if anything, issue/patch text has the low-margin structure that the raw-prose result predicts. The caveat is scope: this is not a multi-turn agent loop, and a production agentic distribution with retrieved repository context could differ. It is one more deployment-like prefix, not a substitute for the coverage theory in Section 3.4.

A.8 The Memory-Only Axis: Weight-Only Quantization

Table A.25: Memory-only axis: real weight-only quantization of the same Qwen3.5-9B target. "ρ_{mem}" is the stored-weight byte ratio (embeddings and, where stated, the LM head stay bf16); "$q^{\star}_{\mathrm{mem}}$" is the break-even threshold at ρ_{mem}, $\gamma = 4$, $\tau = 1$. The quality columns are top-1 rejection (top-2 certificate in parentheses). W4 is not excluded by the memory roofline, but our packed INT4 kernel is not faster than bf16 at batch 1; W8 is near-lossless but the INT8 kernel is slower still. This is a cost-only, implementation-dependent result, not a structural no-go. The W8 quality cell uses 30 WikiText sequences because it is a near-lossless boundary check; the W4 cells use 100 sequences (MATH 500). The optimized QSpec/QuantSpec/ML-SpecQD kernels do achieve 1.2–2.7× speedups; this table is a boundary check with an unoptimized kernel.

setting	ρ_{mem}	$q^{\star}_{\mathrm{mem}}$	Wiki	FineWeb	C4	MATH	ρ_1	cost	measured
W4, LM head bf16	0.433	0.308	0.141 (0.100)	0.113 (0.079)	0.134 (0.095)	0.039 (0.032)	1.009	5.34	fail
W4, all linears	0.350	0.378	0.156 (0.110)	0.135 (0.097)	–	–	0.986	5.24	fail
W8, LM head bf16	0.614	0.186	0.022[†] (0.013)	–	–	–	2.105	9.72	fail

[†] 30 WikiText sequences (15,360 tokens); other cells are 100 sequences (51,200 tokens; MATH 500 sequences). Cost is $\gamma\rho_1 + \tau_1$; "fail" means cost > 5 in this kernel. The W4 roofline threshold is passed on every measured corpus, and the INT4 microbenchmark is $\approx 2\times$ faster than bf16 per linear layer; the end-to-end ratio is dominated by the unchanged linear-attention, normalization, and launch path.

Quantization plus SD is a different, memory-only axis. QSpec [43] uses a W4A4 draft with a W4A16 verifier and reports up to 1.64× speedup without quality degradation; QuantSpec [38] uses 4-bit weights and a hierarchical 4-bit KV cache with $> 90\%$ acceptance and up to $\sim 2.5\times$ speedup at long context; ML-SpecQD [12] uses MXFP4 weight-only drafts with 80–90% acceptance and up to 2.72×; Quasar [14] uses a W8A8 verifier and reports 1.28×; and the W4A16 hierarchical framework of Zhang et al. [42] reports 2.78× on W4

Llama-3-70B. These methods keep the architecture, depth, width, and head count and change numerical precision (and sometimes the KV cache). They are therefore orthogonal to the FLOP-ratio-reducing operators in this paper. Our theory and no-go do not cover them, and the measurements below are a boundary check, not a claim that quantization fails. Our W4 top-1 acceptance (86–96%) is in the same range as the literature's 80–98%; the difference is kernel efficiency, not acceptance.

Why this axis is outside the FLOP no-go. Weight-only quantization changes the number of bytes, not the matmul FLOP count: $\rho_{\text{FLOP}} = 1$ and the roofline bound is $\rho_{\text{roof}} = \rho_{\text{mem}}$. The FLOP break-even argument therefore says nothing about it. We measure the axis directly with packed INT4/INT8 weights and the fused dequantization kernels of PyTorch (`_weight_int4pack_mm`, `_weight_int8pack_mm`), not with fake quantization: an earlier control keeps bf16 master weights and cannot support a memory claim. The target cache is the same Qwen3.5-9B bf16 model, so the quality columns are quantization-induced top-1 flips, not a capacity gap. The earlier capacity control's W4 $q = 0.266$ on WikiText was measured against the 35B target and mixes the 9B–35B capacity gap with quantization; against the same 9B target the W4 draft adds $q = 0.141$.

Results. A W4 draft with the LM head in bf16 has $\rho_{\text{mem}} = 0.433$ and $q^\star_{\text{mem}} = 0.308$; its measured rejection is 0.141 on WikiText, 0.113 on FineWeb, 0.134 on C4, and 0.039 on MATH, so it *passes* the memory-roofline break-even. Quantizing the LM head too lowers ρ_{mem} to 0.350 ($q^\star_{\text{mem}} = 0.378$) and raises rejection only to 0.156/0.135 on WikiText/FineWeb. W8 is near-lossless ($q = 0.022$ on a 30-sequence WikiText check) and has $\rho_{\text{mem}} = 0.614$ ($q^\star_{\text{mem}} = 0.186$), also a roofline pass. At the roofline these cells would give $S = 1.39$–$1.53\times$ for W4 and $1.38\times$ for W8; the measured kernels give $S < 1\times$. The gap is entirely kernel efficiency, not acceptance. The top-2 certificate is non-vacuous throughout (e.g. W4 WikiText $q_{\text{lb}} = 0.100$).

The bytes ratio is weight-only at the 512-token measurement prefix. The KV cache is architecture-shared and adds less than 1% of the target's weight bytes at this length; at very long context it would grow and push both ratios toward one, which is the standard memory-bound caveat. The *footprint* of a quantized-draft deployment is larger than the target alone because the bf16 verifier and the packed draft coexist (1.35–$1.43\times$); the cost model uses per-pass traffic, not total footprint. This is a capacity trade-off, not a per-token cost. The embedding matrix is kept bf16 in the first two rows; quantizing it would lower ρ_{mem} further but is not needed to make the roofline point.

The measured batch-1 cost tells a different story. Our INT4 kernel gives $\rho_1 = 1.009$, $\tau_1 = 1.300$, cost 5.34 (LM head bf16) and $\rho_1 = 0.986$, cost 5.24 (all linears); the INT8 kernel gives $\rho_1 = 2.105$, cost 9.72. Thus our unoptimized implementation is a cost-only no-go even though the roofline is not; the Marlin/MXFP4/QSpec kernels cited above do achieve real speedups. The margin is small: W4 all-linears is 5.24, only 5% above the cost-only threshold, so a modest kernel improvement could move it below 5 and make the quality axis decisive. A per-layer microbenchmark shows the packed INT4 matmul is $\approx 2\times$ faster than bf16, so the end-to-end ratio is dominated by the unchanged linear-attention mixer, normalization, embedding, CPU dispatch, and launch path, not by the quantized matmul. A production W4 kernel or CUDA-graph engine or a W4A8 tensor-core path could move ρ_1 toward ρ_{mem} and reopen the window. We therefore do *not* claim a structural no-go for quantization; it is a memory-only axis whose status is kernel-dependent. This is the same distinction as the zero-MLP kernel ceiling: the roofline is the optimistic limit, and the measured kernel is an upper bound.

Composing width and quantization. A natural question is whether the no-go covers the composed operator. Table A.26 reports real packed INT4 applied after width reduction. At 50% width the composed draft has $q = 0.526$ on WikiText/FineWeb against $q^\star_{\text{FLOP}} = 0.140$, with $q_{\text{lb}} = 0.280/0.277$ (certified); at 90% and 98% width (effective 90.6% and 99.0% after the INT4 padding) it has $q = 0.246/0.171$ against $q^\star = 0.023/0.0025$, also certified. The in-domain MATH 50% cell is 0.1405 versus 0.1400, point-above but not lower-bound-certified; its measured batch-1 cost closes the window anyway. On the memory axis $\rho_{\text{mem}} = 0.361$–$0.432$ ($q^\star_{\text{mem}} = 0.368$–$0.309$); the 50% point estimate is above but the lower bound is not, while the 90/98% cells pass the memory roofline. The FLOP axis is unchanged by quantization, so composition cannot rescue the compute-reducing claim; it is a memory-only operator stacked on a compute-reducing one.

Table A.26: Combined operator: structured width reduction followed by real packed INT4 (group 128, LM head bf16). The FLOP ratio is the width ratio; quantization only changes ρ_{mem}. The composed draft is a certified no-go on the FLOP axis at every raw-text width; the in-domain MATH 50% cell is point-above (0.1405 versus 0.1400) but its lower bound is below, so it is not certified. On the memory axis the point rejection is above $q^{\star}_{\text{mem}}$ at 50% width but the lower bound is not, so we do not claim a memory-axis certificate; the optimized QSpec/QuantSpec kernels are a different, memory-only path. Widths marked * are padded to a multiple of 128 for the INT4 layout; the effective width is shown.

Width	Eff.	Set	q	q_{lb}	ρ_{FLOP}	$q^{\star}_{\text{FLOP}}$	ρ_{mem}	$q^{\star}_{\text{mem}}$	cert. FLOP
0.50	0.500	WikiText	0.5255	0.2796	0.696	0.1400	0.361	0.3675	True
0.50	0.500	FineWeb	0.5261	0.2768	0.696	0.1400	0.361	0.3675	True
0.50	0.500	MATH	0.1405	0.0939	0.696	0.1400	0.361	0.3675	False
0.90*	0.906	WikiText	0.2459	0.1579	0.943	0.0234	0.420	0.3184	True
0.90*	0.906	FineWeb	0.2413	0.1446	0.943	0.0234	0.420	0.3184	True
0.98*	0.990	WikiText	0.1706	0.1158	0.994	0.0025	0.432	0.3090	True
0.98*	0.990	FineWeb	0.1532	0.0992	0.994	0.0025	0.432	0.3090	True

Relation to prior quantization work. Quasar [14] uses a W8A8 quantized verifier and reports that quantization preserves acceptance length while structural pruning does not; our W8 result is consistent. Chen et al. [6] define the top-1 flip rate Flip@1 of a quantized model and report 8.3–12.7% for NVFP4 on Qwen3-32B, in the same range as our W4 top-1 flips. Arai and Ichikawa [1] give a Lipschitz propagation bound on the output error of layer-wise PTQ; combined with the margin certificate of Lemma 1, it bounds the rejection of a quantized draft by the realized quantization residual.

Concretely, let δ_L be the final hidden-state error of the quantized copy, R_k the per-layer quantization residual, W_s the layer weights, and γ_s the activation Lipschitz constants. The QEP propagation bound is

$$\|\delta_L\|_F \;\le\; U := \sum_{k=1}^{L} \Big(\prod_{s=k+1}^{L} \gamma_s \|W_s\|_2 \Big) \gamma_k \|R_k\|_F. \tag{38}$$

Since $\Delta z = W_{\text{LM}} \delta_L$, the margin perturbation obeys $|\Delta_m| \le \|W_{\text{LM}}\|_2 U$ (or the corresponding top-2 row difference). Composing with Lemma 1 gives

$$q_g \;\ge\; \Pr\big(m(x) < \|W_{\text{LM}}\|_2 U\big). \tag{39}$$

This bound is valid for PTQ and for a QAT copy; QAT changes the residuals R_k, not the form of the bound. It is loose because it multiplies spectral norms across depth, but it is the general statement that survives training. We are not aware of a distribution-free lower bound on top-1 disagreement of a QAT draft that is independent of the evaluation distribution: QAT can in principle learn a quantized model that matches the teacher on the evaluated tokens, and the QAT bounds we found concern loss/generalization [3, 9], not top-1 agreement with a fixed bf16 teacher. QAT also updates weights, so it is a trained operator under our taxonomy and outside the post-hoc claim. The measurement-based certificate above is the only general statement we can make for a deployed QAT copy.

A.9 Certificate Tightness and Standard Sampling

Table A.27: Status of the top-2 certificate. q_{lb} is the point certificate and the bracket is its 95% sequence-cluster interval; q_{lb}/q is its tightness. $q^{\star}_{\mathrm{FLOP}}$ uses $\tau = 1$ and is therefore an upper bound on the admissible rejection for every batch size. The certificate is a measurement-based lower bound, not a parameter-free prediction.

Width	$q^{\star}_{\mathrm{FLOP}}$	q Wiki	q_{lb} Wiki	q_{lb}/q	q_{lb} FineWeb	certified
0.10	0.2933	0.7782	0.3607 [0.355,0.366]	0.46	0.3617	✓
0.30	0.2102	0.6565	0.3319 [0.324,0.340]	0.51	0.3159	✓
0.50	0.1400	0.5017	0.2690 [0.262,0.276]	0.54	0.2674	✓
0.70	0.0791	0.3618	0.2124 [0.206,0.218]	0.59	0.2071	✓
0.90	0.0250	0.2080	0.1337 [0.130,0.138]	0.64	0.1254	✓
0.98	0.0049	0.0968	0.0636 [0.061,0.066]	0.66	0.0574	✓
0.99	0.0024	0.0677	0.0455 [0.043,0.048]	0.67	0.0381	✓

Table A.27 reports the certificate status. The point certificate rises from 0.36–0.39 at 10% width to 0.038–0.046 at 99% width; its ratio to the observed q is 0.46–0.67 on raw text. It is a measurement-based lower bound, not a parameter-free prediction: computing it requires the compressed draft's logits on the evaluation tokens. Its value is that it is non-vacuous and model-free in the sense that it does not require a TV power law or a Markov relaxation. A third token overtaking the target's top-1 can cause rejection without the top-2 flip, which is why the bound is conservative.

Standard speculative sampling. Table A.28 replaces the greedy rejection by the measured $q_s = \mathbb{E}[\mathrm{TV}]$ and keeps the same cost model. Every compressed width has $q_s > q^{\star}_{\mathrm{FLOP}}$ on both raw text sets. By Lemma 8, q_s is a lower bound on the rejection of any exact token-level protocol using this draft, so the conclusion is not specific to greedy verification. The standard-SD loop on raw text at 98% width gives high acceptance ($\tau = 4.32/4.19$) but the measured cost is still a cost-only no-go at batch 1 (Table A.29). The reference HF engine re-runs the full target prefix each block and its target-only baseline does not use a KV cache; its wall-clock ratio is therefore not a fair throughput benchmark, and we do not use it for the cost claim. A fair KV-cache rollback is nontrivial for the hybrid linear-attention layers of Qwen3.5, so we do not report an end-to-end throughput number for this model; fair KV-cache loops for Llama-3.1-8B, Qwen3-4B, and K2-3.7B are reported in Appendix A.21. The paper's claim is the roofline no-go, not a wall-clock guarantee for every engine. The loop here is used only for acceptance diagnostics.

Table A.28: Standard-SD sampling rejection $q_s = \mathbb{E}[\mathrm{TV}]$ from the same caches. The greedy certificate is replaced by the measured q_s; the cost model is unchanged. Every compressed width has $q_s > q^{\star}_{\mathrm{FLOP}}$ on both raw OOD sets; measured $\tau \geq 1.3$ lowers the true threshold.

Width	ρ	$q^{\star}_{\mathrm{FLOP}}$	q_s MATH	q_s Wiki	q_s FineWeb	certified
0.90	0.939	0.0250	0.0395	0.1809	0.1773	✓
0.95	0.970	0.0123	0.0253	0.1271	0.1251	✓
0.98	0.988	0.0049	0.0151	0.0803	0.0756	✓
0.99	0.994	0.0024	0.0106	0.0576	0.0511	✓

Table A.29: Standard SD loop on raw OOD at 98% width (30 prompts, $\gamma = 4$, 96 new tokens). The loop acceptance is high, but the measured cost is a cost-only no-go (Table A.20); the reference engine's wall-clock ratio is not a fair throughput benchmark because it recomputes the target prefix each block.

Set	τ	α	$\mathbb{E}[\mathrm{TV}]$ loop	q_{greedy} loop	cache q_s	prompts
WikiText raw	4.32	0.831	0.0770	0.0977	0.0803	30
FineWeb raw	4.19	0.798	0.0879	0.1205	0.0756	30

A narrow caveat: q_s is measured on the target-generated cache, while a real loop visits harder prefixes. On generated OOD a 30-prompt loop gives an $\mathbb{E}[\mathrm{TV}]$ 1.6–3.4× the cache and a greedy rejection 2.0–4.2× the cache; on raw text the loop is close to the cache. This makes the cached OOD numbers optimistic, not conservative; the raw-text certificates are unaffected because they are computed on raw text. The sampling-matched cache closes most of the generated-OOD gap; see Table A.30.

Table A.30: Standard-SD real loop on generated OOD (30 prompts/cell, $\gamma = 4$, 96 new tokens). The loop samples from the target and therefore visits harder prefixes than the target-greedy cache: its $\mathbb{E}[\mathrm{TV}]$ is 1.6–3.4× the cache estimate, especially on FineWeb gen. Within the loop, greedy and TV rejection still track each other (Pearson 0.987), and the capacity exponent remains shallow (0.13–0.18).

Set	Draft	τ	$\mathbb{E}[\mathrm{TV}]$ loop	q_{greedy} loop	$\mathbb{E}[\mathrm{TV}]$ cache	ratio
WikiText gen.	0.8B	2.59	0.339	0.342	0.178	1.90×
WikiText gen.	2B	2.76	0.291	0.291	0.157	1.85×
WikiText gen.	4B	3.04	0.259	0.276	0.157	1.65×
WikiText gen.	9B	3.11	0.253	0.274	0.160	1.58×
WikiText gen.	27B	3.32	0.206	0.208	0.115	1.80×
FineWeb gen.	0.8B	2.61	0.336	0.355	0.103	3.26×
FineWeb gen.	2B	2.81	0.287	0.308	0.090	3.18×
FineWeb gen.	4B	3.22	0.226	0.233	0.071	3.17×
FineWeb gen.	9B	3.35	0.199	0.201	0.059	3.35×
FineWeb gen.	27B	3.41	0.184	0.205	0.057	3.21×

A.10 Additional OOD Domains

Table A.31 lists the additional domains.

Table A.31: Additional OOD domains at 98% width: code (The Stack) and first-turn MT-Bench, target-generated from the 9B target. Both are certified no-go.

Set	q	q_{lb}	$q_s = \mathbb{E}[\mathrm{TV}]$	$q^\star_{\mathrm{FLOP}}$	certified
Code	0.0145	0.0102	0.0173	0.0049	✓
MT-Bench	0.0629	0.0483	0.0564	0.0049	✓

At 98% width, code has $q = 0.048$ and MT-Bench $q = 0.063$, against $q^\star_{\mathrm{FLOP}} = 0.0049$; the lower bounds are 0.026 and 0.048, both certified. We extended MT-Bench to five widths (Table A.32): rejection is 0.397/0.147/0.098/0.063/0.043 at 50/90/95/98/99% width, above $q^\star$ at every compressed width. The chat/instruction prefix is therefore not an easier deployment distribution than raw prose. The same measured-cost failure applies at these widths; independently, the cells are roofline-bound.

Table A.32: MT-Bench chat/instruction prefix (80 first-turn conversations) across widths. The raw-text threshold $q^\star_{\mathrm{FLOP}}$ uses the 9B MLP share. Every compressed width is above the threshold; at 50% width rejection is 0.397, so the chat prefix is not an easier deployment distribution than raw prose.

Width	ρ_{FLOP}	$q^\star_{\mathrm{FLOP}}$	q	$\mathbb{E}[\mathrm{TV}]$
0.50	0.696	0.1400	0.3972	0.4378
0.90	0.939	0.0250	0.1465	0.1388
0.95	0.970	0.0123	0.0984	0.0935
0.98	0.988	0.0049	0.0629	0.0564
0.99	0.994	0.0024	0.0431	0.0382

Table A.33: Post-hoc repair attempts at 98% width ($q^\star_{\mathrm{FLOP}} = 0.0049$). "No flip" means no certified raw cell changes status. No cheap post-hoc fix moves the raw-text frontier; the best-of-budget global search is reported separately in Section 5.5 and also does not reopen a tested raw cell. Evidence: Appendices A.11, A.12, and A.19.

Repair attempt	Objective / information	Raw q (Wiki/FineWeb)	Outcome
Layer-wise ridge refit	Layer-output MSE, held-out calibration	0.097/0.096 (MSE −35%)	no flip
Gradient fit of narrow matrices	Adam on layer MSE, 20 steps	0.716/0.710	worse
Top-1 margin calibration	CE + top-2 margin hinge	0.137/0.114	worse
Same-corpus held-out selection	Select on 50 chunks, test on 50	0.056/0.049	no flip
Oracle selection	Select on the evaluation corpus (upper bound)	0.053/0.043	no flip
Calibration size 50k/100k/200k	Channel ranking only	0.103/0.101/0.097	stable
Down-projection adaptation	0.5M raw tokens, next-token loss	0.448/0.201	no flip
Independent SVD / shared rotation	Non-coordinate bases at matched FLOP	0.82/0.77; 1.00/1.00	worse

A.11 Reconstruction Details

Table A.34: Post-pruning layer-wise reconstruction. The pruned down projection is refitted by residual ridge regression to the original layer output on a held-out calibration split; the selected ridge penalty minimises held-out reconstruction MSE. MSE is relative to the original layer-output energy, averaged over layers; every layer improves. q is greedy rejection before/after reconstruction; paired Δq intervals are reported in the text. Reconstruction reduces MSE by 34–36% but leaves q unchanged (all TOST tests equivalent within ±0.005).

Width	MSE red.	q MATH	q MATH′	q Wiki	q Wiki′	q FineWeb	q FineWeb′
0.90	36.0%	0.0399	0.0389	0.2080	0.2116	0.2085	0.2128
0.95	35.3%	0.0260	0.0253	0.1502	0.1530	0.1551	0.1569
0.98	34.9%	0.0157	0.0156	0.0968	0.0974	0.0959	0.0960
0.99	34.2%	0.0112	0.0111	0.0677	0.0685	0.0649	0.0661

After reconstruction the perturbation is no longer exactly the deleted-channel sum of Lemma 3: the down projection changes, so Δy includes a readout correction. Lemma 6 gives the exact decomposition for the deployed residual-ridge readout:

$$\Delta y(x) = W_{d,D}\Big[a_D(x) - \Sigma_{DK}(\Sigma_K + \lambda I)^{-1} a_K(x)\Big], \tag{40}$$

with the empirical moments used by the fit. The implementation uses unnormalized sums and scales λ by the mean diagonal; the identity is unchanged. The first term is the deleted-channel sum; the second is the linear part of a_D that the refitted readout can absorb. At $\lambda = 0$ the correction is the population regression of a_D on a_K; at $\lambda \to \infty$ it vanishes and the pure-deletion identity returns. Because Δ_m is measured from the reconstructed draft's logits, the certificate remains valid for the composed map; Lemma 6 explains the operator rather than replacing the measurement. The reconstruction is selected by held-out layer MSE, not by top-1 rejection. The TOST equivalence margin of ±0.005 was fixed before these runs as a practically negligible change in rejection; the paired 95% CIs are reported in the main text. The full table shows that the selected ridge penalty reduces MSE at every layer and every width while leaving q within the TOST equivalence margin. The gradient-fit control in Appendix A.12 is the aggressive alternative; it is worse on full-model rejection.

A.12 Compression Baselines

Table A.35: Operator zoo on Qwen3.5-9B at comparable nominal FLOP reduction. All rows except quantization are compute-reducing. "Width" is activation-aware channel deletion (the main operator); "weight norm" deletes the same fraction by weight magnitude; "independent SVD" factorizes each FFN matrix; "shared rotation" mixes intermediate coordinates before the elementwise nonlinearity and gives $q = 1$ at every rank. Weight-only quantization reduces memory but not FLOPs; the 9B W4 boundary is in Table A.25 (WikiText 0.141, MATH 0.039). The earlier 0.236/0.266 values were a 35B-target capacity control, not a 9B quantization result.

Operator	Fraction / rank	ρ_{FLOP}	MATH q	Wiki raw q
Activation width	0.10	0.452	0.4350	0.7782
Activation width	0.20	0.513	0.2808	0.7033
Activation width	0.50	0.696	0.1297	0.5017
Weight norm	0.50	0.696	0.6793	0.8893
Independent SVD	0.10	0.472	0.9503	0.9600
Independent SVD	0.20	0.554	0.9500	0.9600
Independent SVD	0.50	0.797	0.7492	0.8201
Shared rotation	0.50	0.696	1.0000	1.0000

Why the shared rotation fails. The shared-rotation operator chooses the top-r eigenvectors of $W_g W_g^\top + W_u W_u^\top$, applies the rotation to the gate/up outputs, and rotates back after the elementwise product. It gives $q = 1.000$ at every rank from 5% to 100% on all three sets. The 100% entry is not a bug: even at full rank a nontrivial orthogonal rotation changes the function because $\sigma(Pz) \odot Pz \neq P(\sigma(z) \odot z)$ for a general orthogonal P, which is exactly Lemma 2. The rotation mixes intermediate coordinates, and the elementwise nonlinearity is not equivariant to a general orthogonal map. The independent-SVD operator keeps the width and factorizes each of the three matrices separately; it is exact at full rank but the gate and up bases drift apart as the rank falls, so the elementwise product is no longer the target's product. Both baselines are worse than structured pruning at the same nominal cost.

Gradient-fit negative control. The main recovery check uses the ridge readout because it is the optimal linear reconstruction for the kept channels and it is selected by held-out MSE. We also tried a per-layer Adam fit of all three narrow matrices (20 steps, learning rate 10^{-6}, gradient clipping, best state selected by held-out layer MSE). It reduces the layer MSE slightly further (from 9.55×10^{-7} to 8.37×10^{-7} on the validation split) but changes full-model rejection to 0.284 on MATH, 0.716 on WikiText, and 0.710 on FineWeb. Full-model validation therefore rejects it. This is a cautionary result: layer-wise MSE is not a sufficient selection criterion for top-1 agreement, and an aggressive fit can be much worse than the conservative ridge readout. We use the ridge readout as the recovery baseline and report the gradient fit only as a negative control.

Margin-sensitive channel score. A natural mechanism-driven repair is to weight calibration tokens by their inverse margin, $w(m) = 1/(1 + m)$, so that low-margin tokens contribute more to the channel score. We recompute the activation-aware score on 100k calibration tokens and resweep. At 98% width the result is $q = 0.0993/0.0965$ on WikiText/FineWeb against 0.0968/0.0959 for the default score; at 50% it is 0.5085/0.5087 against 0.5017/0.5047. The margin-aware score is slightly worse, not better. This is consistent with the family-tightness argument: within the uniform-width activation-ranking family, selection is bounded by oracle selection, and the ridge readout is already the optimal linear reconstruction for a fixed kept set. The score is still post-hoc admissible (it uses target activations and top-1/top-2 labels, not target responses).

Calibration-size sensitivity. Recomputing the channel-importance scores on 50k, 100k, and 200k calibration tokens and evaluating at 98% width gives WikiText rejection 0.103/0.101/0.097 and FineWeb 0.098/0.096/0.096; all are far above $q^\star = 0.0049$ and certified. The selection is therefore not sensitive to the calibration size over this range; the 200k point is the main sweep, and the 50k/100k artifacts are in `out/rank_scan/calib_size_50000` and `calib_size_100000`.

Table A.36: Calibration-size sensitivity of the channel-importance scores at 98% width on Qwen3.5-9B. Rejection changes by at most 0.006 across 50k–200k tokens; every cell remains far above $q^\star = 0.0049$.

Calibration tokens	WikiText q	FineWeb q	certified
50k	0.1033	0.0981	✓
100k	0.1012	0.0963	✓
200k	0.0968	0.0959	✓

Relation to SparseGPT/OBS and SliceGPT. SparseGPT/OBS update individual weights using a layer-input Hessian; the structured analogue for channel deletion is a Hessian-weighted channel score followed by the ridge readout update. The ridge readout is already the optimal linear reconstruction for a fixed kept set, and the gradient-fit control updates all three narrow matrices and is worse on full-model rejection. A full unstructured SparseGPT update is not directly comparable because it does not delete a structured channel: it changes individual weights while keeping the width. The oracle-selection control (Table A.58) bounds any improvement within the uniform-width activation-ranking family, and it still leaves the 98%-width raw rejection at 0.043–0.056, far above $q^\star$. The shared-rotation baseline does not apply the SliceGPT RMSNorm scale folding; with folding the operator is still non-coordinate, so the $q = 1$ entry is a bound on the unfused rotation, not on a fully optimized SliceGPT pipeline.

A.12.1 Global post-hoc search details

Search space and budget. All selectors use the same 200k-token calibration text as the main activation score, the same 192-token target caches, and the same audit. The global selector ranks all $32 \times 12{,}288$ FFN channels by a layer-normalised score and keeps the top-R channels under the total budget (a top-R knapsack, not an exhaustive subset search); the per-layer DP minimises $\sum_l [m_l(r_l) + \beta m_l(r_l)^2]$ on a 32-channel grid, where $m_l(r_l)$ is the fraction of layer l's score mass removed at width r_l and $\beta = 1$; the sensitivity-weighted DP replaces the score mass by the measured q increase from deleting 2% of one layer at a time, measured on the first 30 evaluation sequences (an oracle diagnostic, not a deployable selector); the random search samples 12 per-layer width vectors with mean equal to the target average and standard deviation 0.03, evaluates them on the first 30 WikiText sequences, and keeps the best. The OBS-style score is the exact marginal output error for independent channel removal, $\mathbb{E}[a_j^2]\|W_{\text{down}}[:, j]\|_2^2$. Selection took under one A100 GPU-hour, excluding the final 100-sequence audits.

Table A.37: Global post-hoc selectors on FineWeb at fixed average FFN width. All rows use the same calibration text and target cache and the same 100-sequence audit. The search rows are pruning-only (no ridge readout); a 98% ridge spot check moves the reference by at most 0.004, but it covers only the uniform and global OBS rows and is not a full ridge-refit test (Appendix A.12.1). $q_{\mathrm{lb}}^{\mathrm{lo}}$ is the lower endpoint of the 95% sequence-cluster interval (single-cell, not family-wise); $q^\star$ is the $c = 1$ FLOP break-even. The selectors are post-hoc choices on the same evaluation cells, not new confirmatory cells. The global allocation rows are budgeted heuristics, not a proof of global optimality. The lowest point q and the largest certified margin need not coincide: $q_{\mathrm{lb}}^{\mathrm{lo}}$ is the comparison the paper uses; ratios use the unrounded $q^\star$. The nonuniform rows are rejection-only: the $q^\star$ column is the $c = 1$ FLOP threshold at the average width, and no measured-cost (R/G/K) verdict is assigned.

Width	Selector	q	q_{lb}	$q_{\mathrm{lb}}^{\mathrm{lo}}$	$q^\star$	$q_{\mathrm{lb}}^{\mathrm{lo}}/q^\star$	Verdict
98%	Uniform, activation-aware	0.0959	0.0574	0.0543	0.0049	11.1×	no-go
	Uniform, OBS second-moment	0.0920	0.0544	0.0513	0.0049	10.5×	no-go
	Global subset, activation-aware	0.1636	0.0828	0.0794	0.0049	16.2×	no-go
	Per-layer DP, activation-aware	0.1578	0.0796	0.0762	0.0049	15.6×	no-go
	Global subset, OBS	0.1637	0.0819	0.0783	0.0049	16.0×	no-go
	Per-layer DP, OBS	0.1626	0.0815	0.0779	0.0049	15.9×	no-go
99%	Uniform, activation-aware	0.0649	0.0381	0.0357	0.0024	14.6×	no-go
	Uniform, OBS second-moment	0.0617	0.0347	0.0324	0.0024	13.3×	no-go
	Global subset, activation-aware	0.1385	0.0652	0.0615	0.0024	25.2×	no-go
	Per-layer DP, activation-aware	0.1350	0.0630	0.0596	0.0024	24.4×	no-go
	Global subset, OBS	0.1337	0.0634	0.0601	0.0024	24.6×	no-go
	Per-layer DP, OBS	0.1338	0.0630	0.0597	0.0024	24.5×	no-go

Table A.38: Global post-hoc selectors on WikiText at fixed average FFN width. All rows use the same calibration text and target cache and the same 100-sequence audit. The search rows are pruning-only (no ridge readout); a 98% ridge spot check moves the reference by at most 0.004, but it covers only the uniform and global OBS rows and is not a full ridge-refit test (Appendix A.12.1). $q_{\mathrm{lb}}^{\mathrm{lo}}$ is the lower endpoint of the 95% sequence-cluster interval (single-cell, not family-wise); $q^\star$ is the $c = 1$ FLOP break-even. The selectors are post-hoc choices on the same evaluation cells, not new confirmatory cells. The global allocation rows are budgeted heuristics, not a proof of global optimality. The lowest point q and the largest certified margin need not coincide: $q_{\mathrm{lb}}^{\mathrm{lo}}$ is the comparison the paper uses; ratios use the unrounded $q^\star$. The nonuniform rows are rejection-only: the $q^\star$ column is the $c = 1$ FLOP threshold at the average width, and no measured-cost (R/G/K) verdict is assigned.

Width	Selector	q	q_{lb}	$q_{\mathrm{lb}}^{\mathrm{lo}}$	$q^\star$	$q_{\mathrm{lb}}^{\mathrm{lo}}/q^\star$	Verdict
30%	Uniform, activation-aware	0.6565	0.3319	0.3245	0.2102	1.5×	no-go
	Uniform, OBS second-moment	0.5682	0.3004	0.2937	0.2102	1.4×	no-go
	Global subset, activation-aware	0.6614	0.3326	0.3253	0.2102	1.5×	no-go
	Per-layer DP, activation-aware	0.6603	0.3232	0.3163	0.2102	1.5×	no-go
	Global subset, OBS	0.6224	0.3119	0.3050	0.2102	1.5×	no-go
	Per-layer DP, OBS	0.6243	0.3136	0.3068	0.2102	1.5×	no-go
	Oracle sensitivity DP	0.6654	0.3321	0.3252	0.2102	1.5×	no-go
50%	Uniform, activation-aware	0.5017	0.2690	0.2617	0.1400	1.9×	no-go
	Uniform, OBS second-moment	0.4622	0.2529	0.2461	0.1400	1.8×	no-go
	Global subset, activation-aware	0.5339	0.2841	0.2762	0.1400	2.0×	no-go
	Per-layer DP, activation-aware	0.5392	0.2852	0.2775	0.1400	2.0×	no-go
	Global subset, OBS	0.5055	0.2734	0.2668	0.1400	1.9×	no-go
	Per-layer DP, OBS	0.5024	0.2727	0.2660	0.1400	1.9×	no-go
	Oracle sensitivity DP	0.5242	0.2809	0.2734	0.1400	2.0×	no-go
90%	Uniform, activation-aware	0.2080	0.1337	0.1298	0.0250	5.2×	no-go
	Uniform, OBS second-moment	0.1969	0.1278	0.1239	0.0250	5.0×	no-go
	Global subset, activation-aware	0.2704	0.1644	0.1593	0.0250	6.4×	no-go
	Per-layer DP, activation-aware	0.2089	0.1331	0.1289	0.0250	5.2×	no-go
	Global subset, OBS	0.2351	0.1475	0.1430	0.0250	5.7×	no-go
	Per-layer DP, OBS	0.2340	0.1478	0.1433	0.0250	5.7×	no-go
	Oracle sensitivity DP	0.1994	0.1297	0.1260	0.0250	5.0×	no-go
98%	Uniform, activation-aware	0.0968	0.0636	0.0608	0.0049	12.4×	no-go
	Uniform, OBS second-moment	0.0863	0.0568	0.0541	0.0049	11.1×	no-go
	Global subset, activation-aware	0.1250	0.0735	0.0705	0.0049	14.4×	no-go
	Per-layer DP, activation-aware	0.1244	0.0725	0.0696	0.0049	14.2×	no-go
	Global subset, OBS	0.1242	0.0702	0.0668	0.0049	13.7×	no-go
	Per-layer DP, OBS	0.1235	0.0697	0.0665	0.0049	13.6×	no-go
99%	Uniform, activation-aware	0.0677	0.0455	0.0432	0.0024	17.7×	no-go
	Uniform, OBS second-moment	0.0635	0.0412	0.0390	0.0024	16.0×	no-go
	Global subset, activation-aware	0.1050	0.0576	0.0549	0.0024	22.5×	no-go
	Per-layer DP, activation-aware	0.1027	0.0567	0.0539	0.0024	22.1×	no-go
	Global subset, OBS	0.0982	0.0532	0.0506	0.0024	20.7×	no-go
	Per-layer DP, OBS	0.0965	0.0523	0.0498	0.0024	20.4×	no-go

Results. Table A.38 gives the full WikiText selector matrix, including the activation-aware global/DP variants and the 90% width; Table A.37 gives the FineWeb cells. The global allocation is worse than uniform OBS at every tested width. At 30% the global/DP OBS variants improve on the activation-aware uniform point rejection, while the activation-aware variants do not; neither improves the certified margin or beats uniform OBS. OBS and random lower point q but not $q_{\mathrm{lb}}^{\mathrm{lo}}$; the default retains a larger certified margin than both. The low-width cells are the closest to reopening but remain above $q^\star$. These are deployable-selection rows: the scores and plans use calibration text, not evaluation tokens. We do not run an evaluation-corpus oracle global search, so this table is a stress test, not an upper bound on every post-hoc selection. Table A.39 reports the per-layer allocation. The global and DP plans remove disproportionately from a few layers (minimum widths 0.59–0.71 at 98%), while the sensitivity-weighted DP stays closer to uniform (minimum 0.80); neither beats the default certified margin.

Cross-family check. On Llama-3.1-8B, the OBS score is worse than activation-aware at 98% width ($q = 0.0956/0.0944$ versus $0.0807/0.0767$), and the global/DP OBS allocations do not beat the activation-aware uniform selector there. At 99% the global OBS allocation lowers point q to 0.0579/0.0537 and the lower-bound ratio to 12.0–13.0× $q^\star$ (from 13.4–14.3× for uniform activation-aware), but it still does not reopen. This is a cross-family check, not a replication of the Qwen3.5-9B search; it uses the same calibration text and audit.

Table A.39: Per-layer width allocation for the global selectors at 98% average width. Uniform keeps 0.980 in every layer. CV is the coefficient of variation across layers. Allocation is a post-hoc choice at a fixed total budget; only the rejection audit decides whether it helps.

Plan	min	max	mean	CV
Uniform	0.980	0.980	0.980	0.000
Global subset, activation-aware	0.618	1.000	0.980	0.071
Per-layer DP, activation-aware	0.714	1.000	0.980	0.056
Global subset, OBS	0.589	1.000	0.980	0.077
Per-layer DP, OBS	0.594	1.000	0.980	0.076
Oracle sensitivity DP	0.798	1.000	0.980	0.045

Table A.40: Random allocation check. Twelve per-layer width vectors are sampled at the same average width, evaluated on the first 30 WikiText sequences, and the best calibration candidate is audited on 50 held-out WikiText and FineWeb chunks. The calibration range is the min–max over candidates; the selected row is the candidate with the lowest calibration q. A search that lowers point q but not the certified lower bound is evidence that the allocation is not the binding variable for the certificate.

Width	Calib. q range	Selected calib. q	Wiki q	Wiki $q_{\mathrm{lb}}^{\mathrm{lo}}$	Fine q	Fine $q_{\mathrm{lb}}^{\mathrm{lo}}$
50%	0.450–0.457	0.450	0.4648	0.2433	0.4518	0.2364
98%	0.075–0.092	0.075	0.0805	0.0545	0.0725	0.0468
99%	0.053–0.065	0.053	0.0573	0.0357	0.0507	0.0301

Random-search null. Table A.40 is a random allocation check, not a new selector. The best-of-12 random allocation is consistent with this reading. At 98% the selected candidate gives $q = 0.0805/0.0725$ on held-out WikiText/FineWeb, but its lower-bound ratio is 11.1 × /9.6× $q^\star$; at 99% it gives $q = 0.0573/0.0507$ with lower-bound ratios 14.6 × /12.3×. The point estimate can fall because the search picks a lucky allocation on the calibration split, but the certified lower bound does not improve consistently, and no cell opens. The calibration range (0.075–0.092 at 98%) shows the candidate-to-candidate spread; the selected candidate is the minimum of a small sample, so it is a best-of-budget upper bound on what this search family can do, not a converged optimum.

Ridge check and cost caveat. The search rows are pruning-only. Adding the ridge readout moves the uniform activation-aware reference from 0.0968 to 0.0974 on WikiText and from 0.0959 to 0.0960 on FineWeb at 98%; for the global OBS allocation it moves 0.1250 to 0.1282 on WikiText, a change of 0.0032. The 98% spot check therefore moves the reference by at most 0.004; it covers the uniform and global OBS rows only and is not a full ridge-refit test of every search row. A per-layer width vector is not the uniform padded narrow-MLP implementation, so we do not assign R/G/K verdicts or transfer the measured 5.004–5.28 cost to these rows; the table is rejection-only. The exact combinatorial optimum over subsets and integer widths remains open, and the random allocation check is a fixed-budget diagnostic, not a convergence proof.

A.13 Qwen3.5-4B Replication

The 4B model has a smaller MLP share than 9B, so the FLOP saving for the same retained width is smaller; the no-go is if anything stronger. In Table A.41, $S_{\mathrm{OOD}}^{\mathrm{FLOP}}$ is the FLOP-model speedup at the worse of the two raw-text rejection rates. In-domain rejection at 70% width is 0.154, close to the 0.18 break-even threshold;

raw text is 0.49–0.51, far above it. At 90% width the in-domain rejection is still 0.08 and raw text is 0.33; the 4B model is more sensitive to channel deletion than 9B. The top-2 certificate lower bound is 0.039–0.375 on both raw text sets against $q^\star = 0.0022$–0.248; every compressed width from 10% to 99% is certified no-go.

Table A.41: Qwen3.5-4B replication (target and compressed draft are the same model). ρ is the nominal FLOP ratio; $q^\star$ uses $\gamma = 4$. The same ordering holds: raw OOD stays above break-even at every compressed width.

Width	ρ	$q^\star_{\text{FLOP}}$	MATH	Wiki raw	FineWeb raw	$S^{\text{FLOP}}_{\text{OOD}}$
0.10	0.515	0.2482	0.5176	0.7807	0.7926	0.42
0.20	0.569	0.2131	0.3395	0.6797	0.7034	0.45
0.30	0.623	0.1806	0.2553	0.6295	0.6334	0.45
0.50	0.731	0.1216	0.1542	0.4862	0.5056	0.51
0.70	0.838	0.0693	0.0973	0.3521	0.3588	0.58
0.85	0.919	0.0334	0.0616	0.2483	0.2524	0.65
0.90	0.946	0.0220	0.0472	0.2043	0.2084	0.70
0.95	0.973	0.0109	0.0317	0.1445	0.1479	0.77
0.99	0.995	0.0022	0.0128	0.0622	0.0621	0.89
1.00	1.000	0.0000	0.0000	0.0000	0.0000	1.00

The layer-wise reconstruction check also replicates: at 98% width it reduces 4B layer MSE by 39.6% but changes MATH q from 0.0188 to 0.0180, WikiText from 0.0883 to 0.0883, and FineWeb from 0.0942 to 0.0947.

A.14 Cross-Family Replication

Table A.42 reports the Qwen3-4B replication.

Table A.42: Cross-family replication: Qwen3-4B target and compressed draft (different family and tokenizer from Qwen3.5). All compressed widths are certified no-go on MATH and both raw OOD sets.

Width	ρ	$q^\star$	MATH q (q_{lb})	Wiki q (q_{lb})	FineWeb q (q_{lb})	certified
0.95	0.967	0.0136	0.036 (0.029)	0.155 (0.106)	0.158 (0.109)	✓
0.98	0.987	0.0054	0.022 (0.018)	0.100 (0.070)	0.104 (0.072)	✓
0.99	0.993	0.0027	0.016 (0.012)	0.068 (0.048)	0.070 (0.049)	✓

To test whether the compression floor is specific to Qwen3.5, we repeat the small-width check on Qwen3-4B-Instruct-2507, a different family with a different tokenizer. The target is Qwen3-4B itself; the evaluation sets are 100 target-generated MATH prompts and 100 retokenized raw WikiText and FineWeb chunks. At 95%, 98%, and 99% width, every set is certified no-go by the lower confidence bound of q_{lb}. At 98% width, $q = 0.022$ on MATH, 0.100 on WikiText, and 0.104 on FineWeb against $q^\star = 0.0054$. A layer-wise ridge reconstruction at 98% width reduces layer MSE by 39.6% but changes MATH q from 0.0224 to 0.0220, WikiText from 0.1005 to 0.1010, and FineWeb from 0.1038 to 0.1035; the floor is unchanged. This addresses the single-family concern, although it is a same-size (4B) replication rather than a full family sweep.

A second cross-family check uses Llama-3.1-8B-Instruct, again a different family and tokenizer. At 95%, 98%, and 99% width every set is certified no-go; at 98% width $q = 0.019$ on MATH, 0.081 on WikiText, and 0.077 on FineWeb against $q^\star = 0.0060$ (Table A.43). The floor is not Qwen-specific.

Table A.43: Second cross-family replication: Llama-3.1-8B-Instruct target and compressed draft (different family and tokenizer). All compressed widths are certified no-go.

Width	ρ	$q^\star$	MATH q (q_{lb})	Wiki q (q_{lb})	FineWeb q (q_{lb})	certified
0.95	0.962	0.0153	0.030 (0.023)	0.123 (0.087)	0.123 (0.084)	✓
0.98	0.985	0.0060	0.019 (0.013)	0.081 (0.056)	0.077 (0.053)	✓
0.99	0.992	0.0030	0.014 (0.010)	0.060 (0.043)	0.058 (0.040)	✓

A.15 Low-Width Cross-Family Check

Table A.44 gives the low-width rows.

Table A.44: Low-width cross-family check. Qwen3-4B-Instruct-2507 and Llama-3.1-8B-Instruct, target and compressed draft the same model; 100 target-generated MATH prompts and 100 raw WikiText/FineWeb chunks each. Raw OOD is certified at every width for both families. MATH has a FLOP window at 50–70% for Llama; Qwen3-4B has no window in this sweep (its point estimates are above the threshold at 50–70%). The measured batch-1 cost control for Qwen3.5-9B closes the analogous window, but we do not have a matching cost measurement for these two models.

Model	Width	MATH q	Wiki q	q_{lb} Wiki	certified
Qwen3-4B	0.3	0.2846	0.6740	0.3369	✓
Qwen3-4B	0.5	0.1675	0.5521	0.2906	✓
Qwen3-4B	0.7	0.1097	0.3920	0.2260	✓
Qwen3-4B	0.9	0.0540	0.2103	0.1398	✓
Llama-3.1-8B	0.3	0.3249	0.7416	0.3609	✓
Llama-3.1-8B	0.5	0.1567	0.5736	0.2970	✓
Llama-3.1-8B	0.7	0.0871	0.3642	0.2124	✓
Llama-3.1-8B	0.9	0.0430	0.1771	0.1182	✓

The original cross-family tables covered only 95–99% width. We extend Qwen3-4B and Llama-3.1-8B to 30%, 50%, 70%, and 90% width on the same 100 MATH prompts and 100 raw WikiText/FineWeb chunks. Raw text is certified no-go at every added width for both families. MATH is model-dependent: Llama has a narrow 50–70% FLOP window, while Qwen3-4B has no window in this sweep (its point estimates are above the threshold at 50–70%). We now have a matched batch-1 cost measurement for Llama at 50/70/90% width: the clean kernel-time cost is 5.067/5.066/5.154 (Table A.45). The 50–70% points are realized-kernel-bound (K): both rooflines pass, but the measured padded kernel does not, and the required shape advantage exceeds the roofline advantage by only 0.4–2.3%, inside the 9% timing spread. The 90% point is roofline-bound (R): $q = 0.0430$ is above the optimistic FLOP threshold $q^{\star}_{\mathrm{FLOP}} = 0.0310$, so rejection closes it and the measured cost is redundant. We do not claim structural closure at 50–70%, and the raw-text no-go is certified independently.

Table A.45: Matched clean kernel-time batch-1 cost for the Llama-3.1-8B low-width MATH window. The FLOP frontier has a window at these widths, but the measured cost (padded narrow-MLP implementation) does not: $\gamma\rho_1 + \tau_1 > 5$ at 50–90% width. The margin over 5 is only 1.3%, while the target-pass timing spread across repeats is up to 9% of the pass, so this is a point-estimate closure, not a confidence-bound closure. A faster production MLP kernel could reopen the window; we do not claim structural closure. Columns are rounded; the cost column uses the unrounded ratios.

Width	ρ_1	τ_1	$4\rho_1 + \tau_1$	measured verdict
0.50	0.973	1.17	5.067	cost-closed (point)
0.70	0.973	1.17	5.066	cost-closed (point)
0.90	0.994	1.17	5.154	cost-closed (point)

For each model, the channel importance scores are computed on that model's own activations with its own tokenizer; we do not reuse the 9B importance file. The evaluation corpora are the same fixed chunks as in the high-width tables.

A.16 K2-Horizon-3.7B Cross-Family Replication

Table A.46: K2-Horizon-3.7B cross-family replication: a standard dense Transformer with 36 layers, $d = 2560$, $h = 10240$ (no hybrid linear attention), target and compressed draft the same model. Evaluation: 30 target-generated MATH prompts and 50 target-generated WikiText chunks. ρ uses $\phi_{\mathrm{MLP}} = 0.75$. The same channel-deletion floor appears at every width; WikiText is certified at all widths, MATH at $\geq 70\%$. The sample is small, so this is a qualitative cross-family check, not a powered sweep.

Width	ρ_{FLOP}	$q^{\star}_{\mathrm{FLOP}}$	MATH q	Wiki q	q_{lb} Wiki	cert. Wiki
0.05	0.288	0.4394	0.9919	0.9982	0.4550	✓
0.10	0.325	0.4014	0.8716	0.9805	0.4295	✓
0.20	0.400	0.3344	0.6062	0.8141	0.3580	✓
0.30	0.475	0.2764	0.5312	0.6291	0.3046	✓
0.50	0.625	0.1794	0.3352	0.4640	0.2487	✓
0.70	0.775	0.0994	0.2169	0.3748	0.2161	✓
0.80	0.850	0.0640	0.1794	0.3443	0.2081	✓
0.90	0.925	0.0309	0.1357	0.3201	0.1976	✓
0.95	0.962	0.0152	0.1214	0.3130	0.1957	✓
0.98	0.985	0.0060	0.1120	0.3061	0.1931	✓
0.99	0.992	0.0030	0.0992	0.3054	0.1939	✓

K2-Horizon-3.7B [17] is a standard dense Transformer (36 layers, $d = 2560$, $h = 10240$, no hybrid linear attention), so it also tests whether the Qwen3.5 hybrid token mixer is responsible for the floor. Target and compressed draft are the same model; we use activation-aware channel importance measured on the K2 WikiText corpus, 30 target-generated MATH prompts, and 50 target-generated WikiText chunks. The qualitative pattern is unchanged. In the 30/50 target-generated check, at 99% width $q = 0.305$ and $q_{\mathrm{lb}} = 0.194$ against $q^{\star}_{\mathrm{FLOP}} = 0.0030$ for that check; every width from 5% to 99% is certified no-go by the lower confidence bound. The raw 100/100 replication is weaker but still R: at 99% width $q = 0.063$ and $q_{\mathrm{lb}} = 0.041$ against $q^{\star}_{\mathrm{FLOP}} = 0.0026$. MATH is certified at widths $\geq 70\%$ and the point estimate is above threshold at every width. The sample is small (30/50 prompts), so this is a qualitative target-generated cross-family check, not a powered sweep; it does show that the floor is not specific to the Qwen3.5 architecture or tokenizer. The confirmatory K2 cells use the 100/100 raw WikiText replication in Table A.66, not this 30/50 target-generated check; the raw cells are weaker ($q = 0.079$ at 98%) but still R.

A.17 Matched-Cost Details

Table A.47 gives the matched-cost rows.

Table A.47: Matched-cost comparison against the Qwen3.5-9B target. ρ_{FLOP} is the nominal FLOP ratio; ρ_{meas} is the measured batch-1 per-token pass ratio (same tokenizer, prefix, batch, GPU, and warm-up). The controlled comparison is at matched ρ_{FLOP} (4B vs. 20% compressed 9B). At high width the compressed draft can have lower in-domain rejection, but it remains above the raw-OOD break-even threshold. Pretrained-draft pretraining data and token counts are not controlled.

Draft	ρ_{FLOP}	$q^{\star}_{\mathrm{FLOP}}$	ρ_{meas}	MATH	Wiki raw	FineWeb raw	source
Qwen3.5-0.8B	0.095	0.723	0.730	0.0958	0.3713	–	pretrained
Qwen3.5-2B	0.237	0.497	0.738	0.0731	0.3030	–	pretrained
Qwen3.5-4B	0.530	0.238	1.012	0.0474	0.2236	–	pretrained
9B, width 0.10	0.452	0.293	0.973	0.4350	0.7782	0.7616	compressed
9B, width 0.20	0.513	0.250	0.983	0.2808	0.7033	0.6914	compressed
9B, width 0.30	0.574	0.210	0.916	0.2098	0.6565	0.6284	compressed
9B, width 0.50	0.696	0.140	0.961	0.1297	0.5017	0.5047	compressed
9B, width 0.98	0.988	0.005	0.983	0.0157	0.0968	0.0959	compressed

Table A.48: Pretrained 4B draft against the 9B target at batch sizes $1, 4, 8, 16$ (512-token prefix, $\gamma = 4$). The nominal FLOP ratio is 0.53, but the 4B draft is memory-bound at every measured batch: $\rho_B \approx 1$ and $\gamma\rho_B + \tau_B > 5$. The matched-FLOP comparison in the main text is therefore a controlled rejection comparison, not a claim that the 4B draft accelerates batch-1 serving.

Batch	ρ_B	τ_B	$\gamma\rho_B + \tau_B$	cost-only no-go
1	0.991	1.274	5.240	✓
4	1.000	1.265	5.266	✓
8	1.001	1.270	5.274	✓
16	1.000	1.284	5.283	✓

The table reports both the nominal FLOP ratio and the measured batch-1 per-token ratio. The controlled comparison is at matched nominal FLOPs, where the pretrained 4B draft and the compressed 9B draft at 20% width are close (0.53 vs. 0.51). At batch 1 both are memory-bound ($\rho_1 \approx 1.01$ vs. 0.99), so the measured ratio compresses the difference; the rejection gap remains. The 4B FineWeb cell is $q = 0.178$ (95% CI $[0.173, 0.184]$) and the WikiText cell is 0.224 ($[0.218, 0.229]$); both are below the 0.238 frontier. At high width the compressed draft can have lower in-domain rejection, but its raw-text rejection is still above the FLOP break-even. The pretrained drafts are off-the-shelf Qwen3.5 checkpoints; their pretraining data and token counts are not controlled, so this is not a scaling law. The Qwen3.5-4B and 9B target share a tokenizer and family; the cross-family checks use their own tokenizers. The controlled comparison therefore supports a narrow statement: a separately pretrained (or adapted) small draft can beat a post-hoc compressed large draft at matched nominal FLOPs on raw text. It does not show that capacity is irrelevant, and it does not isolate the effect of training data from the effect of capacity.

The missing 20%-width batch-1 point is measured: $\rho_1 = 0.983$, $\tau_1 = 1.349$, so $\gamma\rho_1 + \tau_1 = 5.28 > 5$. The pretrained 4B draft is also memory-bound in this implementation (Table A.48): at $B = 1, 4, 8, 16$ its measured ρ_B is 0.99–1.00 and its cost is 5.24–5.28. Thus neither draft is a batch-1 serving speedup at these costs; the matched-FLOP comparison is about rejection at controlled nominal cost. A serving engine that makes the 4B draft compute-bound (larger batch, more aggressive batching, or a kernel that exploits the smaller weight set) could change the wall-clock picture, but that is outside the controlled comparison and would not change the compressed draft's raw-text rejection.

A.18 Pretrained-Draft Capacity Scan

Setup and local exponents. The target is Qwen3.5-35B-A3B (35B total, 3.07B active). Drafts are Qwen3.5-0.8B/2B/4B/9B/27B. Table A.49 reports the local exponent α of q_{greedy} versus capacity over 0.8B–27B, with sequence-bootstrap CIs. The exponent is 0.17–0.26 on task/raw sets; halving q needs 14–43× more capacity on task/raw sets (58× on generated FineWeb). We do not extrapolate to a deployment capacity.

Table A.49: Local scaling of the greedy rejection rate, Qwen3.5 target. α is the slope of $\log q_{\text{greedy}}$ vs. $\log D$ over 0.8B–27B, with a sequence-bootstrap 95% CI; $D_{1/2} = 2^{1/\alpha}$ is the capacity multiple needed to halve q; the last column is $q(0.8\text{B})/q(27\text{B})$. The local exponent is descriptive, not a fitted asymptote.

Evaluation set	α [95% CI]	$D_{1/2}$ [95% CI]	q(0.8B)	q(27B)	ratio
MATH-500	0.264 [0.247, 0.282]	13.9× [11.7, 16.5]	0.0958	0.0388	0.405
GSM8K	0.255 [0.242, 0.269]	15.1× [13.1, 17.6]	0.0973	0.0407	0.418
HumanEval	0.195 [0.151, 0.240]	34.9× [17.9, 98.5]	0.0374	0.0191	0.511
MT-Bench	0.209 [0.159, 0.260]	27.4× [14.4, 78.4]	0.2497	0.1216	0.487
WikiText-2 raw	0.184 [0.175, 0.193]	42.8× [36.1, 52.4]	0.3805	0.1998	0.525
FineWeb-Edu raw	0.195 [0.184, 0.208]	34.7× [27.9, 43.3]	0.3328	0.1721	0.517
FineWeb-Edu generated	0.171 [0.113, 0.227]	58.0× [21.3, 458.7]	0.0870	0.0492	0.565

Table A.50 reports the capacity axes. The 27B draft is 8.34× the target's active parameters and forward FLOPs but 0.78× its total parameters. All super-capacity statements in the main text therefore refer to the active/FLOPs axis; the total-parameter control is Table A.51.

Table A.50: Capacity definitions for the draft series and the target. Total parameters are checkpoint tensors (text backbone); compute parameters exclude the embedding lookup; active parameters are the matmul weights actually executed per token (top-8 experts for the MoE target); forward FLOPs are $2N_{\text{active}}$ plus the attention term at sequence length 512. The 27B draft is 8.34× the target's active parameters and 8.30× its forward FLOPs, but only 0.78× its total parameters.

Model	Total (B)	Compute (B)	Active (B)	Fwd GFLOP/token
Qwen3.5-0.8B	0.752	0.752	0.752	1.52
Qwen3.5-2B	1.882	1.882	1.882	3.78
Qwen3.5-4B	4.206	4.206	4.206	8.45
Qwen3.5-9B	8.954	7.937	7.937	15.91
Qwen3.5-27B	26.896	25.625	25.625	51.35
Qwen3.5-35B-A3B (target)	34.661	34.152	3.072	6.19

Table A.51: Total-parameter super-capacity control. Target is Qwen3.5-9B; the 27B draft has 3.0× the target's total parameters and 3.2× its compute parameters. A 27B draft does not improve over the 4B draft, so total-parameter super-capacity does not rescue the OOD residual in this control. The 9B draft is the target itself and is omitted (q=0 by construction).

Draft	Total (B)	MATH-500 q	WikiText raw q	τ_∞ (MATH / Wiki)
0.8B	0.75	0.096	0.371	10.4 / 2.7
2B	1.88	0.073	0.303	13.7 / 3.3
4B	4.21	0.047	0.224	21.1 / 4.5
27B	26.90	0.048	0.226	20.7 / 4.4

Equivalence test for the 9B-target control. The 27B draft is 3.0× the 9B target's total parameters. It does not improve over the 4B draft: MATH $q = 0.0483$ vs 0.0474 (paired $\Delta = +0.0008$, 95% CI $[-0.0001, +0.0017]$), WikiText $q = 0.2263$ vs 0.2236 (paired $\Delta = +0.0027$, 95% CI $[-0.0013, +0.0067]$). With a pre-specified equivalence margin of 0.01, both TOST tests reject the null of a difference at $p < 0.001$ (MATH) and $p = 0.00015$ (WikiText). The correct statement is "equivalent within ±0.01", not "27B improves" or "27B is identical".

Robustness. Table A.52 reports the full 500-prompt MATH regeneration at 768 tokens. The 512-token generation is conservative: q falls by 0.0011–0.0031 and the local exponent changes from 0.2596 to 0.2614. The top-1024 cache check is summarized in Appendix A.3. The K2 Horizon replication gives local exponents 0.13–0.14 on 30 MATH prompts and 50 WikiText chunks; it is qualitative, not a larger-sample replication.

Table A.52: Full MATH-500 regeneration at 768 tokens (500/500 prompts) versus the original 512-token generation. The 512-token values are slightly conservative: q falls by 0.0011–0.0031, and the local exponent is essentially unchanged (0.2596 at 512 vs. 0.2614 at 768).

Draft	q @ 512	q @ 768	Δ
0.8B	0.09577	0.09265	−0.00312
2B	0.07278	0.07060	−0.00218
4B	0.05253	0.05108	−0.00145
9B	0.04696	0.04584	−0.00113
27B	0.03878	0.03718	−0.00160

A.19 Training and Adaptation Controls

Relation to trained/adapted drafts. T2D [32], DistillSpec [44], and SD2 [20] report that pruning or distilling the target and then adapting on target responses yields strong in-domain speculative decoding. We do not dispute those results: they are in the trained/adapted regime, whereas our operator is post-hoc. Their objective is empirical risk on a target-response dataset drawn from μ, which controls the target only on μ; a

raw-text evaluation is a different π_{dep}, and (15) explains why in-domain gains need not transfer. The two claims are orthogonal.

Open question for distilled drafts. A natural conjecture is that if the minimal post-hoc perturbation already exceeds the frontier, then a smaller draft distilled from the target with no access to target logits cannot do better. This is not a theorem. Distillation can in principle recover the target's top-1 on the measured distribution, and a pretrained 4B draft is far better than a compressed 9B draft at comparable nominal cost. A capacity-aware lower bound would need an assumption that limits how much of the target's decision margin a smaller student can recover; we do not have one. The coverage bound (15) makes the open problem precise: a positive distillation result must cover the deployment component, not only the adaptation set.

Narrow distillation. Target-data distillation improves in-domain fit while worsening OOD at matched in-domain risk: 2B@10M and 4B@20M both reach $q_{\text{MATH}} \approx 0.009$, but their raw-text q is 1.25–2.07× the off-the-shelf draft; 9B@40M reaches $q_{\text{MATH}} = 0.0067$ but 0.42/0.39 raw text. At equal 10M tokens, 4B/9B remain at $q_{\text{MATH}} = 0.057/0.050$ and are undertrained; they are excluded from the causal claim. The equal-token series is one corpus and one seed, so it is reported as a bounded control, not a scaling law.

Adaptation budget curve. The post-hoc/trained boundary is not only definitional. At a 0.5M-token down-projection adaptation budget, raw rejection moves to 0.448/0.201 (Wiki/FineWeb) rather than down; at 10–20M tokens, 2B and 4B distillation reach in-domain MATH $q \approx 0.009$ but their raw rejection is 1.25–2.07× the off-the-shelf draft; the native MTP head reaches raw $q = 0.304/0.241$ at $\rho \approx 0.126$. The trained path has a slope on in-domain fit, but it does not close the raw-text frontier, and the cheap post-hoc repairs are flat. This is the empirical content behind the admissibility criterion in Section 4.

Table A.53: Adaptation-budget curve. The post-hoc repairs are flat or worse on raw text; the trained path buys in-domain fit at the cost of raw rejection. "Raw" is WikiText/FineWeb unless stated. The 2B/4B distillation rows report raw rejection relative to the off-the-shelf draft because the absolute value depends on the student capacity.

Method	Budget	MATH q	Raw q	Post-hoc?
Ridge readout refit	closed form	0.0157	0.097/0.096	yes
0.5M down-proj. adaptation	0.5M tokens	0.024	0.448/0.201	no
2B distillation	10M tokens	0.009	1.25–2.07× off-the-shelf	no
4B distillation	20M tokens	0.009	0.42/0.39	no
9B distillation	40M tokens	0.0067	0.42/0.39	no
Native MTP head	trained head	–	0.304/0.241	no

Positive adaptation. Table A.54 reports the positive-control grid.

Table A.54: Positive control: continue-training the 2B draft on held-out target-generated FineWeb. Cells are greedy q (mean over seeds) with the change from the off-the-shelf 2B in parentheses (negative is better). The 5e-6/0.5M cell uses three seeds; the other cells use two. The gentle/low-lr runs improve OOD and leave MATH essentially unchanged; 2e-5 overfits. This separates the learning-rate axis from the token-budget axis.

Run	Tokens	FineWeb raw	FineWeb gen.	WikiText raw	WikiText gen.	MATH-500
off-the-shelf	–	0.2769	0.0750	0.3174	0.1348	0.0728
gentle lr 5e-6	0.50M	0.2712 (-0.0057)	0.0731 (-0.0019)	0.3007 (-0.0168)	0.1297 (-0.0051)	0.0731 (+0.0003)
gentle lr 5e-6	1.00M	0.2707 (-0.0063)	0.0733 (-0.0017)	0.2993 (-0.0181)	0.1307 (-0.0041)	0.0732 (+0.0004)
lr 1e-5	0.50M	0.2722 (-0.0047)	0.0738 (-0.0012)	0.3010 (-0.0165)	0.1324 (-0.0024)	0.0736 (+0.0008)
lr 1e-5	1.00M	0.2721 (-0.0048)	0.0739 (-0.0011)	0.3024 (-0.0150)	0.1336 (-0.0012)	0.0736 (+0.0008)
aggressive lr 2e-5	0.50M	0.2817 (+0.0048)	0.0770 (+0.0020)	0.3120 (-0.0054)	0.1383 (+0.0035)	0.0755 (+0.0027)
aggressive lr 2e-5	1.00M	0.2825 (+0.0056)	0.0778 (+0.0028)	0.3178 (+0.0004)	0.1407 (+0.0058)	0.0774 (+0.0046)

The lr×token grid shows the effect is learning-rate driven: 5×10^{-6} and 10^{-5} improve OOD, while 2×10^{-5} overfits. On the larger held-out evaluation (100 WikiText and 200 FineWeb prompts), the 2B gentle deltas are -0.0065 $[-0.0094, -0.0038]$ and -0.0057 $[-0.0072, -0.0042]$; all three seeds are in the same direction. This is a small, seed-stable positive signal, not a fix for the residual. The 4B adaptation is larger on WikiText ($0.139 \to 0.110$) and small on FineWeb; the 0.8B adaptation improves only FineWeb. The effect is capacity- and corpus-dependent.

Certificate status after adaptation. The post-hoc no-go does not classify trained or adapted drafts, but for completeness the 9B 0.5M-token down-projection adaptation at 98% width gives WikiText $q = 0.448$, FineWeb 0.201, and MATH 0.024, with $q_{\mathrm{lb}} = 0.248/0.142/0.020$, all above $q^\star = 0.0049$; the raw and MATH cells remain certified. Thus the gentle adaptation does not flip a certificate cell. The positive 2B control is a different trained operator and is outside the post-hoc claim.

A.20 Target-Native MTP as a Trained Draft

Table A.55: Target-native MTP head as a trained draft. The MTP head shares the target backbone and its LM head; $\rho \approx 0.126$ counts one MTP layer, the fusion projection, and the draft's LM head, while the post-hoc rows use the nominal width ratio. The MTP draft is much cheaper than any post-hoc width point in the table and has lower overall rejection than the 50%-width draft. Its low-margin-quintile rejection remains high (0.59/0.53), and its mean $|\Delta_m|$ is larger than the near-identity 98%-width post-hoc draft's, which costs about 8× more. Training improves the high-margin bulk but does not remove the low-margin bottleneck.

Draft	ρ	Wiki q	FineWeb q	Wiki low-q	FineWeb low-q	Wiki $\lvert\Delta_m\rvert$	FineWeb $\lvert\Delta_m\rvert$
post-hoc width 0.50	0.696	0.5017	0.5047	0.7306	0.7171	2.263	1.974
post-hoc width 0.90	0.939	0.2080	0.2085	0.5169	0.4905	0.914	0.738
post-hoc width 0.98	0.988	0.0968	0.0959	0.3271	0.2923	0.405	0.308
target-native MTP	0.126	0.3039	0.2405	0.5905	0.5327	1.579	1.103

Setup. The Qwen3.5 checkpoint ships a native MTP head (`mtp.*`) that the text-only causal-LM loader ignores. We load the head manually and use the NEXTN formulation: at position p, the draft reads the target hidden at $p-2$ and the embedding of the current token $p-1$, and predicts p. The head is one full-attention decoder layer plus a fusion projection and the shared LM head; its nominal ratio is $\rho \approx 0.126$ including the draft's LM head (or 0.027 if the LM head is counted as shared with verification). We evaluate 100 WikiText and 100 FineWeb sequences with the same target model, so the comparison is against the same target distribution.

Result. The MTP draft has $q = 0.304$ on WikiText and 0.241 on FineWeb, with top-2 lower bounds 0.195/0.161. It is much cheaper than any post-hoc width point and has lower overall rejection than the 50%-width draft (0.502/0.505 at $\rho = 0.696$). However, its low-margin-quintile rejection is 0.590/0.533, higher than the near-identity 98%-width post-hoc draft (0.327/0.292 at $\rho = 0.988$), and its mean $|\Delta_m|$ is 1.58/1.10 versus 0.405/0.308. Training therefore improves the high-margin bulk but does not remove the low-margin bottleneck; the post-hoc near-identity draft is milder on low-margin tokens but costs about 8× more.

Caveat. This is a one-layer next-token MTP head sharing the target backbone, not an EAGLE-2/Medusa-style multi-layer trained draft. We use it only as a low-cost control for the low-margin bottleneck; it is not a matched-latency comparison against other draft families, and we draw no ranking or deployment recommendation from it.

A.21 Additional Raw Corpora, Operators, Calibration, and Serving

Raw-text breadth. Table A.56 extends the 9B sweep to four Wikipedia languages, Chinese web text (FineWeb-2), raw code (CodeParrot), books (PG-19), and scientific prose (arXiv). All corpora are 100 chunks of 1024 tokens (512 prompt / 512 scored), tokenized with the Qwen3.5 tokenizer and scored against a 9B target cache. At 99% width the rejection ranges from 0.036 (code) to 0.093 (FineWeb-2 zh), against $q^\star_{\mathrm{FLOP}} = 0.0024$; the top-2 lower bound is above the threshold in every cell except the 50%-width code cell, which is point-above but not lower-bound-above. The floor is therefore not specific to English prose. Code and scientific prose have lower rejection than encyclopedic prose, but still one to two orders of magnitude above the cost frontier.

Table A.56: Raw-text breadth: greedy rejection q at 90%, 98%, and 99% retained FFN width on Qwen3.5-9B, with the target median top-1/top-2 log-margin and the fraction of tokens below 0.5 at 98% width. The last two columns give the post-hoc top-2 lower bound at 99% width and whether its confidence bound exceeds $q^\star_{\mathrm{FLOP}}$ ($\tau = 1$, $\gamma = 4$). All but one low-width code cell are certified; the code 50% cell is point-above but not lower-bound-above. Code has a larger median margin and fewer low-margin tokens, which explains its lower rejection; multilingual prose has 20–28% low-margin tokens, like the English corpora.

Corpus	q_{90}	q_{98}	q_{99}	med. m_{98}	% $m_{98} < 0.5$	$q_{\mathrm{lb},99}$	cert.
WikiText (en)	0.2080	0.0968	0.0677	1.38	21.9%	0.0455	✓
FineWeb (en)	0.2085	0.0959	0.0649	1.12	25.2%	0.0381	✓
C4 (en)	0.2228	0.1047	0.0717	1.00	28.4%	0.0411	✓
Wikipedia zh	0.2778	0.1308	0.0893	1.00	27.7%	0.0617	✓
Wikipedia de	0.2093	0.0997	0.0791	2.50	21.7%	0.0518	✓
Wikipedia fr	0.2275	0.0993	0.0663	1.44	22.9%	0.0471	✓
Wikipedia ja	0.2433	0.1099	0.0731	1.37	22.7%	0.0529	✓
FineWeb-2 zh	0.2738	0.1332	0.0933	1.06	27.8%	0.0645	✓
Code (raw)	0.0947	0.0477	0.0359	4.38	9.2%	0.0236	✓
PG-19 (books)	0.2273	0.1019	0.0730	1.00	27.3%	0.0489	✓
arXiv (science)	0.1379	0.0603	0.0422	1.63	19.4%	0.0260	✓

Table A.57 reports the 4k long-context check.

Table A.57: Long-context variant: 4096-token sequences (2048 scored continuation) built by concatenating four adjacent PG-19 or arXiv chunks. Rejection and the top-2 certificate are essentially unchanged from the 1024-token version; all cells are certified.

Corpus	Width	q	q_{lb}	$q^\star$	cert.
arxiv	0.9	0.1366	0.0879	0.0250	✓
arxiv	0.98	0.0654	0.0403	0.0049	✓
arxiv	0.99	0.0478	0.0301	0.0024	✓
pg19	0.9	0.2298	0.1457	0.0250	✓
pg19	0.98	0.1027	0.0677	0.0049	✓
pg19	0.99	0.0721	0.0471	0.0024	✓

Oracle / same-corpus calibration. Table A.58 asks whether calibration coverage is the only reason for the raw-text failure. Selecting channels on the first 50 chunks of the same corpus and evaluating on the last 50 lowers 98%-width rejection from 0.097/0.096/0.105 to 0.056/0.049/0.046 (WikiText/FineWeb/C4), and oracle selection on all 100 evaluation chunks lowers it further to 0.053/0.043/0.046. Both remain 9–11× above $q^\star = 0.0049$. Coverage helps but does not rescue the operator; the failure is not merely a calibration-distribution artifact. This is the post-hoc analogue of the uncovered-region bound: even with the best tested activation-score selection at uniform width for the evaluation corpus, the perturbation floor remains.

Table A.58: Oracle / same-corpus calibration upper bound. "Target-aligned" uses the mixed calibration corpus; "held-out" selects channels on the first 50 chunks and evaluates on the last 50 of the same corpus; "oracle" selects channels on all 100 evaluation chunks and evaluates on them (an upper bound, not a deployable method). Even the activation-score oracle remains above $q^\star$ at every width; the global subset/DP search is a separate stress test (Section 5.5). Calibration coverage reduces rejection somewhat but does not rescue the post-hoc operator.

Corpus	Selection	q_{50}	q_{90}	q_{98}
WikiText	target-aligned	0.5017	0.2080	0.0968
WikiText	held-out	0.4177	0.1508	0.0561
WikiText	oracle	0.3782	0.1269	0.0528
FineWeb	target-aligned	0.5047	0.2085	0.0959
FineWeb	held-out	0.4154	0.1213	0.0491
FineWeb	oracle	0.3911	0.1104	0.0433
C4	target-aligned	0.5538	0.2228	0.1047
C4	held-out	0.4168	0.1271	0.0463
C4	oracle	0.3967	0.1173	0.0462

Figure A.4 summarizes the eleven-corpus breadth check and the operator zoo at comparable cost.

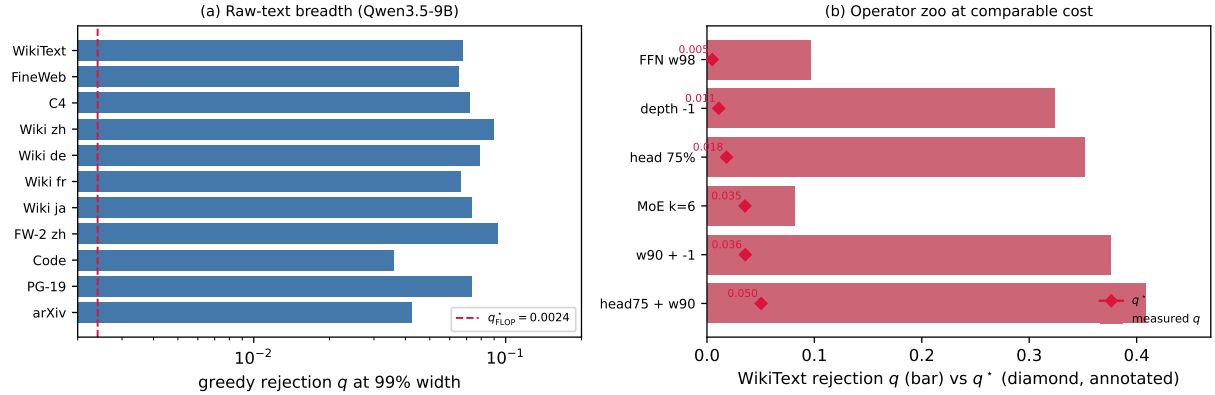


Figure A.4: Left: greedy rejection at 99% width across eleven raw corpora (log scale). The dashed line is the optimistic FLOP break-even $q^\star_{\mathrm{FLOP}} = 0.0024$. Every corpus is one to two orders of magnitude above it. Right: operator zoo at comparable cost on WikiText; bars are greedy rejection q_g and black ticks are the corresponding $q^\star$. FFN width, depth, attention heads, MoE active experts, and combined operators all remain above their own frontiers. The certificate q_{lb} is a lower bound on q_g, so the certified gap is at most the plotted gap.

Attention heads and combined operators. Table A.59 reports physical head pruning on Llama-3.1-8B. We remove whole KV groups (and their query heads), so the nominal FLOP ratio is realized rather than masked. Head-only reduction is much more destructive than FFN-width reduction at a comparable FLOP ratio: 25% head removal ($\rho = 0.866$) gives WikiText $q = 0.771$ and FineWeb $q = 0.757$, while 50% removal gives $q = 0.730/0.725$. The combined FFN-width+head scan does not open a window either: 70% width plus 25% head removal ($\rho = 0.730$) gives WikiText $q = 0.547$ and FineWeb $q = 0.595$. This closes the gap that attention heads were covered by the principle but not measured.

Table A.59: Attention-head pruning on Llama-3.1-8B. We physically remove whole KV groups (and their query heads) so the nominal FLOP ratio is realized. Head-only and combined FFN-width+head reductions are certified no-go on raw text; even 25% head removal leaves WikiText rejection at 0.77 against $q^\star \approx 0.009$.

Variant	ρ_{FLOP}	$q^\star$	MATH q	Wiki q	FineWeb q	q_{lb} Wiki	cert.
h=0.25	0.866	0.0568	0.6440	0.7706	0.7570	0.3765	✓
h=0.5	0.911	0.0371	0.3432	0.7302	0.7250	0.3534	✓
h=0.75	0.955	0.0182	0.1276	0.3513	0.3854	0.2106	✓
h=0.5, w=0.7	0.685	0.1456	0.3541	0.7675	0.7693	0.3714	✓
h=0.5, w=0.9	0.835	0.0706	0.3392	0.7468	0.7480	0.3600	✓
h=0.75, w=0.7	0.730	0.1220	0.1690	0.5467	0.5946	0.2925	✓
h=0.75, w=0.9	0.880	0.0504	0.1384	0.4078	0.4394	0.2358	✓

Table A.60 combines FFN width reduction with depth reduction on Qwen3.5-9B. Dropping one layer and keeping 90% width ($\rho = 0.914$) gives WikiText $q = 0.376$; dropping two layers and keeping 50% width ($\rho = 0.660$) gives $q = 0.634$. Every combined cell is above the frontier, and the raw-text lower bound is above the threshold in every cell. Combining two post-hoc operators does not create a profitable region.

Table A.60: Combined FFN-width and depth reduction on Qwen3.5-9B. ρ_{FLOP} scales the MLP share by the retained width and the decoder-layer share by the retained depth. Every combined cell is above the FLOP frontier on raw text; the raw-text lower bound is above the threshold except where noted. Combining two post-hoc operators does not open a window.

Width / drop	ρ_{FLOP}	$q^\star$	MATH q	Wiki q	FineWeb q	q_{lb} Wiki	cert.
0.5 / 1	0.678	0.1496	0.1590	0.6133	0.5545	0.3028	✓
0.5 / 2	0.660	0.1594	0.3688	0.6339	0.5997	0.3073	✓
0.7 / 1	0.796	0.0893	0.1213	0.4830	0.4314	0.2651	✓
0.7 / 2	0.774	0.0997	0.3741	0.5373	0.5009	0.2791	✓
0.9 / 1	0.914	0.0357	0.0965	0.3757	0.3303	0.2338	✓
0.9 / 2	0.888	0.0468	0.3789	0.4595	0.4408	0.2632	✓
0.98 / 1	0.961	0.0159	0.0873	0.3362	0.3018	0.2192	✓
0.98 / 2	0.934	0.0271	0.3785	0.4366	0.4286	0.2619	✓

Table A.61 measures the batch-1 cost of the non-FFN-width operators. Depth drop 1 gives $\rho_1 = 0.973$, $\tau_1 = 1.289$, cost 5.18; width 0.9 plus depth 1 gives 5.11; physical head pruning on Llama gives $\rho_1 = 1.018$ and cost 5.22; width 0.9 plus head 0.75 gives 5.38. MoE active-expert reduction to $k = 4$ gives $\rho_1 = 0.988$, $\tau_1 = 1.227$, cost 5.18. Cross-family measured costs are also above 5 in the padded implementation: Qwen3-4B at 50%/98% width gives 5.07/5.12, and Llama-3.1-8B gives 5.09/5.09. These are acceptance-independent implementation diagnostics (point estimates); the corresponding raw cells are roofline-bound independently (Appendix A.5). Head pruning is not cheaper in this implementation because the smaller attention GEMMs are still memory-bound; the nominal FLOP saving does not translate into a wall-clock saving. The MoE draft is also nearly as expensive as the target at batch 1, because the router, shared expert, attention, and dispatch remain. The measured ρ_1 is an implementation upper bound, not the nominal ρ_{FLOP}: a better kernel can narrow the gap, but the *rejection* columns are already above the corresponding $q^\star$ in every cell, so the no-go does not rely on this kernel efficiency.

Table A.61: Batch-1 measured cost for non-FFN-width and combined operators. ρ_1 and τ_1 are measured with KV caches on one A100 (512-token prefix, $\gamma = 4$); "cost" is $\gamma\rho_1 + \tau_1$, and a cost above 5 is a cost-only no-go even at $q = 0$. The table shows that depth, head, MoE active-expert, and combined operators are also cost-only no-go in a real implementation, not only in the nominal roofline model.

Operator	ρ_{FLOP}	ρ_1	τ_1	cost	cost-only no-go	Wiki q	evidence
FFN width 0.98	0.988	0.983	1.344	5.277	✓	0.0968	measured
Depth drop 1	0.973	0.973	1.289	5.183	✓	0.3237	measured
Width 0.9 + depth 1	0.914	0.957	1.285	5.114	✓	0.3757	measured
Head 0.75 (Llama)	0.955	1.018	1.151	5.225	✓	0.3513	measured
Width 0.9 + head 0.75	0.880	1.056	1.152	5.377	✓	0.4078	measured
MoE k=4	0.829	0.988	1.227	5.180	✓	0.1762	measured
Qwen3-4B width 0.5	0.669	0.989	1.116	5.072	✓	0.5521	measured
Qwen3-4B width 0.98	0.987	1.001	1.117	5.122	✓	0.1005	measured
Llama width 0.5	0.751	0.985	1.151	5.093	✓	0.5736	measured
Llama width 0.98	0.995	0.985	1.149	5.090	✓	0.0807	measured

Margin-aware calibration. Table A.62 tests whether the MSE failure is simply the wrong objective. We optimize the down projections with a top-1 cross-entropy plus a target-margin hinge on the calibration set. The calibration signal is the target's top-1/top-2 decision, not a target-generated response, so this is a post-hoc calibration baseline rather than distillation. A gentle run (100 steps, lr 2×10^{-6}, target-aligned corpus) raises WikiText/FineWeb rejection from 0.097/0.096 to 0.137/0.114; an aggressive run (300 steps, lr 10^{-5}, raw FineWeb) raises them to 0.84/0.89 and also damages MATH. A top-1-aware objective on the calibration distribution still does not transfer to raw text: the problem is the margin mass and coverage of the calibration distribution, not the MSE loss alone.

Table A.62: Post-hoc calibration and adaptation controls at 98% width. The first four rows optimize the down projections with a top-1 CE plus target-margin hinge using target top-1/top-2 labels, not target responses. "Target-aligned" uses 100 steps at lr 2×10^{-6} on the mixed calibration corpus; "raw FineWeb" uses the same budget on raw FineWeb training text; "aggressive" uses 300 steps at lr 10^{-5} on raw FineWeb. The last row is a 0.5M-token next-token adaptation of the down projections on raw WikiText (an adaptation-budget control, not distillation). None improves the raw-text frontier: the target-aligned run raises WikiText/FineWeb rejection from 0.097/0.096 to 0.137/0.114; the raw-FineWeb run raises them to 0.477/0.461; the aggressive run reaches 0.84/0.89; the down-projection adaptation reaches 0.448/0.201. This is an objective/coverage failure, not an MSE-only artifact.

Variant	MATH q	Wiki q	FineWeb q
Pruned (no recovery)	0.0157	0.0968	0.0959
Margin calibration, target-aligned	0.0339	0.1374	0.1141
Margin calibration, raw FineWeb	0.0219	0.4766	0.4613
Margin calibration, aggressive	0.0618	0.8387	0.8924
Down-proj adaptation (0.5M raw WikiText)	0.0237	0.4483	0.2006

Certificate tightness. Table A.63 reports the nested top-k bound at 98% width. The top-2 event explains 0.60 of WikiText rejection; top-3 explains 0.72, top-5 0.80, top-8 0.84, and the curve saturates at 0.86 by $K = 16$. The remaining rejection is caused by draft tokens outside the target top-256: the compressed draft promotes low-target-probability tokens. This is why the top-2 certificate is conservative but still non-vacuous; the unexplained mass is not a flaw in the event but the tail of the draft's own errors.

Table A.63: Nested top-k certificate at 98% width on Qwen3.5-9B. $q_{\mathrm{lb}}^{(K)}$ counts the event that any of the target's top-K tokens overtakes its top-1 token in the draft. It is monotone in K and reaches q_g at the full vocabulary. Top-3/top-5 events close part of the gap; the remaining gap is rejection by tokens outside the cached top-K or by the target's top-1 being displaced by a token that was not in the target top-K.

Set	$K=2$	$K=3$	$K=5$	$K=8$	$K=16$	$K=32$	$K=64$	$K=128$	$K=256$
math	0.013 (0.80)	0.013 (0.85)	0.014 (0.86)	0.014 (0.86)	0.014 (0.86)	0.014 (0.86)	0.014 (0.86)	0.014 (0.86)	0.014 (0.86)
wikitext	0.064 (0.66)	0.074 (0.76)	0.080 (0.83)	0.082 (0.85)	0.084 (0.87)	0.084 (0.87)	0.084 (0.87)	0.084 (0.87)	0.084 (0.87)
fineweb	0.057 (0.60)	0.069 (0.72)	0.077 (0.80)	0.080 (0.84)	0.082 (0.85)	0.082 (0.86)	0.082 (0.86)	0.082 (0.86)	0.082 (0.86)

Values are $q_{\mathrm{lb}}^{(K)}$ (coverage $q_{\mathrm{lb}}^{(K)}/q_g$).

Table A.64: Clean GPU-kernel zero-FFN decomposition for Qwen3.5-9B. "Zero-MLP ρ" replaces the MLP path with a zero path; "zero-MLP cost" is $4\rho_{\mathrm{zero}} + \tau_1$. Under the clean kernel-time protocol the zero-MLP cost is 3.80–3.83 < 5, so a free MLP would reopen the cost condition; this is a counterfactual about a fully removed MLP, not a width-only kernel. The measured clean kernel-time cost (padded narrow-MLP implementation) is 4.50 at 50%, 4.97 at 90%, and 5.03–5.05 at 98–99%; the 10%–50% no-go is rejection, and the high-width cells are roofline-bound with the measured cost as an additional independent failure. Earlier wall-clock ceilings (5.10 on 9B) were launch-bound and are superseded by this kernel-time table.

Width	ρ_1	zero-MLP ρ	τ_1	cost	zero-MLP cost	cost-only no-go
0.5	0.868	0.693	1.025	4.497	3.796	–
0.9	0.981	0.688	1.044	4.970	3.797	–
0.98	0.998	0.696	1.043	5.033	3.828	✓
0.99	1.002	0.696	1.041	5.051	3.827	✓

Kernel ceiling. Table A.64 uses the module decomposition to bound what a fused MLP kernel could achieve. The protocol pre-clones the 512-token prefix cache outside the timed region and reports GPU kernel time from windows of back-to-back passes; the wall-clock Python loop is launch-bound at batch 1 and is not used for the decomposition (Appendix A.6). On hybrid Qwen3.5-9B at 98% width the realized kernel cost is $5.03 > 5$, while replacing the MLP with a zero-cost path gives $\rho_0 = 0.696$ and a zero-FFN cost of $3.83 < 5$. The same pattern holds on every measured target: realized costs are 5.004–5.28, zero-FFN costs 3.1–4.1. A hypothetical free FFN would therefore put the cost below 5 on every target; a *cost-only* window would reopen on all six; even K2, whose raw $q = 0.079$ is below $q_0^\star = 0.124$, is R rather than R-free. The zero-FFN counterfactual is not a ceiling on any architecture, but it is a fully removed MLP, not a width-only kernel. The measured no-go at high width is roofline-bound (R) by rejection, with the realized-kernel cost as an additional independent failure; a production MLP kernel is portable (Proposition 8) and cannot reopen an R cell. Earlier wall-clock ceilings (5.10 on 9B, 3.35 on Llama) were launch-bound and are superseded.

Fused-kernel prototype. We also implemented a Triton prototype that fuses the gate/up projections and SiLU into one GEMV kernel and the down projection into a second GEMV kernel. On one 9B layer it ran at 0.42× the speed of cuBLAS at 50% width and 0.72× at 98% width, achieving 24%–41% of the A100 HBM roofline. A more carefully tuned kernel could close the gap to cuBLAS, but the zero-FFN decomposition above shows that the residual non-MLP path is still substantial, but it is below the break-even on most targets, so we do not claim a kernel ceiling. We therefore do not rely on a particular kernel implementation for the no-go.

End-to-end serving. Table A.65 reports fair greedy speculative loops with exact KV-cache rollback on three dense models (batch 1, 256-token prefix, 64 generated tokens, $\gamma = 4$). At 98% width the compressed draft reaches 0.38× target-only tokens/s on Llama-3.1-8B, 0.35× on Qwen3-4B, and 0.32× on K2-Horizon-3.7B; the 50%-width Llama draft is 0.19×. The loops are consistent with the roofline no-go across three model families, with the caveat that only the 9B identity loop is a control: the draft is nearly as expensive as the target, and four draft passes plus verification cost more than target-only decoding. Qwen3.5's hybrid linear-attention cache does not support exact rollback, so we do not claim a Qwen3.5 end-to-end number.

Table A.65: Fair end-to-end greedy speculative decoding with exact KV-cache rollback on three dense models (Llama-3.1-8B, Qwen3-4B, K2-Horizon-3.7B; batch 1, 256-token prefix, 64 generated tokens, $\gamma = 4$). "Speedup" is speculative tokens/s over target-only tokens/s on the same prompts and GPU. All compressed drafts are below 1×; the 98%-width drafts are 0.32–0.38×, and the 50%-width Llama draft is 0.19×. The 9B identity control reaches only 0.607× (0.673× at $\alpha = 1.0$), so the harness is launch-bound; the dense models have no identity control. This is a reference KV-cache loop, not a production throughput claim and not the paper's general cost claim.

Draft	γ	Target tok/s	SD tok/s	Speedup	α	$\bar{\ell}$
K2-3.7B, 98% width	4	38.8	12.4	0.320	0.246	1.99
Llama-3.1-8B, 50% width	4	47.4	8.9	0.188	0.020	1.10
Llama-3.1-8B, 98% width	4	47.9	18.3	0.381	0.347	2.40
Qwen3.5-9B identity, $\gamma = 4$	4	25.9	15.7	0.607	0.892	4.57
Qwen3.5-9B identity, 24-token drift	4	26.1	16.5	0.633	0.975	4.90
Qwen3.5-9B identity, short generation	4	24.3	16.3	0.673	1.000	5.33
qwen35_9b_reprefill, 50% width	4	25.2	12.8	0.506	0.603	3.41
Qwen3.5-9B exact re-prefill	4	25.3	16.5	0.652	0.825	4.47
qwen35_9b, 50% width	4	27.1	12.1	0.448	0.532	3.16
qwen35_9b, 98% width	4	26.2	13.0	0.495	0.632	3.56
Qwen3-4B, 98% width	4	34.2	12.0	0.351	0.287	2.18

Powered K2 replication. Table A.66 repeats the K2-Horizon-3.7B check with 100 MATH prompts and 100 WikiText chunks (the earlier check used 30/50). All widths from 50% to 99% are certified no-go on both sets, including the low-width MATH cells that were only point-above in the smaller sample.

Table A.66: K2-Horizon-3.7B powered replication (100 MATH prompts, 100 WikiText chunks; target and compressed draft are the same model). Raw WikiText is certified no-go at every compressed width; the larger sample confirms the earlier 30/50-prompt qualitative check.

Width	ρ_{FLOP}	$q^\star$	MATH q	Wiki q	q_{lb} Wiki	cert.
0.5	0.679	0.1487	0.3211	0.4500	0.2414	✓
0.7	0.808	0.0836	0.1896	0.3019	0.1809	✓
0.9	0.936	0.0263	0.0919	0.1579	0.1041	✓
0.95	0.968	0.0130	0.0678	0.1160	0.0776	✓
0.98	0.987	0.0052	0.0450	0.0791	0.0523	✓
0.99	0.994	0.0026	0.0327	0.0630	0.0407	✓

A.22 Reproducibility and Model Versions

The acceptance pipeline, compressed-draft sweep, certificate, cost frontier, and latency benchmarks are deterministic given the recorded seeds. All confidence intervals use 2,000 sequence-cluster bootstrap resamples with seed 0; the certificate is computed per token and its interval uses the same resampling. Channel importance is estimated on 200k target-aligned tokens whose token IDs are fixed by the seed-0 sampler and retained in the artifact (they are not listed in the paper because they are 200k integers); layer-wise ridge reconstruction uses 16k calibration tokens and a 4k held-out split, with the ridge scale selected from $\{0, 10^{-6}, \ldots, 1\}$ times the mean activation covariance diagonal by held-out layer MSE. The margin-aware calibration uses 100k tokens, two sequences per step, seed 0, and only target top-1/top-2 labels (no target responses); the gentle and aggressive budgets are stated in Table A.62. Evaluation is deterministic teacher forcing on fixed sequences. Per-token match vectors, post-hoc bound outputs, and raw result JSON/NPZ files are retained with the project. The cost taxonomy of Appendix A.5 is regenerated by `paper/tools/generators/make_cost_layer_assets.py` from `out/paper_stats/review12.json`, `out/paper_stats/kernel_efficiency.json`, and the Llama MATH clean measurement `out/latency/clean_llama31-8b_math_widths.json`; the script writes `out/paper_stats/cost_layer.json` and the three threshold tables. The protocol is a small set of deterministic forward passes with fixed seeds; private cluster paths are omitted, and the public checkpoints are identified below. Table A.67 maps each paper-facing table to its script and result file; the mapping is the check we use before a submission build.

Table A.67: Artifact map. Each paper-facing table is regenerated by a script from the listed result file; paths are relative to the repository root. The mapping is the minimum needed to re-derive a table without guessing which result file it came from.

Artifact	Script	Result file	Paper table
Width sweep, certificate	`infra/joint_scan.py`	`out/rank_scan/`	7, A.4
High-width floor	`paper/tools/generators/make_review13_assets.py`	`out/paper_stats/review13.json`	9, 8
Cost decomposition	`infra/latency_clean.py`	`out/latency/clean/`	A.64, A.20
Cost taxonomy	`paper/tools/generators/make_cost_layer_assets.py`	`out/paper_stats/cost_layer.json`	A.11–A.13
Screen calibration	`infra/loco_calibration.py`	`out/paper_stats/loco_calibration.json`	A.79, A.82
Family statistics	`infra/family_stats.py`	`out/paper_stats/family_stats.json`	A.86, A.87
Serving loop	`infra/serve_sd_loop.py`	`out/serve/`	A.65, A.92
Quantization boundary	`infra/quant_kernel.py`	`out/quant/`	A.25, A.26
Global post-hoc search	`paper/tools/generators/make_global_baseline.py`	`out/paper_stats/global_posthoc.json`	10, A.37

Confirmatory freeze and cost protocol. The $36 + 9$ confirmatory family is frozen in `out/paper_stats/family_stats.json` and `out/paper_stats/confirm_family.json`; its two-level family statistics are regenerated by `infra/family_stats.py`. This is a dated lab-notebook freeze, not a public pre-registration. Cost rows report the median of 20 interleaved draft/target timings after warm-up at a 512-token prefix, batch 1, on one A100-SXM4-80GB; the per-run minimum and maximum are retained in the measurement JSON, and the Python-loop wall clock is reported separately as an upper bound. The screen coefficients and per-cell predictions are regenerated from `out/paper_stats/loco_calibration.json` by `paper/tools/generators/make_review7_assets.py`. Code and per-token caches are not attached to this submission; the artifact map and protocol are the reproducibility contract.

Model versions. Qwen3.5-9B and Qwen3.5-4B are the public Qwen3.5 checkpoints; Qwen3-4B is Qwen3-4B-Instruct-2507; Llama-3.1-8B is Llama-3.1-8B-Instruct; K2-Horizon-3.7B is the public K2-Horizon checkpoint. All use their released tokenizers. Table A.68 records the config hash prefix and the CUDA stack used for the reported runs. Full weight-shard hashes are available in the checkpoint release manifests where provided; the config hash is the compact identifier used here. The Qwen3.5 pretraining data and token counts are not released, so the off-the-shelf capacity scan is not a controlled scaling law.

Table A.68: Checkpoint and environment identifiers. The config SHA-256 prefix identifies the released `config.json` used for the parameter accounting; it is not a hash of the weight shards. The runtime is one A100-SXM4-80GB with the listed CUDA stack.

Model	Config SHA-256 (first 16)	Role
Qwen3.5-9B	`d0883072e01861ed`	main target
Qwen3.5-27B	`f8d190c5b89c1521`	cross-scale target
Qwen3.5-4B	`ddc63e1c717afa86`	cross-scale target
Qwen3-4B-Instruct-2507	`5beea1a4a34c6278`	cross-family target
Llama-3.1-8B-Instruct	`29e4c210b0d6ac17`	cross-family target
K2-Horizon-3.7B	`d08ee6a1798db2a0`	cross-family target
Runtime: PyTorch 2.13.0+cu130; CUDA 13.0; cuDNN 92000; driver 580.159.04.		

A.23 MoE Compression Details

We use Qwen3.5-35B-A3B (40 layers, 256 routed experts, top-8, expert width 512, one shared expert per layer). The target cache is rebuilt from the uncompressed 35B model on the same fixed sequences, so the full model has $q = 0$. The active-expert scan reduces k from 8 to $6, 4, 3, 2, 1$; the expert-deletion control

removes the lowest-calibration-usage 50% and 75% of experts while keeping $k = 8$. The latter does not reduce per-token FLOPs and is reported as a memory-only axis. All intervals are sequence-cluster bootstrap intervals.

Table A.69: Active-expert (top-k) reduction on Qwen3.5-35B-A3B. The target and draft are the same MoE; the full model ($k = 8$) has $q = 0$ on the rebuilt cache. ρ_{FLOP} is the active-expert FLOP ratio from (13); $q^{\star}_{\mathrm{FLOP}}$ uses $\tau = 1$ and $\gamma = 4$. Raw-text cells are certified when the lower post-hoc bound exceeds the threshold. The cost saving is modest because the active routed-expert set is about one third of active FLOPs.

k	ρ_{FLOP}	$q^{\star}_{\mathrm{FLOP}}$	MATH q	Wiki q	FineWeb q	q_{lb} Wiki	q_{lb} FineWeb	cert. raw
8	1.000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	–
6	0.915	0.0354	0.0178	0.0815	0.0741	0.0581	0.0496	✓
4	0.829	0.0735	0.0385	0.1762	0.1606	0.1209	0.1106	✓
3	0.786	0.0938	0.0598	0.2509	0.2283	0.1621	0.1478	✓
2	0.744	0.1150	0.1184	0.3826	0.3406	0.2231	0.2026	✓
1	0.701	0.1371	0.4956	0.7924	0.7382	0.3629	0.3461	✓

Table A.70 reports the expert-deletion control.

Table A.70: Expert deletion at fixed $k = 8$ on Qwen3.5-35B-A3B. This reduces parameters and expert memory (column ρ_{mem}) but not the dominant per-token expert FLOPs, so it is a memory-only control, not a compute-reducing operator. Rejection is higher than active-expert reduction at comparable memory.

Keep	ρ_{mem}	MATH q	Wiki q	FineWeb q	q_{lb} Wiki	q_{lb} FineWeb
0.5	0.535	0.0279	0.3540	0.3427	0.2087	0.2015
0.25	0.303	0.0759	0.5116	0.5191	0.2766	0.2796

The active-expert rows are certified: the lower certificate bound exceeds $q^{\star}_{\mathrm{FLOP}}$ on both raw corpora at every $k' < 8$. The cost saving is modest because the active routed-expert set is about one third of the target's active FLOPs; this is why the MoE frontier is less steep than the FFN-width frontier. The nominal ratio uses the active-expert FLOP share ϕ_E; a $\pm 20\%$ variation in ϕ_E changes $q^{\star}$ by less than the observed gap (for example, $k = 4$ has $q = 0.176$ against $q^{\star} = 0.074$), so the qualitative no-go is robust to this share. Expert deletion at fixed k reduces parameters and expert memory (to 53.5% and 30.3% of total parameters) but leaves the dominant per-token expert FLOPs unchanged; its rejection is higher than the active-expert rows at comparable memory, and it is not a compute-reducing operator.

A.24 Depth Reduction

We drop the last $1, 2, 4, 8, 16$ of the 32 Qwen3.5-9B decoder layers and evaluate the same caches. The nominal FLOP ratio is the decoder-layer share times the retained depth. Every dropped-depth cell is above the FLOP frontier, and the certificate is above the threshold at every depth on raw text. This is a second compute-reducing operator outside the FFN-width family; it is much more destructive than width reduction at a comparable FLOP ratio, consistent with the support-selection principle: a layer is not a coordinate that can be removed without changing the residual stream.

Table A.71: Depth reduction on Qwen3.5-9B: drop the last L of 32 decoder layers. ρ_{FLOP} is the decoder-layer FLOP share times retained depth; $q^{\star}$ uses $\tau = 1$ and $\gamma = 4$. Every dropped-depth raw-text cell is certified no-go.

Drop	ρ_{FLOP}	$q^{\star}_{\mathrm{FLOP}}$	MATH q	Wiki q	FineWeb q	q_{lb} Wiki	cert. raw
1	0.973	0.0110	0.0852	0.3237	0.2940	0.2139	✓
2	0.946	0.0223	0.3784	0.4306	0.4228	0.2595	✓
4	0.891	0.0456	0.6228	0.6173	0.6059	0.3196	✓
8	0.782	0.0960	0.8320	0.8436	0.8011	0.3756	✓
16	0.564	0.2163	0.8913	0.9613	0.9326	0.4220	✓

A.25 Full Sweeps of Qwen3.5-27B, Qwen3-4B, and Llama-3.1-8B

The main text reports the 9B sweep. Here we give the full width sweeps that fill the low-width region for the larger and cross-family models. The 27B sweep uses the same fixed evaluation sequences as the 9B sweep and its own target cache; its MATH continuations were generated by the 35B target, so it is a fixed-continuation control rather than a 27B-generated set. The Qwen3-4B and Llama sweeps use their own target caches and tokenizers. All three sweeps use the same activation-aware channel score; the 27B grid has 11 widths (Table A.72), and the two finest 9B widths are not run on 27B. The raw-text certificate is above the frontier at every compressed width in all three sweeps; the in-domain MATH window is narrow and model-dependent.

Table A.72: Qwen3.5-27B full width sweep on the same fixed evaluation sequences. The MATH continuations were generated by the 35B target, so this is a fixed-continuation control. Raw-text cells are certified at every compressed width; all $q^\star$ values use $\tau = 1$ and $\gamma = 4$.

Width	ρ_{FLOP}	$q^\star_{\mathrm{FLOP}}$	MATH q	Wiki q	FineWeb q	q_{lb} Wiki	q_{lb} FineWeb	cert.	raw
0.05	0.366	0.3638	0.5444	0.8441	0.8348	0.3971	0.3937	✓	
0.1	0.399	0.3352	0.3389	0.7365	0.7463	0.3550	0.3549	✓	
0.2	0.466	0.2832	0.1918	0.6312	0.6415	0.3246	0.3183	✓	
0.3	0.533	0.2367	0.1374	0.5538	0.5520	0.2921	0.2878	✓	
0.5	0.666	0.1561	0.0881	0.4119	0.4163	0.2318	0.2316	✓	
0.7	0.800	0.0875	0.0563	0.2835	0.2889	0.1737	0.1734	✓	
0.85	0.900	0.0418	0.0337	0.1948	0.2035	0.1246	0.1241	✓	
0.9	0.933	0.0275	0.0257	0.1597	0.1713	0.1028	0.1066	✓	
0.95	0.967	0.0135	0.0166	0.1213	0.1281	0.0778	0.0781	✓	
0.98	0.987	0.0054	0.0098	0.0802	0.0847	0.0497	0.0484	✓	
0.99	0.993	0.0027	0.0069	0.0566	0.0598	0.0343	0.0317	✓	

Table A.73: Qwen3-4B full width sweep (different family and tokenizer). Raw-text cells are certified at every compressed width; the in-domain MATH window is narrow. All $q^\star$ values use $\tau = 1$ and $\gamma = 4$.

Width	ρ_{FLOP}	$q^\star_{\mathrm{FLOP}}$	MATH q	Wiki q	FineWeb q	q_{lb} Wiki	q_{lb} FineWeb	cert.	raw
0.05	0.365	0.3645	1.0000	0.9982	0.9904	0.4738	0.4662	✓	
0.1	0.398	0.3358	0.8425	0.9404	0.9335	0.4448	0.4438	✓	
0.2	0.465	0.2836	0.4029	0.7607	0.7851	0.3653	0.3862	✓	
0.3	0.532	0.2371	0.2846	0.6740	0.6989	0.3369	0.3519	✓	
0.5	0.666	0.1563	0.1675	0.5521	0.5787	0.2906	0.3090	✓	
0.7	0.799	0.0876	0.1097	0.3920	0.3960	0.2260	0.2345	✓	
0.8	0.866	0.0566	0.0849	0.3048	0.3069	0.1869	0.1934	✓	
0.85	0.900	0.0418	0.0720	0.2601	0.2665	0.1666	0.1733	✓	
0.9	0.933	0.0275	0.0540	0.2103	0.2185	0.1398	0.1455	✓	
0.95	0.967	0.0136	0.0356	0.1547	0.1583	0.1056	0.1093	✓	
0.98	0.987	0.0054	0.0224	0.1005	0.1038	0.0705	0.0718	✓	
0.99	0.993	0.0027	0.0155	0.0676	0.0697	0.0477	0.0488	✓	

Table A.74: Llama-3.1-8B full width sweep (different family and tokenizer). Raw-text cells are certified at every compressed width; the in-domain MATH window is narrow. All $q^\star$ values use $\tau = 1$ and $\gamma = 4$.

Width	ρ_{FLOP}	$q^\star_{\text{FLOP}}$	MATH q	Wiki q	FineWeb q	q_{lb} Wiki	q_{lb} FineWeb	cert. raw
0.05	0.286	0.4405	0.9895	1.0000	0.9960	0.4769	0.4758	✓
0.1	0.324	0.4023	0.9157	0.9906	0.9772	0.4614	0.4500	✓
0.2	0.399	0.3351	0.5830	0.8613	0.8984	0.4092	0.4364	✓
0.3	0.474	0.2770	0.3249	0.7416	0.7763	0.3609	0.3809	✓
0.5	0.624	0.1797	0.1567	0.5736	0.6013	0.2970	0.3080	✓
0.7	0.775	0.0996	0.0871	0.3642	0.3638	0.2124	0.2163	✓
0.8	0.850	0.0641	0.0639	0.2664	0.2622	0.1680	0.1672	✓
0.85	0.887	0.0472	0.0525	0.2225	0.2172	0.1451	0.1421	✓
0.9	0.925	0.0310	0.0430	0.1771	0.1752	0.1182	0.1170	✓
0.95	0.962	0.0153	0.0299	0.1234	0.1225	0.0872	0.0845	✓
0.98	0.985	0.0060	0.0189	0.0807	0.0767	0.0557	0.0532	✓
0.99	0.992	0.0030	0.0142	0.0597	0.0582	0.0430	0.0403	✓

27B self-generated MATH control. The main 27B sweep uses MATH continuations generated by the 35B target. To check whether this affects the in-domain conclusion, we generate 100 MATH continuations with the 27B model itself and repeat widths 0.50, 0.90, and 0.98. Rejection is 0.0945, 0.0262, and 0.0094, against $q^\star = 0.1561$, 0.0275, and 0.0054: the 50–90% widths pass the FLOP frontier, while 98% is above it. This confirms that the in-domain window is model- and continuation-dependent; it does not affect the raw-text no-go. The clean kernel-time cost is 4.23–4.84 at 50–90%, with implied break-even rejection 0.083/0.050/0.016 below the measured 0.088/0.056/0.026: the 50–90% cells are K at the clean cost, and 98% by the realized cost $5.07 > 5$. The 27B zero-MLP cost is 3.50, so the zero-FFN counterfactual would reopen a window ($q = 0.080 < q_0^\star = 0.179$); it is not a ceiling. This replaces the earlier fixed-continuation caveat with a direct self-generated control.

A.26 Third Raw Corpus: C4

To test whether the raw-text floor is specific to WikiText/FineWeb, we repeat the 9B sweep on 100 raw C4 chunks (the same 1024-token continuation format). C4 is rejected at 0.105 at 98% width and 0.072 at 99%, against $q^\star = 0.0049$ and 0.0024; the lower certificate bound is above the threshold at every tested width (Table A.75). This is a third independent raw corpus, although all three are English prose and we do not claim coverage of code, multilingual, or long-context raw distributions.

Table A.75: Third raw corpus: C4 (100 raw chunks, Qwen3.5-9B target and compressed draft). The same channel-deletion floor appears: raw C4 is rejected at 10.5% at 98% width and 7.2% at 99%, against $q^\star = 0.0049/0.0024$ ($\tau = 1$, $\gamma = 4$); every tested width is certified.

Width	ρ_{FLOP}	$q^\star_{\text{FLOP}}$	C4 q	q_{lb}	cert.
0.05	0.422	0.3168	0.8375	0.3961	✓
0.1	0.452	0.2933	0.7954	0.3738	✓
0.2	0.513	0.2498	0.7263	0.3550	✓
0.3	0.574	0.2102	0.6728	0.3363	✓
0.5	0.696	0.1400	0.5538	0.2906	✓
0.7	0.817	0.0791	0.3899	0.2188	✓
0.8	0.878	0.0513	0.3070	0.1808	✓
0.85	0.909	0.0379	0.2663	0.1595	✓
0.9	0.939	0.0250	0.2228	0.1370	✓
0.95	0.970	0.0123	0.1644	0.1001	✓
0.98	0.988	0.0049	0.1047	0.0619	✓
0.99	0.994	0.0024	0.0717	0.0411	✓

A.27 Margin Mechanism Analysis

Table A.76 quantifies the objective mismatch discussed in the main text. At 98% width, the target median top-1/top-2 log-margin is 11.9 on MATH but only 1.4 on raw WikiText and 1.1 on raw FineWeb. The fraction of tokens with margin below 0.5 is 3.3% on MATH versus 21.9% and 25.2% on raw text. Consequently, the lowest-margin quintile has rejection 7.8% on MATH but 32.7% and 29.2% on raw text, even though the mean perturbation in that quintile is similar (0.45 versus 0.36 and 0.30). Ridge reconstruction changes the layer MSE but not the margin distribution: q and q_{lb} move by less than 0.001. This is the mechanism behind the negative result: raw text has much more probability mass near the top-1 decision boundary, and an MSE criterion that weights tokens by output energy does not move that mass.

Table A.76: Mechanism analysis at 98% width on Qwen3.5-9B. "Low quintile" is the lowest target top-1/top-2 log-margin quintile. Raw text has much more mass near the decision boundary than MATH: the same channel deletion produces a similar mean $|\Delta_m|$ but a much higher rejection rate in that quintile. Ridge reconstruction changes neither the margin distribution nor the rejection.

Variant	Set	Median m	% $m < 0.5$	Low-q reject	Low-q $\lvert\Delta_m\rvert$	q	q_{lb}/q
pruned	MATH	11.9	3.3%	7.8%	0.454	0.0157	0.80
pruned	Wiki	1.4	21.9%	32.7%	0.365	0.0968	0.66
pruned	FineWeb	1.1	25.2%	29.2%	0.302	0.0959	0.60
recon	MATH	11.9	3.3%	7.8%	0.446	0.0156	0.81
recon	Wiki	1.4	21.9%	32.6%	0.363	0.0974	0.66
recon	FineWeb	1.1	25.2%	29.2%	0.303	0.0960	0.61

Entropy composition. Table A.77 separates composition from shape. The target margin and target entropy are strongly negatively correlated on every corpus (Spearman −0.80 WikiText, −0.74 FineWeb, −0.69 C4, −0.99 MATH). The raw/MATH gap in low-margin mass is mostly a difference in the entropy distribution: MATH is concentrated at very low entropy, while raw text has a long high-entropy tail. Reweighting MATH tokens to the WikiText entropy histogram raises its low-margin mass from 0.033 to 0.343; reweighting WikiText to the MATH histogram lowers it from 0.219 to 0.008. At fixed entropy MATH has equal or higher low-margin mass, so the raw-text result is not a raw-specific calibration failure. A second-feature screen that adds corpus mean entropy does not improve the leave-one-corpus-out MAE (0.0618 versus 0.0466 at 50% width), which is why the deployment tool remains the one-parameter mass screen: entropy is the mechanism, but mass is the quantity that enters the rejection bound.

Table A.77: Margin mass is an entropy-composition effect at 98% width. Top: median margin, low-margin mass, mean target entropy, and Spearman correlation between margin and entropy. Bottom: reweighting one corpus to the other's entropy histogram. At fixed entropy MATH has equal or higher low-margin mass.

Corpus	med. m	$\Pr(m < 0.5)$	mean entropy	Spearman(m, H)
WikiText	1.38	0.219	1.556	-0.796
FineWeb	1.12	0.252	1.815	-0.743
C4	1.00	0.284	2.050	-0.689
MATH	11.87	0.033	0.136	-0.995
MATH reweighted to Wiki entropy: 0.343; Wiki reweighted to MATH: 0.008				

Cross-model and per-token checks. Table A.78 repeats the WikiText check at 98% width on three families. The low-margin mass is 0.20–0.25, the margin–entropy Spearman correlation is −0.78 to −0.83, and the low-quintile rejection is 0.29–0.35 on the Qwen/Llama models; K2 has a higher overall rejection but the same mechanism. A per-token logistic fit on standardized $(m, |\Delta_m|)$ gives 5-fold cross-validated AUC 0.914 (WikiText), 0.897 (FineWeb), and 0.990 (MATH); the margin coefficient is large and negative and the perturbation coefficient is positive, so the low-margin mass is the dominant variable after controlling for the perturbation.

Table A.78: Cross-model mechanism check on WikiText at 98% width. The low-margin mass, the margin-entropy correlation, and the low-quintile rejection have the same sign and rough magnitude across the Qwen families and the dense Llama model. K2 raw cells are reported separately in Table A.66; the 30/50 target-generated K2 check is not used for this raw comparison.

Model	q	median m	$L = \Pr(m < 0.5)$	Spearman(m, H)	low-quintile q
Qwen3.5-9B	0.097	1.38	0.219	−0.796	0.327
Qwen3.5-4B	0.088	1.25	0.230	−0.778	0.312
Qwen3-4B	0.101	1.63	0.204	−0.833	0.354
Llama-3.1-8B	0.081	1.25	0.248	−0.778	0.286

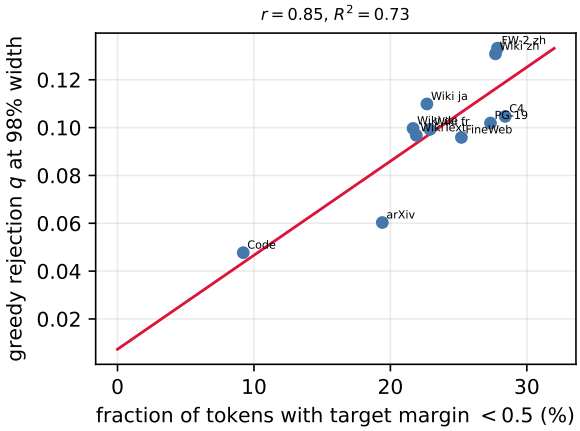


Figure A.5: Raw-text rejection at 98% width versus the fraction of tokens with target top-1/top-2 margin below 0.5, across eleven corpora. The low-margin mass predicts the cross-corpus ordering: Pearson $r = 0.85$, Spearman 0.75; a one-parameter linear fit has $R^2 = 0.73$ and MAE 0.011. The margin distribution is therefore a quantitative diagnostic, not only a narrative explanation.

A.28 Leave-One-Corpus-Out and Deployment Diagnostic

The $r = 0.85$ correlation in Figure A.5 is an in-sample fit. We therefore refit the one-parameter model on ten raw corpora and predict the held-out corpus. Table A.79 reports the result. At 98% width the leave-one-corpus-out MAE is 0.0138 (max error 0.027), and all eleven held-out corpora are predicted above $q^\star = 0.0049$; at 50% width the MAE is 0.0466 and all eleven are again predicted above $q^\star = 0.140$, including code (predicted 0.231, measured 0.221, $q^\star = 0.140$). No raw-text classification failure occurs at any measured width. Adding the mean absolute margin perturbation $|\Delta_m|$ as a second feature lowers the MAE to 0.027 at 50% width and 0.011 at 98% (Table A.79).

We then hold out two deployment-like suites and fit on the eleven raw corpora only. Table A.80 shows MT-Bench (chat/instruction) and SWE-bench (agentic issue-to-patch). At 98% width the mass-only predictions are 0.042 (MT-Bench) and 0.053 (SWE) against measured 0.063 and 0.095; both are above $q^\star$. At 50% width MT-Bench is the largest absolute error (0.214 predicted versus 0.397 measured), but the prediction is still above $q^\star = 0.140$. The only cell predicted below break-even is in-domain MATH at 50% width (0.102 versus $q^\star = 0.140$), where the measurement is also below (0.130); this is the in-domain window, not a raw-text failure. A small-prefix simulation ($n = 1024$ tokens, 200 draws) predicts above $q^\star$ in 100% of draws for every raw-text and deployment cell at every measured width; the MATH 50% cell is below in 100% of draws. The operational recipe is therefore: draw a small deployment prefix, estimate the margin mass (and optionally $|\Delta_m|$), compare the predicted q with $q^\star$, and if the screen is below $q^\star$ measure the certificate directly on that sample before declaring a window.

Table A.79: Leave-one-corpus-out validation of the margin-mass diagnostic. For each width, a linear model $q = a_w + b_w L$ is fit on ten raw corpora and predicts the eleventh; L is the fraction of target tokens with top-1/top-2 margin below 0.5. "2-feat" adds the mean absolute margin perturbation $|\Delta_m|$ measured on the same tokens. At every width all eleven raw corpora are predicted above $q^\star_{\mathrm{FLOP}}$, including code at 50% width (predicted 0.231, measured 0.221, $q^\star = 0.140$).

Width	$q^\star_{\mathrm{FLOP}}$	MAE	RMSE	max	MAE (2-feat)	pred $> q^\star$	actual $> q^\star$
0.50	0.1400	0.0466	0.0601	0.1152	0.0270	11/11	11/11
0.90	0.0250	0.0254	0.0287	0.0489	0.0150	11/11	11/11
0.95	0.0123	0.0181	0.0204	0.0384	0.0145	11/11	11/11
0.98	0.0049	0.0138	0.0154	0.0270	0.0114	11/11	11/11
0.99	0.0024	0.0101	0.0114	0.0198	0.0073	11/11	11/11

Table A.80: Deployment-like held-out suites, predicted by the eleven-corpus margin-mass fit (no fitting on the suite). MT-Bench is chat/instruction, SWE-bench is an agentic issue-to-patch prefix, and MATH is the in-domain control. The mass-only screen under-predicts the two deployment suites at 50% width (MT-Bench -0.183) but is still above $q^\star$; the two-feature refinement is closer at high widths. The in-domain MATH cell at 50% is the only case predicted below $q^\star$, and the measurement agrees.

Width	Suite	L	q	q_{pred}	q_{pred} (2-feat)	$q^\star$	pred $> q^\star$
0.50	MT-Bench	0.0873	0.3972	0.2138	0.5038	0.1400	True
0.50	SWE-bench	0.1152	0.3050	0.2710	0.3417	0.1400	True
0.50	MATH	0.0325	0.1297	0.1019	0.4017	0.1400	False
0.90	MT-Bench	0.0873	0.1465	0.0888	0.1438	0.0250	True
0.90	SWE-bench	0.1152	0.1530	0.1127	0.1593	0.0250	True
0.90	MATH	0.0325	0.0399	0.0420	0.0466	0.0250	True
0.95	MT-Bench	0.0873	0.0984	0.0625	0.0919	0.0123	True
0.95	MATH	0.0325	0.0260	0.0279	0.0256	0.0123	True
0.98	MT-Bench	0.0873	0.0629	0.0416	0.0550	0.0049	True
0.98	SWE-bench	0.1152	0.0946	0.0526	0.0829	0.0049	True
0.98	MATH	0.0325	0.0157	0.0201	0.0110	0.0049	True
0.99	MT-Bench	0.0873	0.0431	0.0321	0.0362	0.0024	True
0.99	SWE-bench	0.1152	0.0669	0.0392	0.0588	0.0024	True
0.99	MATH	0.0325	0.0112	0.0182	0.0063	0.0024	True

Calibration, coverage, and decision errors. Table A.81 reports the deployment-relevant properties of the screen. Pooled over the 55 leave-one-corpus-out folds, the calibration slope of actual on predicted rejection is 0.989, the intercept is +0.0023, and $R^2 = 0.964$. The 95% residual interval covers 92.7% of held-out cells. At the decision threshold there are no false negatives (no raw cell is predicted below $q^\star$ and measured above) and no false positives. On the external suites the mass-only screen is noisier: the pooled MAE is 0.0355, the slope is 1.40 because MT-Bench at 50% is under-predicted by 0.183, but there are still no decision errors. The operational rule is therefore asymmetric: a screen above $q^\star$ is strong evidence of a no-go; a screen below $q^\star$ must be followed by a direct lower-bound measurement.

Table A.81: Calibration and decision errors of the margin-mass screen. The screen is $q = a_w + b_w L$ with $L = \Pr(m < 0.5)$, fitted leave-one-corpus-out on the eleven raw corpora. Coverage is the fraction of held-out cells inside the 95% residual interval. FN is "predicted below $q^\star$, actually above"; FP is the reverse. The external row pools 14 cells: nine raw/deployment cells (five MT-Bench, four SWE-bench) and five in-domain MATH cells. The nine raw/deployment cells have zero decision errors; the MATH cells are broken out in Table A.80.

Sample	n	MAE	Bias	Slope	R^2	Cov.	FN	FP
Raw LOCO	55	0.0228	+0.0001	0.989	0.964	0.927	0	0
External	14	0.0355	+0.0333	1.404	0.894	–	0	0

Table A.82: Margin-mass screen $q = a_w + b_w L$ with $L = \Pr(m < 0.5)$, fitted leave-one-corpus-out on the eleven raw corpora. Panel A gives the pooled coefficients (the LOCO prediction refits on ten corpora), MAE, and maximum absolute error. Panel B varies the low-margin threshold at 98% width; the decision is unchanged for 0.1, 0.25, 0.5, and 1.0. Across all eleven raw corpora the minimum predicted q at these thresholds is 0.013/0.025/0.044/0.076, all above $q^\star = 0.0049$, so the 0.5 cutoff is not tuned. The fitted screen is a deployment diagnostic; below-threshold predictions must be followed by a direct certificate measurement (Appendix A.29).

Width	a_w	b_w	LOCO MAE	max error	$q^\star_{\mathrm{FLOP}}$
0.50	0.0354	2.0446	0.0466	0.1152	0.1400
0.90	0.0141	0.8556	0.0254	0.0489	0.0250
0.95	0.0074	0.6305	0.0181	0.0384	0.0123
0.98	0.0073	0.3932	0.0138	0.0270	0.0049
0.99	0.0099	0.2548	0.0101	0.0198	0.0024

Threshold sensitivity at 98% width.

Corpus	$L(0.1)$	$L(0.25)$	$L(0.5)$	$L(1.0)$	predicted q at 0.1/0.25/0.5/1.0
WikiText	0.037	0.105	0.219	0.399	0.022 / 0.049 / 0.093 / 0.164
FineWeb	0.041	0.120	0.252	0.451	0.024 / 0.055 / 0.106 / 0.185
MATH	0.005	0.014	0.033	0.065	0.009 / 0.013 / 0.020 / 0.033

Table A.83: Raw-side boundary control on a synthetic high-predictability corpus (templated JSON records, repetitive code, and key-value logs; 100 chunks). The target is highly confident (median margin 8.75, low-margin mass 4.3%), yet rejection remains above $q^\star$ at every width because the deletion perturbation is large (mean $|\Delta_m| = 1.2$–3.4). The mass-only screen under-predicts 50% width (0.123 versus measured 0.152 and $q^\star = 0.140$); the two-feature screen with mean $|\Delta_m|$ corrects the decision (0.355). This is a boundary case, not a window.

Width	q	$q^\star$	med. m	low mass	mean $\lvert\Delta_m\rvert$	mass pred.	2-feat pred.
0.50	0.152	0.1400	8.75	0.043	3.41	0.123	0.355
0.90	0.169	0.0250	8.75	0.043	2.29	0.051	0.208
0.98	0.138	0.0049	8.75	0.043	1.50	0.024	0.136
0.99	0.116	0.0024	8.75	0.043	1.17	0.021	0.112

A.29 Screen-to-Audit Procedure for Deployment

Algorithm 1: Screen-to-audit for a new deployment prefix (FFN-width family).

Input: target, prefix sample X, threshold $q^\star$ from (ρ_1, τ_1, γ).

1. Estimate $L = \Pr(m < 0.5)$ on X; predict $q_{\mathrm{pred}} = a_w + b_w L$.
2. If $q_{\mathrm{pred}} > q^\star$, report no-go.
3. Otherwise measure $q_{\mathrm{lb}} = \Pr(m + \Delta_m < 0)$ on X; report a candidate window only if $q_{\mathrm{lb}} < q^\star$, and scale X before deploying. If $q_{\mathrm{lb}} \geq q^\star$, report a no-go.

The screen is a triage rule, not a certificate; a below-threshold prediction must be measured. The template/tool-call control is the boundary case.

The diagnostic is a two-stage screen, not a substitute for measurement.

1. Draw $n = 10^3$–10^4 tokens from the deployment prefix distribution π_{dep}.
2. Estimate the low-margin mass $L = \Pr(m < 0.5)$ and, if draft logits are available, the mean $|\Delta_m|$.
3. Predict q with the leave-one-corpus-out fit of Appendix A.28; compare with $q^\star_{\mathrm{FLOP}}$ (and, if the memory axis is relevant, $q^\star_{\mathrm{mem}}$).

4. If the prediction is below $q^\star$, measure the lower bound directly on the sample ($q_{\mathrm{lb}} = \Pr(m + \Delta_m < 0)$) before declaring a window. The screen under-predicts MT-Bench at 50% width, so a below-threshold prediction is not sufficient evidence.

5. If $q_{\mathrm{lb}} < q^\star$ on the sample, a candidate window remains: scale the sample and re-check before deployment. If $q_{\mathrm{lb}} \geq q^\star$, report a no-go and use a trained/adapted draft or memory-only quantization.

On the measured distributions, stage 4 never opened a raw-text window: the mass-only screen predicted above $q^\star$ for all eleven raw corpora at every width, and the only below-threshold prediction was in-domain MATH at 50% width, where the measurement also stayed below.

Limitations. The main sweep is Qwen3.5-9B; other model/operator checks are smaller (K2 is 100/100). Serving is a reference KV-cache loop, not a production engine; the 9B claim is roofline-type; the agentic check is a static single-turn prefix; quantization uses one INT4/INT8 kernel family on one A100. MT-Bench (multi-turn chat), SWE-bench (agentic issue-to-patch), and the synthetic template/tool-call control are the structurally different deployment distributions. The certificate is in-sample and requires draft logits on the measured tokens. Point q_g and q_s support the no-go as well as q_{lb}.

A.30 Operational OOD Diagnostics, FWER, and Certificate Audit

Statistical analysis plan. The confirmatory family is the 9B English raw-text sweep: 13 widths $(0.05, 0.10, 0.20, 0.30, 0.50, 0.70, 0.80, 0.85, 0.90, 0.95, 0.98, 0.99, 1.00)$ × three corpora (WikiText, FineWeb, C4), 39 tests; the 36 compressed cells use Bonferroni-adjusted 99.86% sequence-cluster intervals ($\alpha = 0.05/36$), and $\alpha = 0.05/39$ is a conservative sensitivity. The three 100%-width cells are identity controls ($q = q_{\mathrm{lb}} = 0$) and are not tests of the no-go; the 36 compressed cells are the confirmatory set. The in-domain MATH sweep is analyzed as a separate 13-test family ($\alpha = 0.05/13$), and the nine Llama/K2 cells form a separate cross-family family ($\alpha = 0.05/9$); the Qwen cells are not counted twice. The fourth-round breadth family (48 cells), the cross-model/operator families, and the 218-cell audit are secondary/exploratory: they use 95% cluster intervals, and the breadth family additionally reports a 99.8% one-sided lower bound. The 200-cell raw-text count in the main text is a descriptive summary, not a family-wise test. The 99.99% one-channel floor and the clean float32 control are post-hoc mechanism checks; they are not members of either confirmatory family and are reported with cluster intervals where available.

Counting conventions. The counts used in the paper are nested or overlapping sets, not separate families. The confirmatory family was fixed in the experiment log before the reported runs; this is a lab-notebook freeze, not a public pre-registration. It is $36 + 9$: 36 compressed Qwen English raw cells (12 widths × three corpora) and 9 independent Llama/K2 raw cells. Each family is controlled at $\alpha = 0.05$; the joint union bound across the two confirmatory families is at most 0.10, and we state this rather than claiming a single 0.05 level for the $36 + 9$ body. The 39-test family is the 36 compressed Qwen cells plus three 100%-width identity controls. The 45-test pooled family is the 36 Qwen cells plus the 9 Llama/K2 cells when the Qwen cells are reused; it is reported only to explain why we do not use it as an independent confirmation. The 55-cell breadth suite is 11 corpora × five widths (50–99%). The 99-cell cross-family point count and the 135-cell union ($36 + 99$) are descriptive. The 200-cell raw count and the 218-cell audit are descriptive; only $36 + 9$ carries the confirmatory claim. Table A.88 gives all 36 confirmatory cells with the adjusted lower endpoints, including C4; Table A.89 collects the counts and their roles in one place so that a reader does not have to reconstruct the family structure from the prose.

All intervals resample whole sequences with replacement (2,000 draws, seed 0), not tokens. With 100 sequences and 512 tokens per sequence, the cluster-bootstrap half-width on q_{lb} is about 0.003 at 98–99% width; the smallest certified margin is +0.0320, so the design is not power-limited at the reported widths. Table A.84 lists the cluster counts and token counts; the smallest raw-text clusters are 25 sequences in the 4k long-context check, which is why that check is reported as qualitative and its cells are not used to carry the main claim.

Table A.84: Evaluation sets and cluster counts. "Seqs" is the number of independent sequences resampled by the cluster bootstrap; "prompt / scored" are token counts per sequence (variable prompts are marked ≤). Raw-text sets use 512/512 chunks unless noted. The cluster bootstrap resamples whole sequences, not tokens.

Set	Seqs	Prompt / scored	Source
MATH-500 (9B)	500	variable / ≤ 512	MATH-500 target-generated
MATH-500 (K2 100)	100	variable / ≤ 256	K2 target-generated
WikiText raw	100	512 / 512	WikiText-2 test
FineWeb raw	100	512 / 512	FineWeb-Edu
C4 raw	100	512 / 512	C4
Wikipedia zh/de/fr/ja	100 each	512 / 512	Wikimedia 20231101
FineWeb-2 zh	100	512 / 512	FineWeb-2 cmn_Hani
Code raw	100	512 / 512	CodeParrot clean
PG-19 / arXiv	100 each	512 / 512	PG-19 / arXiv abstracts
PG-19 / arXiv 4k	25 each	2048 / 2048	four concatenated chunks
SWE-bench-Lite	100	≤1024 / ≤512	issue + hints → patch
MT-Bench chat	80	variable / ≤512	first-turn chat/instruction
K2-Horizon-3.7B raw	100	512 / 512	WikiText raw (100/100)
Llama cross-family	100 each	variable / ≤ 512	Llama-3.1-8B target

Target NLL and calibration overlap. Table A.85 reports the target NLL/perplexity on the scored continuation and the fraction of evaluation word 8-grams that occur in the calibration corpus. Raw WikiText and FineWeb have much higher target NLL than target-generated text, and zero 8-gram overlap with calibration. MATH has low NLL and 21% overlap because the calibration corpus contains the MATH evaluation prompts; Appendix A.4 removes them and shows that the in-domain window is unchanged. These are operational checks, not a contamination certificate: the target's pretraining data are not public. If the target checkpoint or its data provenance changes, the certificate must be recomputed on the new evaluation tokens; the margin-mass screen is a deployment-time check, not a property of a corpus name.

Table A.85: Operational OOD diagnostics on Qwen3.5-9B. NLL/perplexity are on the scored continuation; overlap is the fraction of evaluation word 8-grams that also occur in the calibration corpus. Raw text has much higher target NLL and zero calibration overlap. This is not a contamination certificate because the target's pretraining data are not public.

Set	NLL	Perplexity	8-gram overlap
MATH-500	0.065	1.07	21.1%
WikiText raw	2.156	8.64	0.0%
FineWeb raw	2.103	8.19	0.0%
WikiText gen	0.187	1.21	0.0%
FineWeb gen	0.208	1.23	0.0%

Family-wise error and certificate failures. The raw family has 39 tests, of which three are identity controls. All 36 compressed raw cells are certified at the Bonferroni level $\alpha = 0.05/36$; the more conservative $\alpha = 0.05/39$ leaves every decision unchanged, and there are no bad raw cells. The in-domain MATH family is separate: 5 of its 12 compressed cells are point-above but adjusted-lower-bound-below (Table A.87), which is the in-domain window discussed in the main text. In the exploratory 218-cell audit (before the fourth-round breadth, head, and combined additions), 26 cells are point-above but adjusted-lower-bound-below; all are in-domain MATH. The fourth-round additions are summarized separately in Appendix A.21. Table A.86 gives the per-family counts and Table A.87 the individual cells. In the descriptive 200-cell raw summary, all 200 cells are point-above and 199 are certified; the only point-above cell that is not certified is code at 50% width (point $q_g = 0.221$, $q_{\mathrm{lb}} = 0.128$ with interval upper endpoint 0.139, $q^\star = 0.140$). The bound is measurement-based and is not claimed to be tight in every cell.

Table A.86: Family-wise certificate audit with explicit family composition. The Qwen raw family is 13 widths $\times$ three raw corpora (WikiText, FineWeb, C4), 39 tests; the three 100%-width identity cells have $q = q_{\mathrm{lb}} = 0$ and are not no-go tests, so the 36 compressed cells are certified at $\alpha = 0.05/36$; $\alpha = 0.05/39$ is a conservative sensitivity. Llama and K2 form a separate 9-test family at $\alpha = 0.05/9$; the two confirmatory families are each at 0.05, so their joint union bound is at most 0.10. The MATH family is analyzed separately: 5 of its 12 compressed cells are point-above but adjusted-lower-bound-below. We do not use a pooled 45-test family; it would reuse the Qwen cells. "Bad" means point rejection above $q^\star$ but adjusted lower bound at or below $q^\star$.

Family	Tests	Compressed	Certified	Bad	Identity	α
Qwen3.5-9B raw (Wiki/FineWeb/C4)	39	36	36	0	3	0.00139
Llama + K2 raw (independent)	9	9	9	0	0	0.00556
Qwen3.5-9B MATH (in-domain)	13	12	5	5	1	0.00385

Table A.87: The 26 cells in the exploratory 218-cell audit where point rejection is above the frontier but the family-adjusted lower bound is not. All are in-domain MATH; the 9B raw family has no bad cells (Table A.86). The five 9B MATH rows use the 13-test MATH family correction; the remaining rows are exploratory cross-model MATH cells with their own 95%/Bonferroni-adjusted intervals.

Family	Set	Width	q	q_{lb}	$q^\star$	adj. q_{lb} lower
k2-horizon-3.7b	math_k2	0.2	0.6062	0.2534	0.2677	0.2225
k2-horizon-3.7b	math_k2	0.3	0.5312	0.2182	0.2245	0.1878
k2-horizon-3.7b	math_k2	0.5	0.3352	0.1742	0.1487	0.1423
llama31-8b	math_llama	0.2	0.5830	0.2616	0.3351	0.2406
llama31-8b	math_llama	0.3	0.3249	0.1661	0.2770	0.1511
llama31-8b	math_llama	0.85	0.0525	0.0407	0.0472	0.0353
llama31-8b	math_llama	0.9	0.0430	0.0338	0.0310	0.0287
qwen3-4b	math_qwen3	0.1	0.8425	0.2960	0.3358	0.2731
qwen3-4b	math_qwen3	0.2	0.4029	0.1754	0.2836	0.1572
qwen3-4b	math_qwen3	0.3	0.2846	0.1396	0.2371	0.1254
qwen3-4b	math_qwen3	0.5	0.1675	0.1039	0.1563	0.0916
qwen3-4b	math_qwen3	0.7	0.1097	0.0770	0.0876	0.0672
qwen3-4b	math_qwen3	0.8	0.0849	0.0632	0.0566	0.0553
qwen3.5-27b	math27b	0.05	0.5444	0.2270	0.3638	0.2216
qwen3.5-27b	math27b	0.1	0.3389	0.1586	0.3352	0.1535
qwen3.5-27b	math27b	0.95	0.0166	0.0139	0.0135	0.0129
qwen3.5-4b	math4b	0.1	0.5176	0.2246	0.2482	0.2190
qwen3.5-4b	math4b	0.2	0.3395	0.1666	0.2131	0.1618
qwen3.5-4b	math4b	0.3	0.2553	0.1379	0.1805	0.1336
qwen3.5-4b	math4b	0.5	0.1542	0.0995	0.1216	0.0958
qwen3.5-4b	math4b	0.7	0.0973	0.0720	0.0693	0.0687
qwen3.5-9b MATH	MATH	0.05	0.5846	0.2415	0.3168	0.2365
qwen3.5-9b MATH	MATH	0.1	0.4350	0.1917	0.2933	0.1870
qwen3.5-9b MATH	MATH	0.2	0.2808	0.1417	0.2498	0.1375
qwen3.5-9b MATH	MATH	0.7	0.0831	0.0629	0.0791	0.0604
qwen3.5-9b MATH	MATH	0.8	0.0637	0.0504	0.0513	0.0481

Table A.88: The 36-cell confirmatory Qwen raw family with Bonferroni-adjusted lower endpoints. Each entry is the greedy rejection q and the lower endpoint of the 99.86% sequence-cluster interval for q_{lb} (family level $\alpha = 0.05/36$; the $\alpha = 0.05/39$ sensitivity leaves every decision unchanged). The cell is certified when the adjusted lower endpoint exceeds $q^{\star}_{\mathrm{FLOP}}$. All 36 cells are certified; the minimum adjusted margin is +0.0320 at FineWeb 99% width.

Width	Wiki q	Wiki lb	FineWeb q	FineWeb lb	C4 q	C4 lb	cert.
0.05	0.8319	0.3789	0.8161	0.3759	0.8375	0.3843	✓
0.10	0.7782	0.3509	0.7616	0.3500	0.7954	0.3616	✓
0.20	0.7033	0.3289	0.6914	0.3241	0.7263	0.3406	✓
0.30	0.6565	0.3186	0.6284	0.3038	0.6728	0.3216	✓
0.50	0.5017	0.2570	0.5047	0.2556	0.5538	0.2769	✓
0.70	0.3618	0.2030	0.3643	0.1959	0.3899	0.2083	✓
0.80	0.2876	0.1699	0.2926	0.1621	0.3070	0.1708	✓
0.85	0.2513	0.1507	0.2518	0.1409	0.2663	0.1505	✓
0.90	0.2080	0.1267	0.2085	0.1180	0.2228	0.1285	✓
0.95	0.1502	0.0931	0.1551	0.0864	0.1644	0.0932	✓
0.98	0.0968	0.0591	0.0959	0.0524	0.1047	0.0569	✓
0.99	0.0677	0.0418	0.0649	0.0344	0.0717	0.0372	✓

Table A.89: Counting conventions. The confirmatory claim is the $36 + 9$ family only. The counts are nested or overlapping sets, not separate families; the other counts are descriptive or exploratory and are not pooled into a family-wise test. "Point-above" means $q_g > q^{\star}$; "certified" means the lower endpoint of the interval for q_{lb} exceeds $q^{\star}$.

Count	Cells	Point-above	Certified	Role / exception
raw36	36	36	36	confirmatory Qwen English raw family
xfam9	9	9	9	confirmatory Llama/K2 raw family
raw39	39	–	–	36 compressed + 3 identity controls (FWER denominator)
pooled45	45	–	–	not used; reuses the 36 Qwen cells
raw200	200	200	199	descriptive; code @ 50% is point-only
audit218	218	–	–	exploratory; 26 bad, all in-domain MATH
breadth55	55	55	54	descriptive; 11 corpora × five widths
breadth48	48	–	–	fourth-round exploratory breadth family
xfam99	99	–	–	descriptive cross-family point count
union135	135	–	–	descriptive union $36 + 99$

Non-overlapping confirmatory families. The 39-test family is Qwen-only. Pooling its 36 compressed cells with six Llama-3.1-8B and three K2-Horizon-3.7B raw WikiText cells into a single 45-test family would double-count the Qwen cells: they are already the main family, so the pooled Bonferroni level is not an independent confirmation. We therefore report two non-overlapping families: the 36 compressed Qwen English raw cells ($\alpha = 0.05/36$) and the nine Llama/K2 raw cells ($\alpha = 0.05/9$). Both are fully certified (Table A.91). The 39-test count includes the three 100%-width identity controls; the correction used for the confirmatory 36 compressed cells is $\alpha = 0.05/36$. Bonferroni is conservative under the positive dependence induced by overlapping widths and corpora, so it is a safe choice here; using the still more conservative $\alpha = 0.05/39$ leaves every decision unchanged. The smallest adjusted margin is +0.0320 on Qwen FineWeb at 99% width and +0.0330 on Llama FineWeb at 99% width. A two-level bootstrap that resamples corpora/models and then sequences gives a descriptive stability check of +0.0333 and +0.0344 on the minimum margin, with a Monte Carlo frequency $\Pr(\text{all cells above } q^{\star}) = 1.000$ in both families. With only three Qwen corpora and two cross-family groups, this bootstrap is not a new family-wise confidence guarantee; the confirmatory intervals are the Bonferroni-adjusted cluster intervals, and the two-level numbers are reported as a stability check. The independent cross-family check therefore survives without reusing the Qwen cells.

Table A.90 reports the q_g/q_{lb} consistency matrix.

Table A.90: Decision consistency of the greedy point criterion ($q_g > q^\star$) and the audit-passed top-2 lower bound (lower endpoint of $q_{\mathrm{lb}}^{(2)} > q^\star$). On the 36 confirmatory Qwen raw cells and the 9 independent Llama/K2 cells the two criteria agree. In the 55-cell breadth suite, the only raw exception is code at 50% width (point-above, not audit-passed). Apart from that code cell, all disagreements are in-domain MATH, where the point criterion is optimistic. The lower bound therefore buys robustness, not a different raw-text verdict.

Cell set	Compressed cells	Both above	Both below	q_g only	q_{lb} only
Qwen3.5-9B raw (Wiki/FineWeb/C4)	36	36	0	0	0
Cross-family raw (4B, 27B, Qwen3-4B, Llama, K2)	99	99	0	0	0
Raw breadth (11 corpora, 50–99%)	55	54	0	1	0
In-domain MATH (five targets)	67	33	11	23	0

Table A.91: Non-overlapping confirmatory families. The 36 Qwen raw cells form the main family; the nine Llama/K2 cells form an independent cross-family family. The previous 45-test family double-counted the Qwen cells. The last two columns are a descriptive two-level bootstrap (corpora/models, then sequences) for the minimum margin; they are a stability check, not a new family-wise confidence level. All raw cells are certified in both families; the MATH family is the in-domain exception.

Family	Tests	Cert.	Bad	α	Min margin	Hier. lower	Pr(all > 0)
Qwen raw (Wiki/FineWeb/C4)	36	36	0	0.00139	+0.0320	+0.0333	1.000
Llama + K2 raw	9	9	0	0.00556	+0.0330	+0.0344	1.000
Qwen MATH (in-domain)	13	5	5	0.00385	–	–	–

A.31 Service Benchmark and Fused-Kernel Limit

What would change the wall-clock conclusion? (i) Compute-bound batching can change absolute tokens/s, but the roofline threshold still falls with γ and raw-text rejection remains above it. (ii) Fused/compiled kernels can move ρ_1 toward the roofline; the clean decomposition shows that a free FFN would put the cost below 5 on most targets, so a production MLP kernel is the main remaining cost-side lever. (iii) If an engine makes the target disproportionately inefficient, the measured ratio can fall below the roofline; that is an engine artifact, not a property of the operator. (iv) Qwen3.5's hybrid linear-attention cache does not support exact rollback in our implementation, so we report fair loops on three dense models instead. (v) Memory-only compression (quantization, expert deletion at fixed k) reduces memory but not the dominant per-token compute. (vi) Distillation/adaptation changes the draft distribution and is outside this audit; we do not rank it against the post-hoc operator. Section 3.4 shows that any positive result from it must cover the deployment distribution, not only the adaptation set.

Measured cost is a service benchmark, not the general claim. The batch-1 measurement uses a padded narrow MLP and ordinary dense GEMMs. It is an upper bound on ρ_1, not a fused-kernel lower bound. A fused kernel can remove padding and launch overhead and approach ρ_{roof} and $\tau = 1$, which is exactly the frontier we evaluate; it cannot go below the frontier. The best possible fused-kernel speedup at measured rejection is therefore the roofline-model speedup, which is below 1 on raw text at every compressed width.

Roofline derivation. Let F_D, F_T be the matmul FLOPs and M_D, M_T the bytes moved per pass. A roofline implementation on hardware (P, W) has

$$T_D^{\mathrm{roof}} = \max(F_D/P, M_D/W), \qquad T_T^{\mathrm{roof}} = \max(F_T/P, M_T/W), \qquad \rho_{\mathrm{roof}} = \min(\rho_{\mathrm{FLOP}}, \rho_{\mathrm{mem}}). \tag{41}$$

If the target is compute-bound, $T_T^{\mathrm{roof}} = F_T/P$ and the ratio is at least $F_D/F_T = \rho_{\mathrm{FLOP}}$; if it is memory-bound, $T_T^{\mathrm{roof}} = M_T/W$ and the ratio is at least ρ_{mem}. Hence $\rho_B^{\mathrm{roof}} \geq \rho_{\mathrm{roof}}$. This is an optimistic same-kernel/equal-efficiency comparison, not a wall-clock theorem for arbitrary engines.

For weight streaming, the compressed FFN moves the same fraction of its weights, while attention, KV cache, embeddings, and the LM head are unchanged. The bytes ratio for the FFN-width family is

$$\rho_{\mathrm{mem}} = 1 - \phi_{\mathrm{MLP}}^{\mathrm{bytes}}(1 - r/h), \tag{42}$$

Table A.92: End-to-end greedy SD throughput (batch 1, $\gamma = 4$, 16–64 generated tokens, same prompts and GPU for the target-only and speculative runs). Speedup is speculative tokens/s over target-only tokens/s. The dense rows use exact KV rollback. The 9B negative-crop rows use the hybrid cache's negative crop; that loop is optimistic and its identity check drifts after several blocks. The exact re-prefill rows remove rollback entirely and are a cost upper bound; they are still below one.

Target	Loop	Width	Speedup	α_{acc}	$\bar{\ell}$
Llama-3.1-8B	dense KV rollback	0.50	0.188	0.020	1.10
Llama-3.1-8B	dense KV rollback	0.98	0.381	0.347	2.40
Qwen3-4B	dense KV rollback	0.98	0.351	0.287	2.18
K2-Horizon-3.7B	dense KV rollback	0.98	0.320	0.246	1.99
Qwen3.5-9B	exact re-prefill (main, upper bd.)	0.50	0.506	0.603	3.41
Qwen3.5-9B	exact re-prefill (main, upper bd.)	0.98	0.652	0.825	4.47
Qwen3.5-9B	hybrid negative-crop (optimistic)	0.50	0.448	0.532	3.16
Qwen3.5-9B	hybrid negative-crop (optimistic)	0.98	0.495	0.632	3.56

where $\phi_{\text{MLP}}^{\text{bytes}}$ is the FFN share of parameter bytes. For Qwen3.5-9B, $\phi_{\text{MLP}}^{\text{bytes}} = 0.540$, slightly below the FLOP share 0.609; the bytes frontier is therefore slightly stricter than the FLOP frontier ($\rho_{\text{mem}} > \rho_{\text{FLOP}}$), so $\rho_{\text{roof}} = \rho_{\text{FLOP}}$. Table A.95 compares the two. Including KV-cache bytes makes the FFN share smaller and the frontier stricter. Thus a memory-bound serving model does not reopen the raw-text window; the FLOP frontier is the optimistic compute-bound limit, and the padded narrow-MLP implementation under the clean protocol is above both.

The 9B hybrid-cache loop. The main model does not support exact rollback because Qwen3.5's linear-attention cache rejects a positive `crop`. We enable `activate_past_recording()` and use the negative crop that the cache documents (remove the last n positions). An identity draft is exact for short generations but drifts after several blocks: at 16 new tokens it gives $\alpha = 1.0$; at 24 tokens it gives $\alpha = 0.975$ on two prompts and 0.892 on four prompts. We therefore take the exact re-prefill loop as the main number: it removes rollback entirely and re-prefills the committed prefix after every block, giving 0.506× at 50% width and 0.652× at 98% width. This is exact but a cost upper bound. The negative-crop loop is an optimistic reference (0.448× and 0.495× at the same widths); its cache drift is small but nonzero. Both are below one, so the 9B result is not an artifact of the approximate cache. The loop acceptance ($\alpha = 0.53$–0.82) is lower than the teacher-forced top-1 agreement because the loop visits draft-generated prefixes; this is the same prefix-shift effect documented in Table A.93. The loops use a 256-token prefix and generate 64 tokens, so prefill is excluded and the numbers are decode-loop sanity checks, not end-to-end serving throughput. Only the 9B identity control is measured; the dense models do not have a separate identity-loop control, and we do not use these loops as the general cost claim.

Table A.93: Real standard-SD loop versus the cache estimate. Each cell uses 15 prompts, $\gamma = 4$, and 96 new tokens. α_{emp} is accepted/proposed; α_{tok} is the mean per-position acceptance probability $1 - \text{TV}$ on reached prefixes; $\bar{\ell}$ is emitted tokens per target call. The cache columns use the target-greedy continuations, so the ratio is a distribution mismatch diagnostic, not an error of the estimator. With 15 prompts the per-cell intervals are wide; the per-prompt JSONs are retained, and we report medians as a sanity check rather than a precision estimate.

Cell	α_{emp}	α_{tok}	$\mathbb{E}$[TV] loop	$\bar{\ell}$ loop	$\mathbb{E}$[TV] cache	$\bar{\ell}$ sim / ratio
math500 / 0.8B	0.652	0.841	0.147	3.61 / 3.88 (0.93)	0.115	1.276
math500 / 2B	0.744	0.886	0.108	3.98 / 4.18 (0.95)	0.083	1.294
math500 / 4B	0.828	0.933	0.070	4.31 / 4.45 (0.97)	0.055	1.291
math500 / 9B	0.836	0.931	0.061	4.34 / 4.52 (0.96)	0.048	1.276
math500 / 27B	0.903	0.953	0.046	4.61 / 4.61 (1.00)	0.039	1.199
wikitext / 0.8B	0.395	0.648	0.352	2.58 / 2.34 (1.10)	0.385	0.914
wikitext / 2B	0.426	0.688	0.308	2.70 / 2.67 (1.01)	0.314	0.982
wikitext / 4B	0.498	0.740	0.261	2.99 / 3.02 (0.99)	0.251	1.040
wikitext / 9B	0.511	0.754	0.246	3.04 / 3.19 (0.95)	0.224	1.101
wikitext / 27B	0.592	0.800	0.200	3.37 / 3.41 (0.99)	0.191	1.052

Table A.94 summarizes the serving regimes.

Table A.94: Regime × assumption map. The rejection layer is the $c = 1$ same-kernel roofline statement; the cost layer is conditional on the realized draft-kernel efficiency c and on the serving regime. The 98–99% cells are an arithmetic cost conditional on a whole-pass draft-specific kernel, which is not realizable when 98–99% of the pass is shared; they are not a deployable window. The mid-width and batch-1 memory-bound cells are rejection audit failures.

Regime	$c = 1$ (same-kernel roofline)	$c = 1.3$–1.5 (faster draft kernel)
Compute-bound (large batch)	Rejection-layer audit failure: $q_{\text{lb}} > q^{\star}_{\text{FLOP}}$ at every compressed width.	50% width needs $c^{\star} \approx 3.0$–6.1 at $\tau = 1$ (≥ 4.6 at τ_1); 98–99% would need 1.2–1.4× on a whole pass, which is not a draft-specific object.
Memory-bound (batch 1)	Measured $4\rho_1 + \tau_1 > 5$ at every compressed width; the clean kernel-time zero-MLP cost is 3.1–$4.1 < 5$, so the audit failure is the realized kernel plus rejection, not a zero-FFN ceiling.	A whole-draft kernel can move ρ_1 toward ρ_{mem}; for this model $\rho_{\text{mem}} \approx \rho_{\text{FLOP}}$, so the same frontier applies. Our fused-MLP and compiled kernels are not faster.

Table A.95: FLOP, weight-streaming bytes, and roofline frontiers for Qwen3.5-9B. The bytes share of the FFN is 0.540, below the FLOP share 0.609, so $\rho_{\text{roof}} = \rho_{\text{FLOP}}$ for every row; the redundant roof columns are omitted. At 30% and 50% width, MATH rejection is 0.2098 and 0.1297, above $q^{\star}_{\text{mem}} = 0.1812$ and 0.1220: the in-domain window is closed on the memory frontier as well, and exists only under the FLOP frontier with $\tau = 1$. The measured raw-text rejection is above all frontiers at every width.

Width	ρ_{FLOP}	$q^{\star}_{\text{FLOP}}$	ρ_{mem}	$q^{\star}_{\text{mem}}$	Wiki q	FineWeb q	MATH q
0.05	0.422	0.3168	0.487	0.2679	0.8319	0.8161	0.5846
0.1	0.452	0.2933	0.514	0.2491	0.7782	0.7616	0.4350
0.2	0.513	0.2498	0.568	0.2139	0.7033	0.6914	0.2808
0.3	0.574	0.2102	0.622	0.1812	0.6565	0.6284	0.2098
0.5	0.696	0.1400	0.730	0.1220	0.5017	0.5047	0.1297
0.7	0.817	0.0791	0.838	0.0695	0.3618	0.3643	0.0831
0.8	0.878	0.0513	0.892	0.0452	0.2876	0.2926	0.0637
0.9	0.939	0.0250	0.946	0.0221	0.2080	0.2085	0.0399
0.95	0.970	0.0123	0.973	0.0109	0.1502	0.1551	0.0260
0.98	0.988	0.0049	0.989	0.0043	0.0968	0.0959	0.0157
0.99	0.994	0.0024	0.995	0.0022	0.0677	0.0649	0.0112